



US012412716B2

(12) **United States Patent**
Okuda et al.

(10) **Patent No.:** **US 12,412,716 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **ELECTROMAGNETIC RELAY**

(56) **References Cited**

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(72) Inventors: **Akihiro Okuda, Mie (JP); Machiko Nishiyama, Mie (JP)**

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(73) Assignee: **Panasonic Intellectual Property Management Co., Ltd., Osaka (JP)**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 216 days.

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(21) Appl. No.: **18/259,592**

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(22) PCT Filed: **Dec. 9, 2021**

International Search Report of PCT application No. PCT/JP2021/045405 dated Mar. 1, 2022.

(86) PCT No.: **PCT/JP2021/045405**

§ 371 (c)(1),

(2) Date: **Jun. 28, 2023**

Primary Examiner — Bernard Rojas

(74) Attorney, Agent, or Firm — Rimon P.C.

(87) PCT Pub. No.: **WO2022/149404**

PCT Pub. Date: **Jul. 14, 2022**

(57) **ABSTRACT**

(65) **Prior Publication Data**

US 2024/0136133 A1 Apr. 25, 2024
US 2024/0234063 A9 Jul. 11, 2024

An electromagnetic relay includes: a housing including a base and a case that opens downward and is provided on the base, the housing having an internal space surrounded by the base and the case; a main fixed contact; a main moving contact that comes into contact with and separates from the main fixed contact; an auxiliary fixed contact; an auxiliary moving contact that comes into contact with and separates from the auxiliary fixed contact; a coil provided in the housing such that a central axis extends in a vertical direction; an iron core disposed so as to be surrounded by the coil; a yoke disposed around the coil; an armature that is disposed to face an upper end of the iron core and swings in response to excitation/non-excitation of the coil; a movable part provided with the main moving contact and configured to move in response to swinging of the armature; and a partition wall that divides the internal space of the housing into a first space and a second space. The main fixed contact and the main moving contact are disposed in the first space, and the auxiliary fixed contact and the auxiliary moving contact are disposed in the second space.

(30) **Foreign Application Priority Data**

Jan. 6, 2021 (JP) 2021-000828

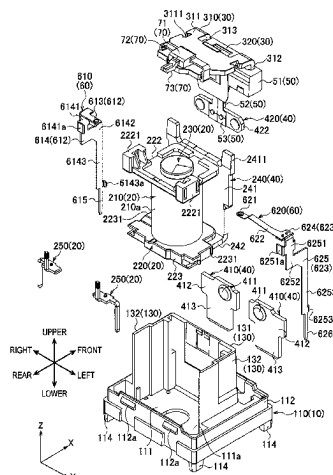
(51) **Int. Cl.**
H01H 50/54 (2006.01)
H01H 50/26 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H01H 50/541** (2013.01); **H01H 50/26** (2013.01); **H01H 50/36** (2013.01); **H01H 50/546** (2013.01); **H01H 2050/446** (2013.01)

(58) **Field of Classification Search**
None

See application file for complete search history.

15 Claims, 44 Drawing Sheets



- (51) **Int. Cl.**
H01H 50/36 (2006.01)
H01H 50/44 (2006.01)

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FIG. 1

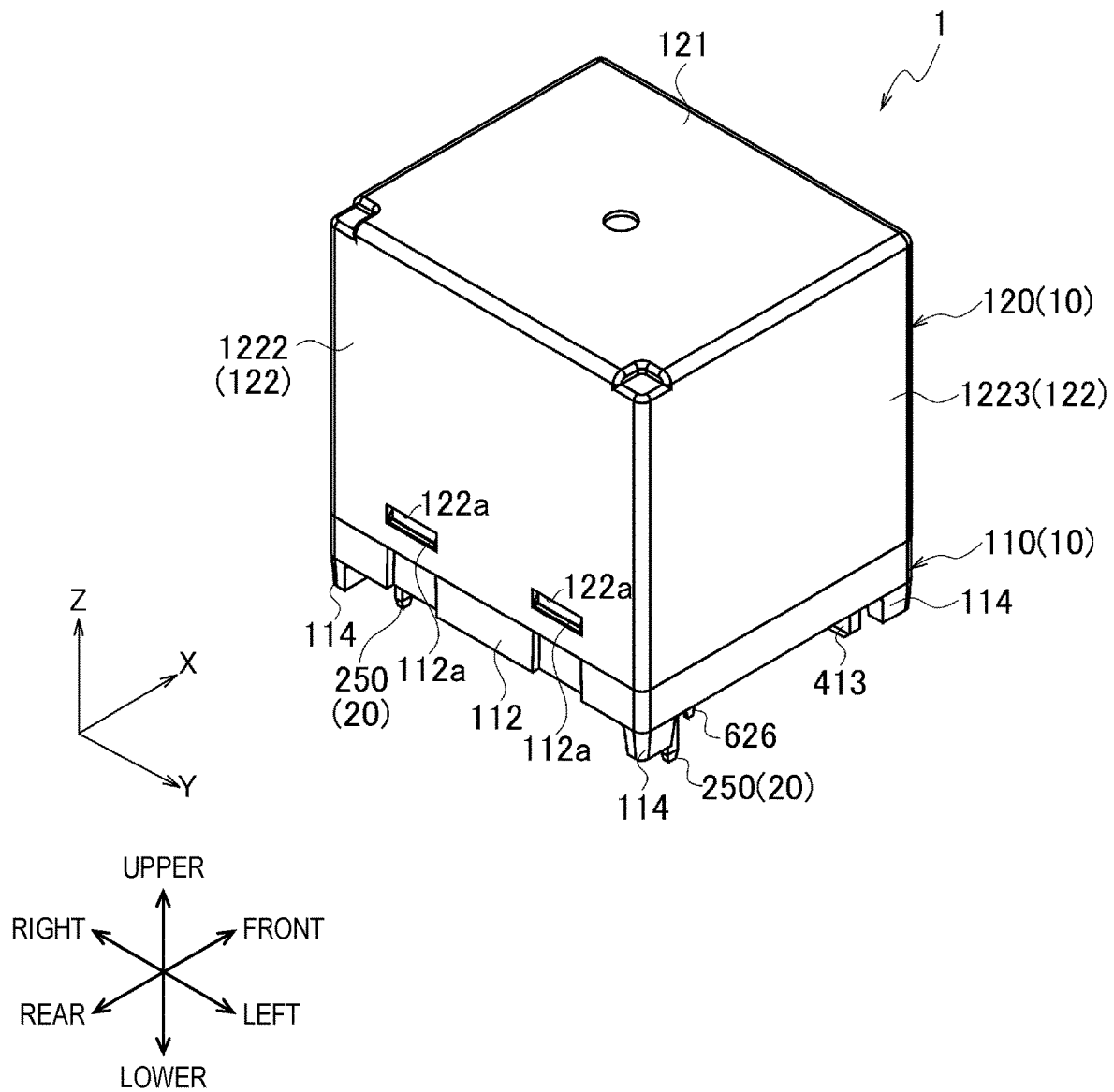


FIG. 2

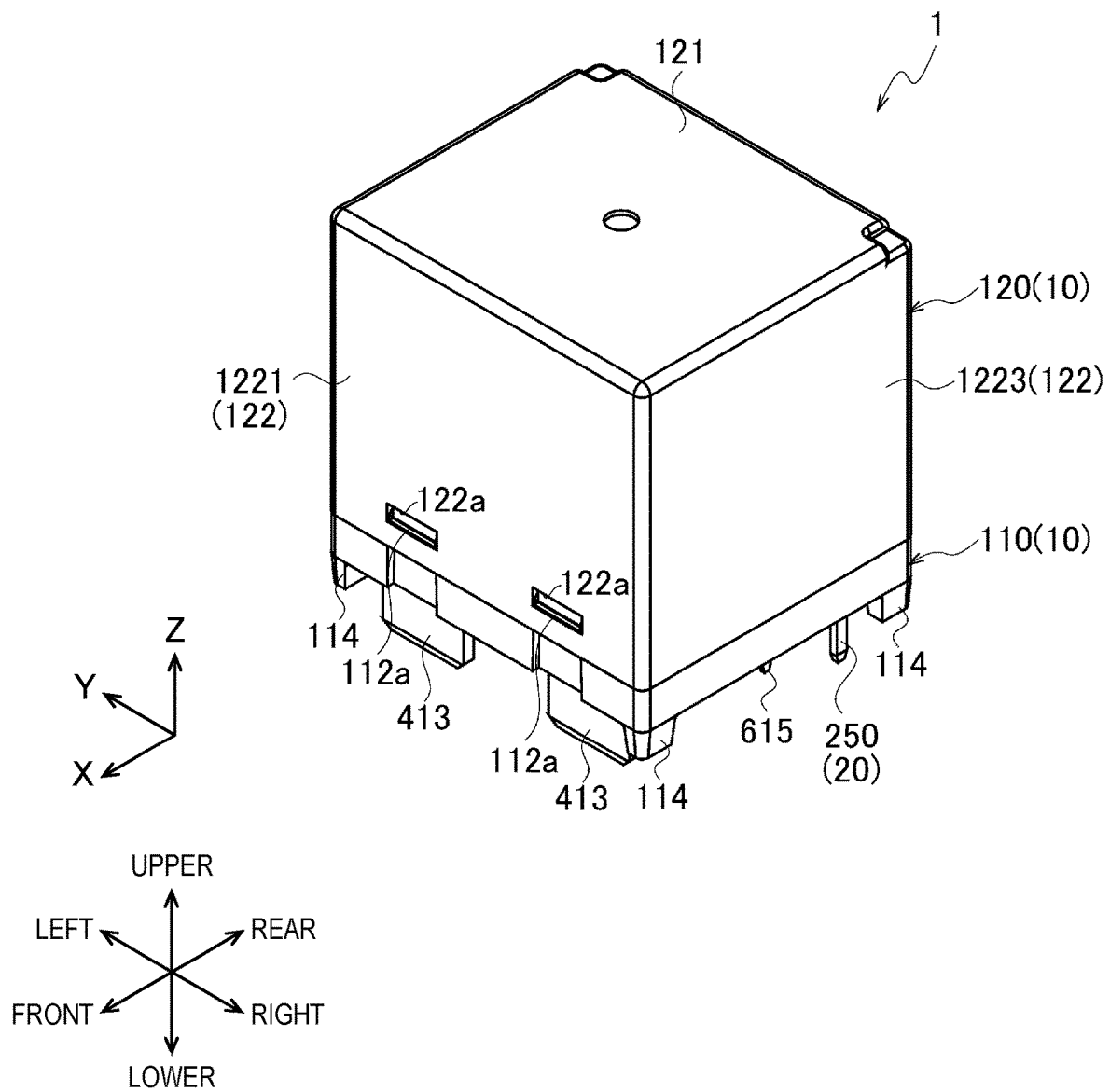


FIG. 3

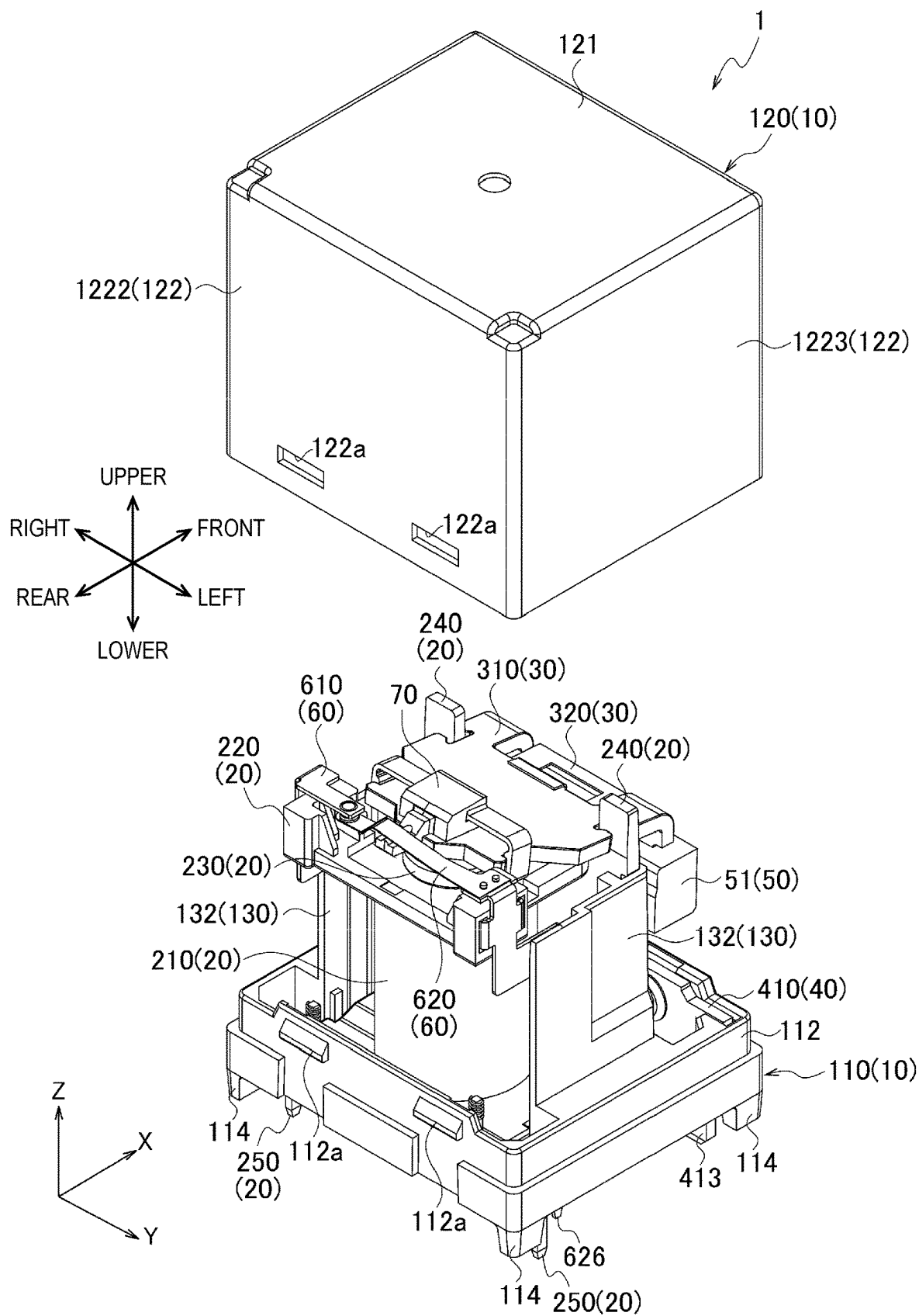


FIG. 4

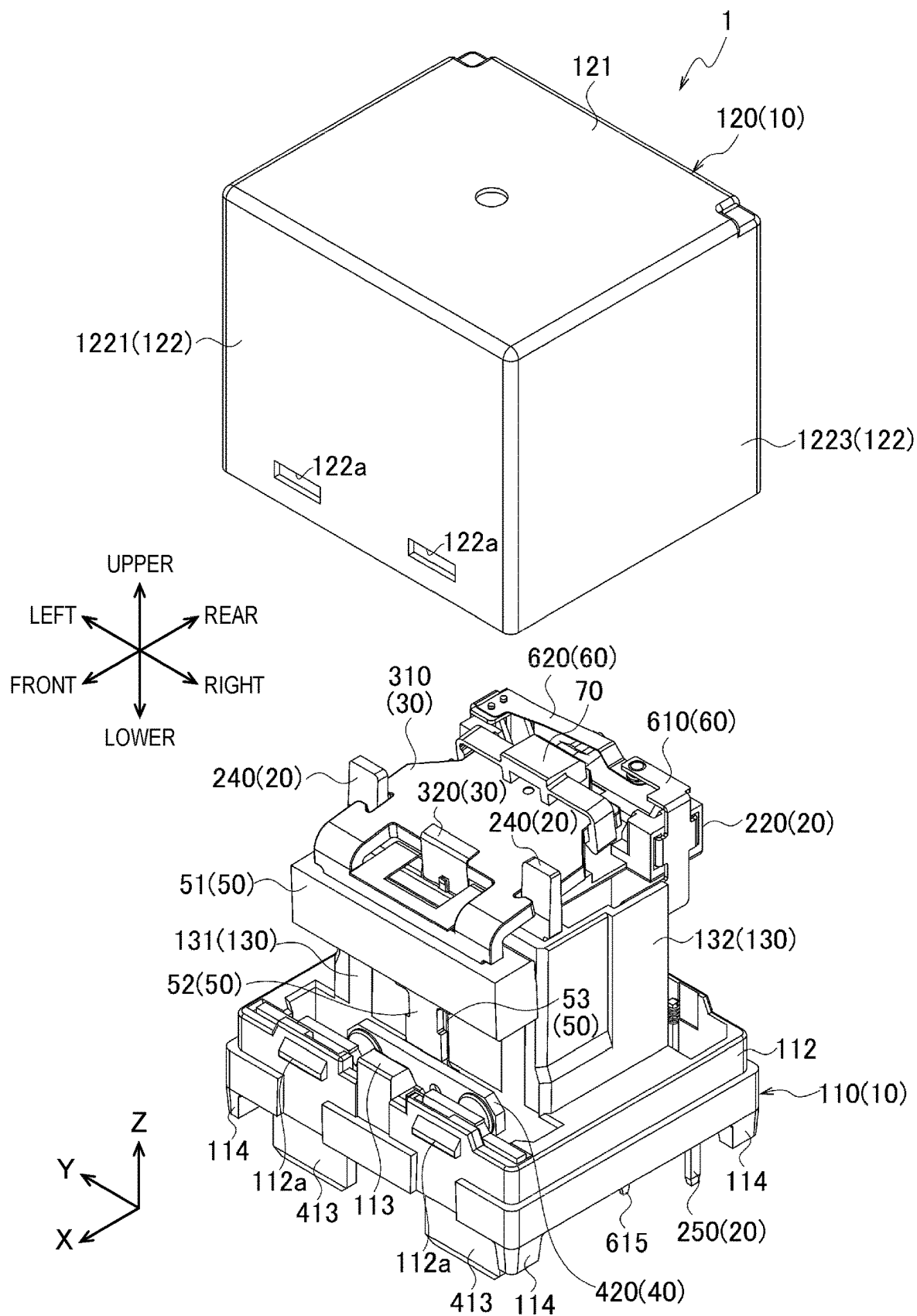


FIG. 5

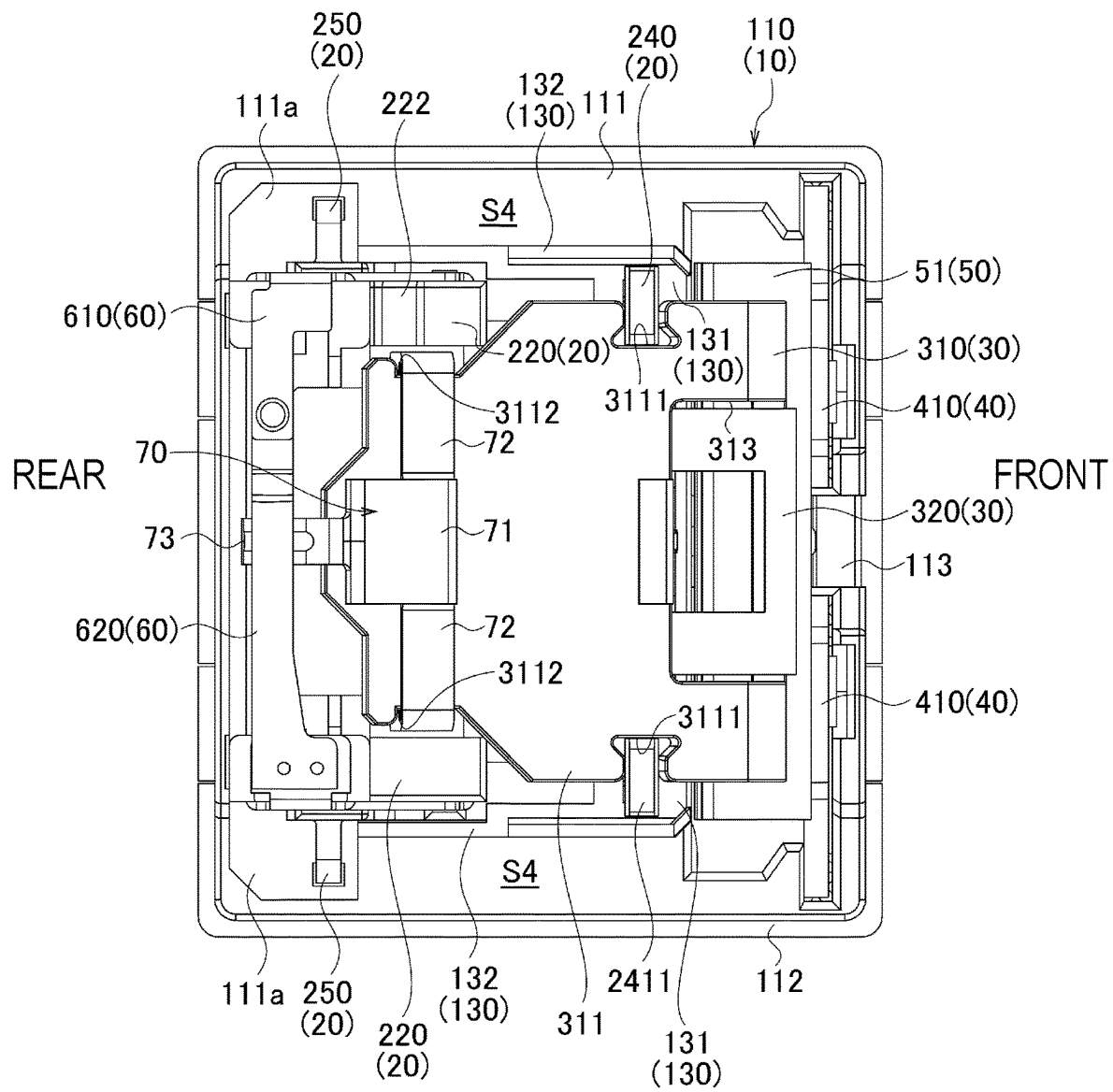


FIG. 6

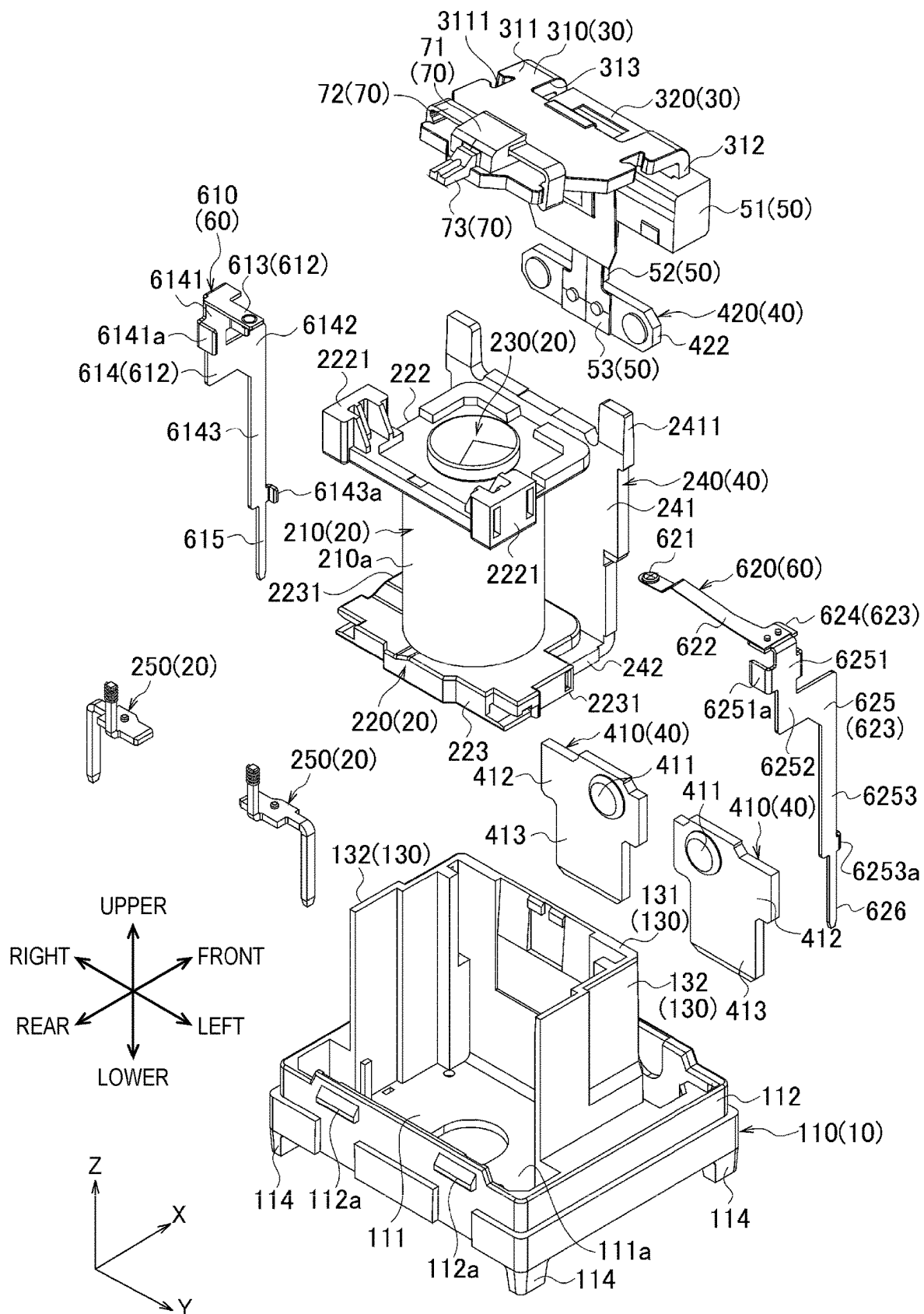


FIG. 7

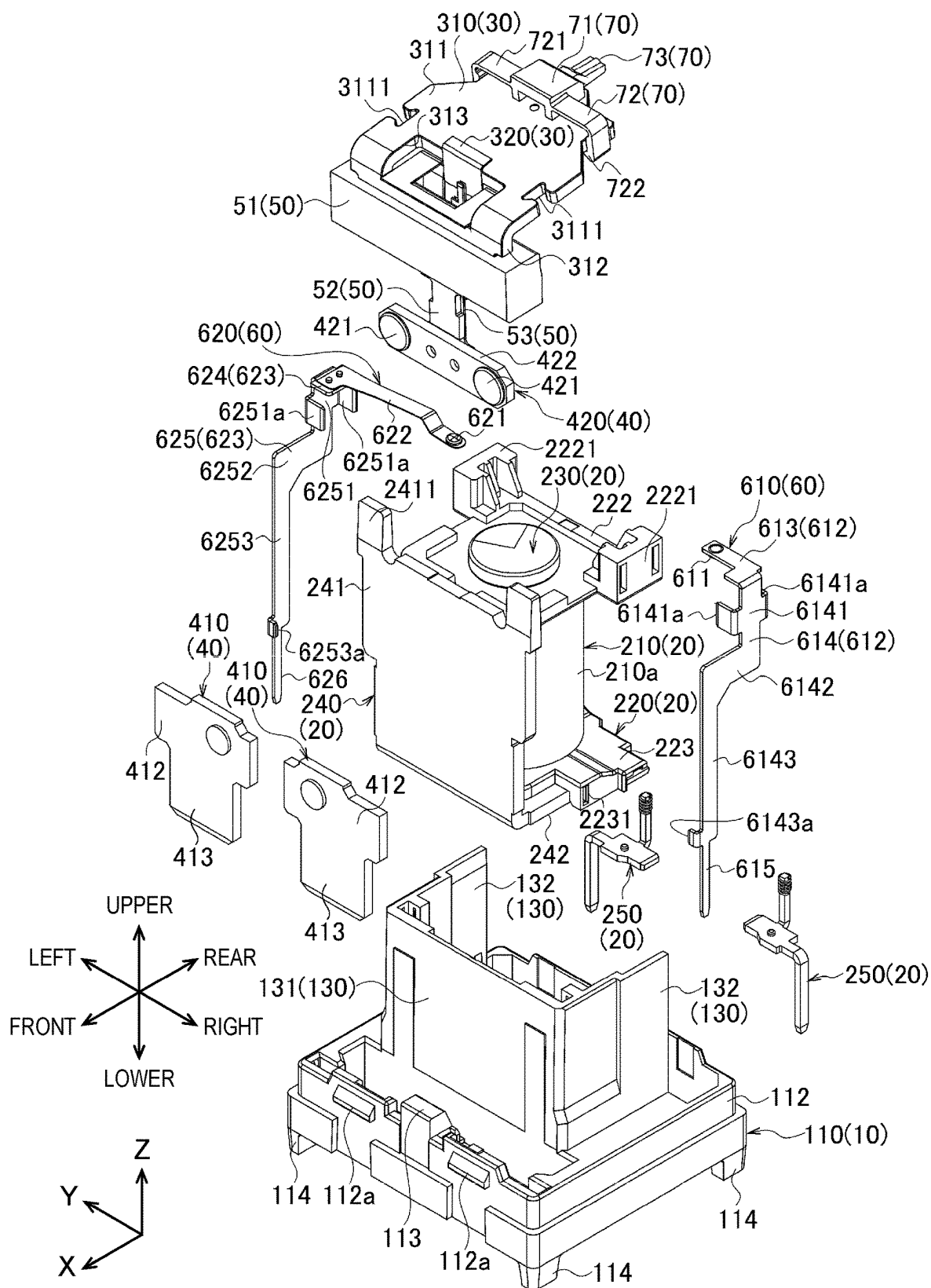


FIG. 8

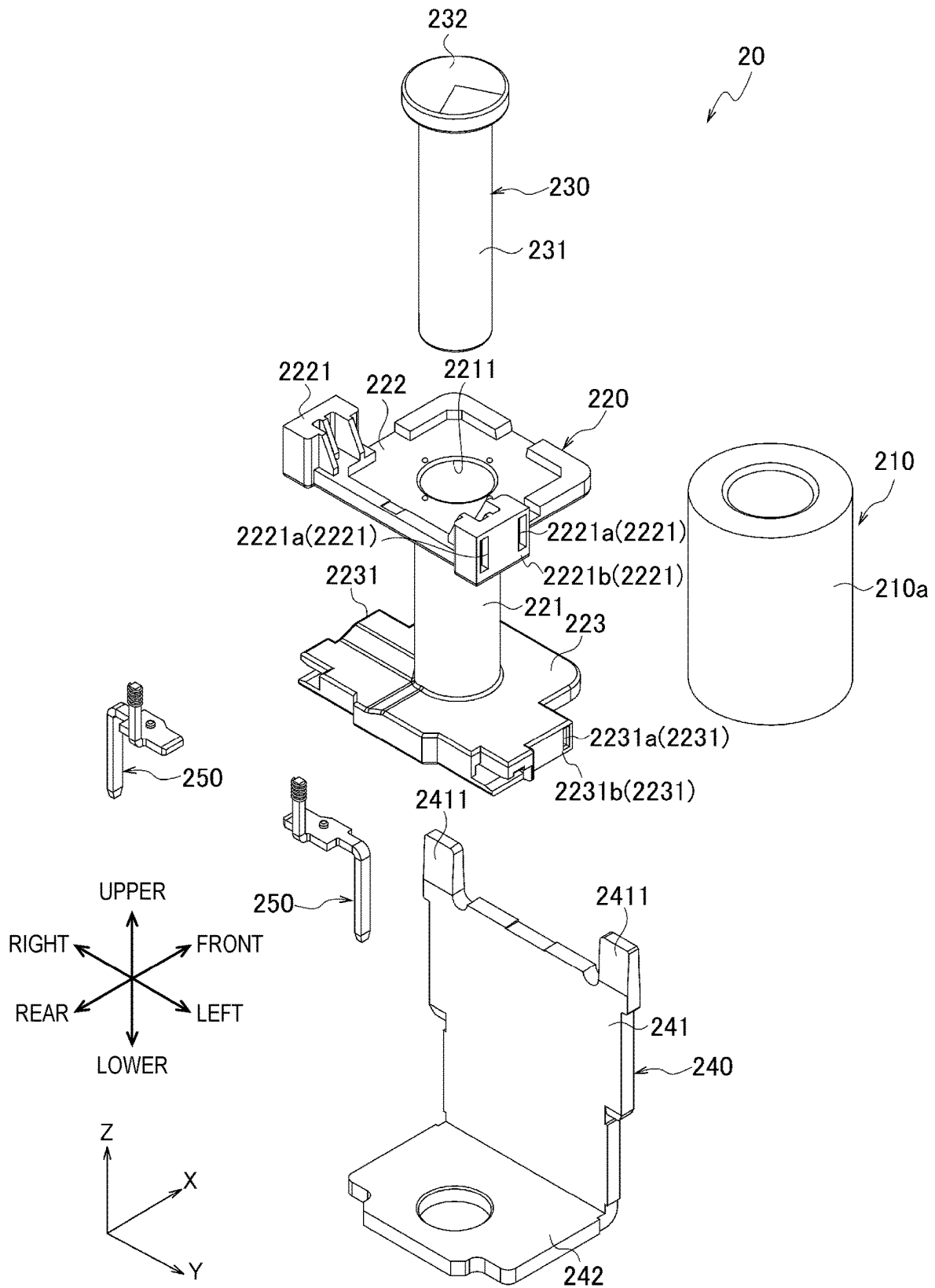


FIG. 9

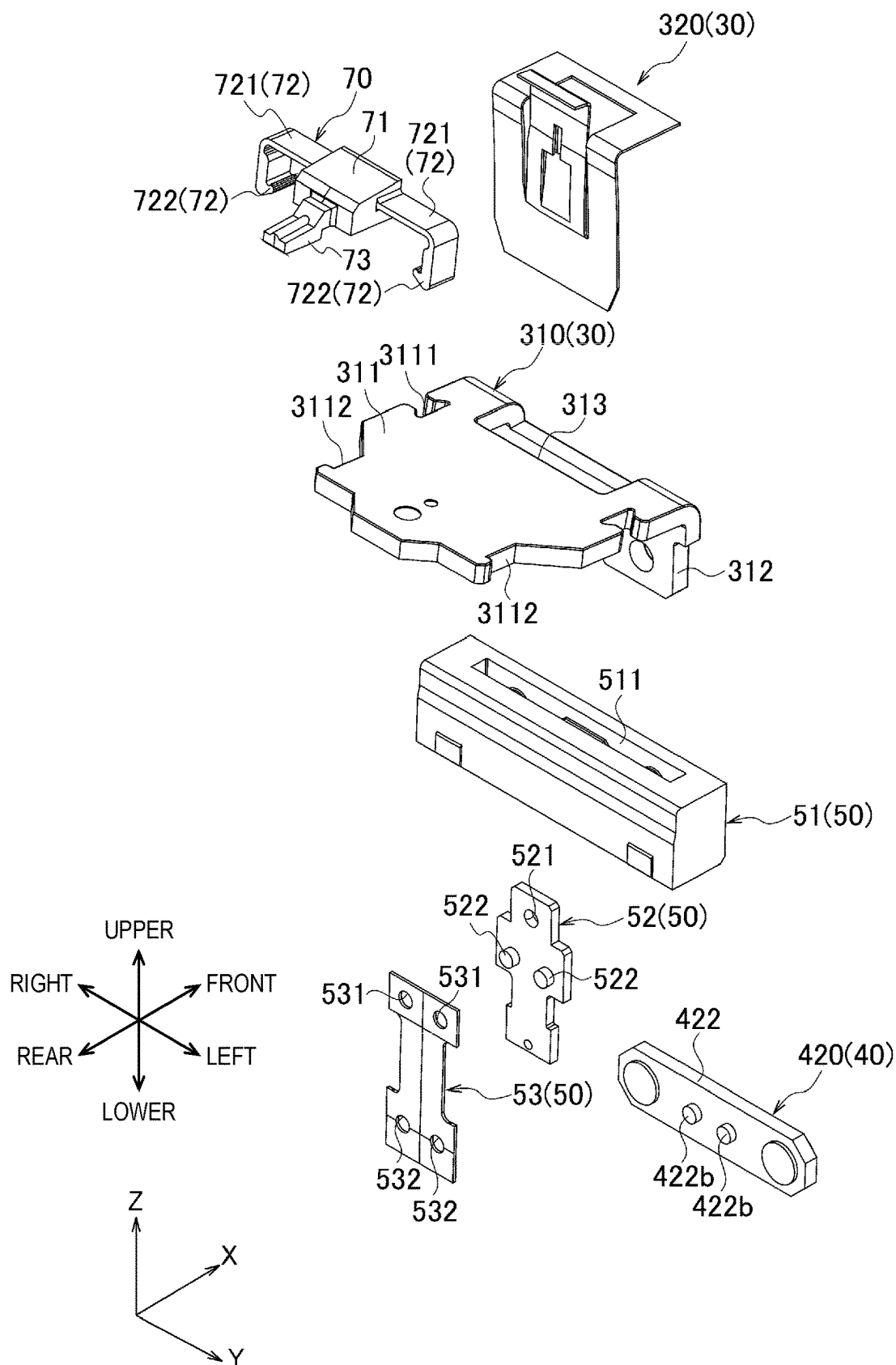


FIG. 10

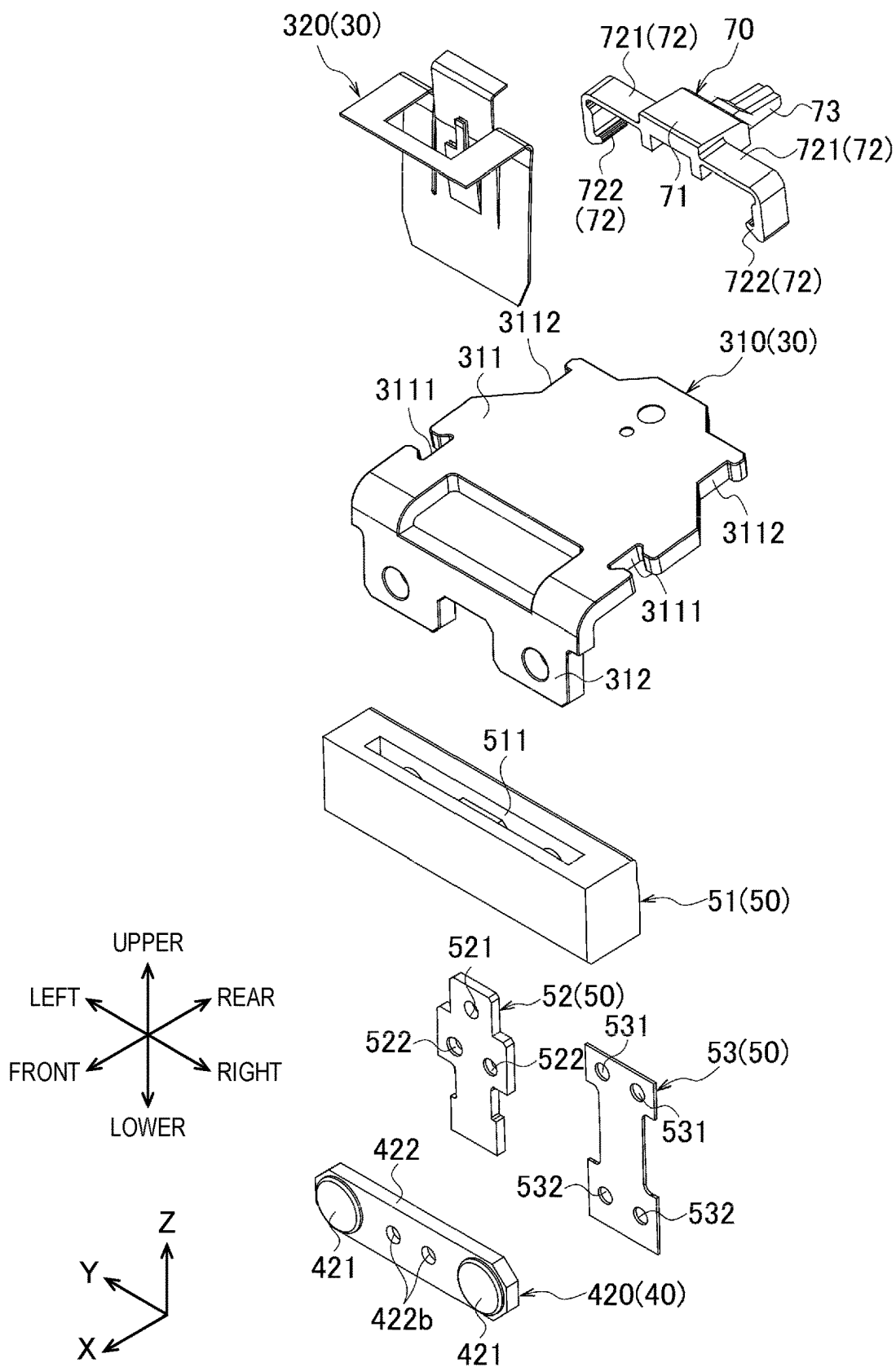


FIG. 11

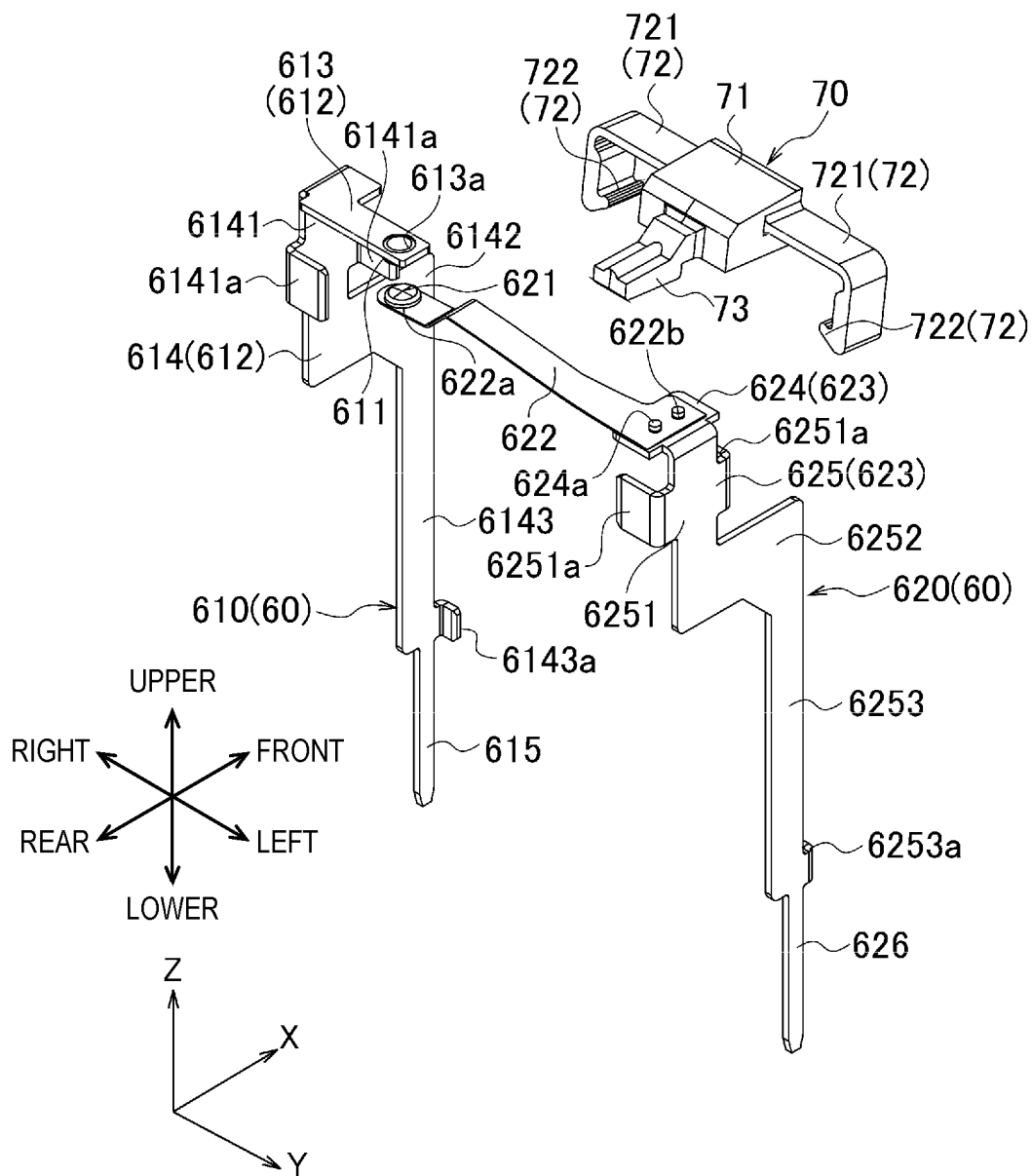


FIG. 12

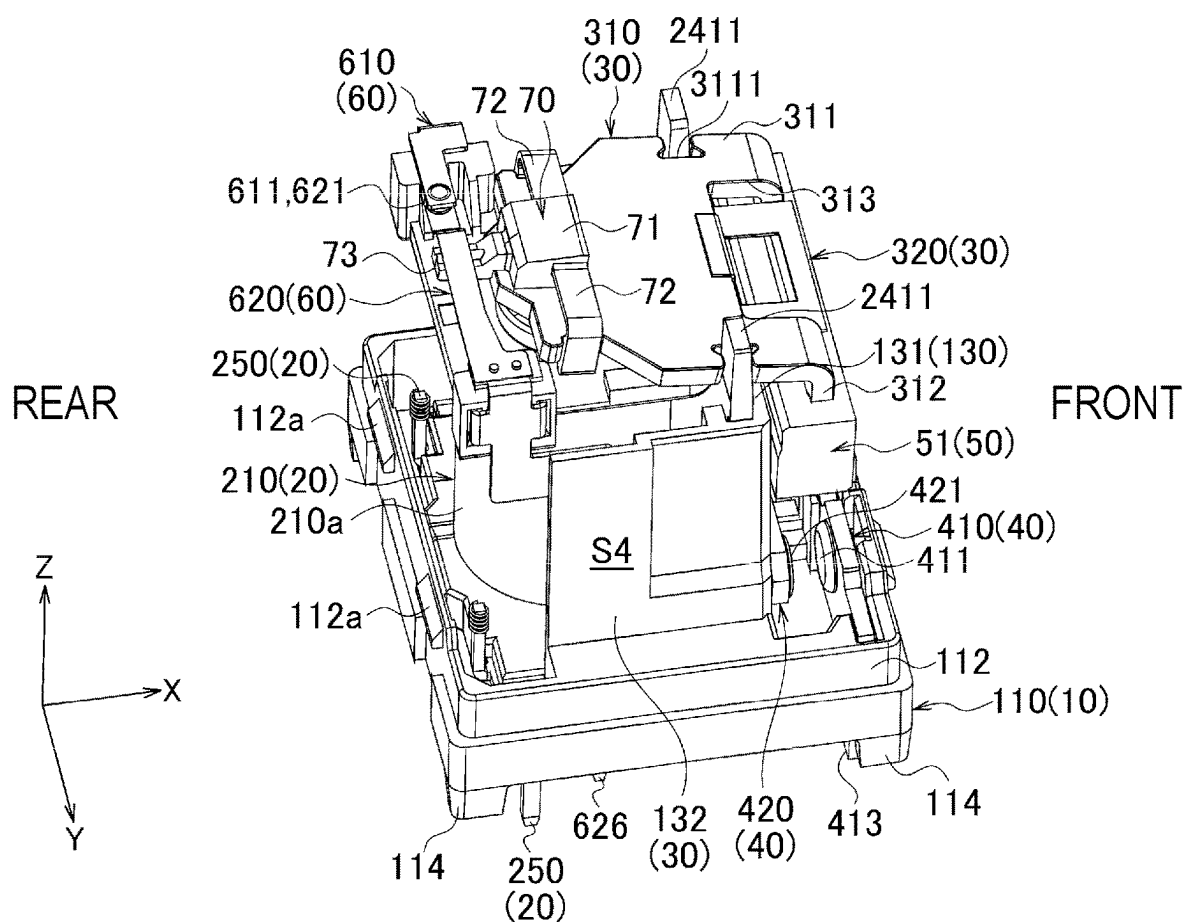


FIG. 13

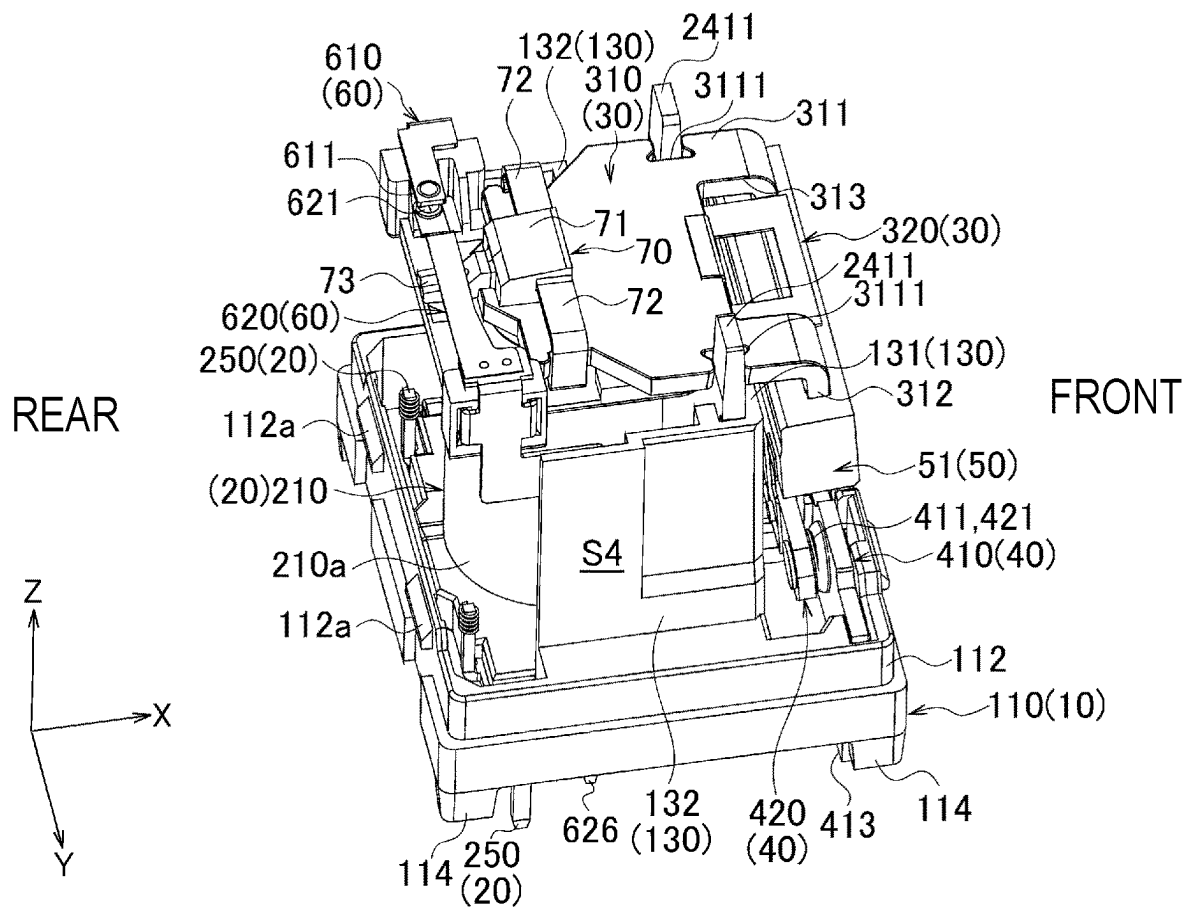


FIG. 14

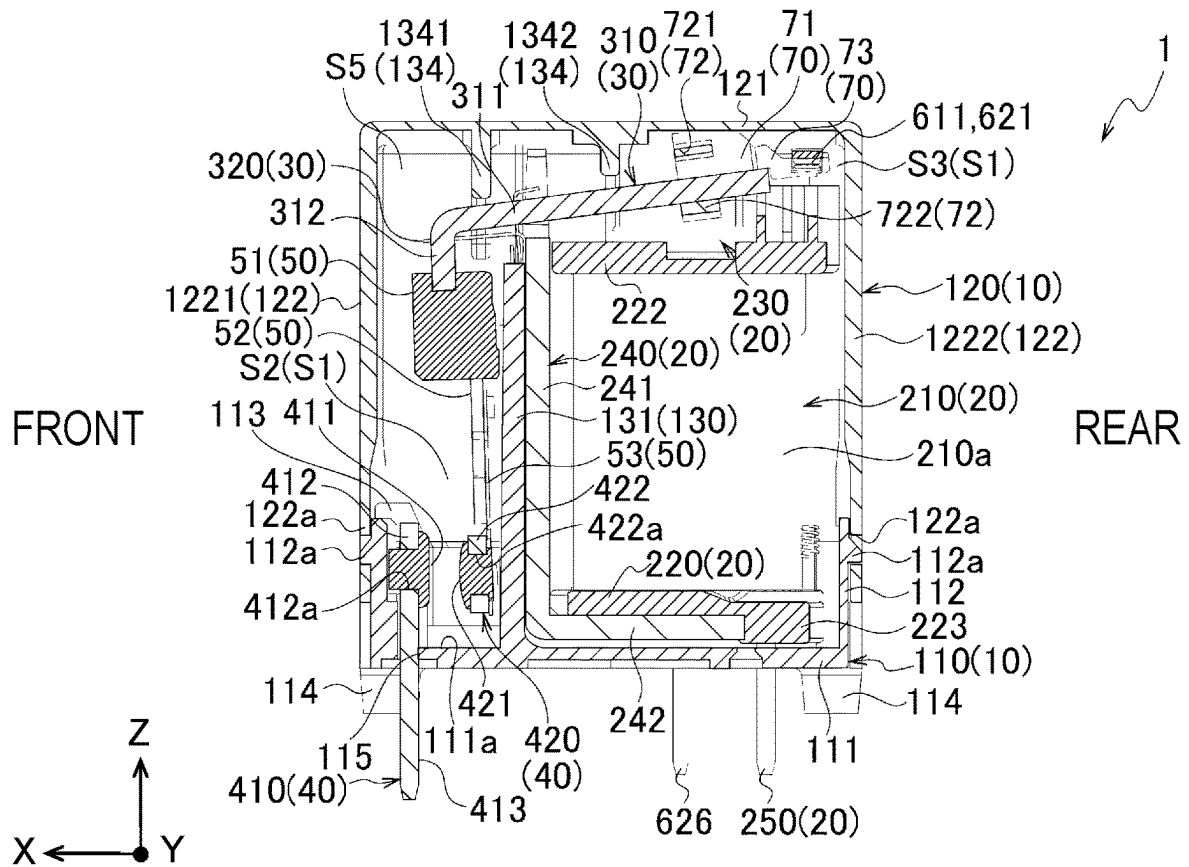


FIG. 16

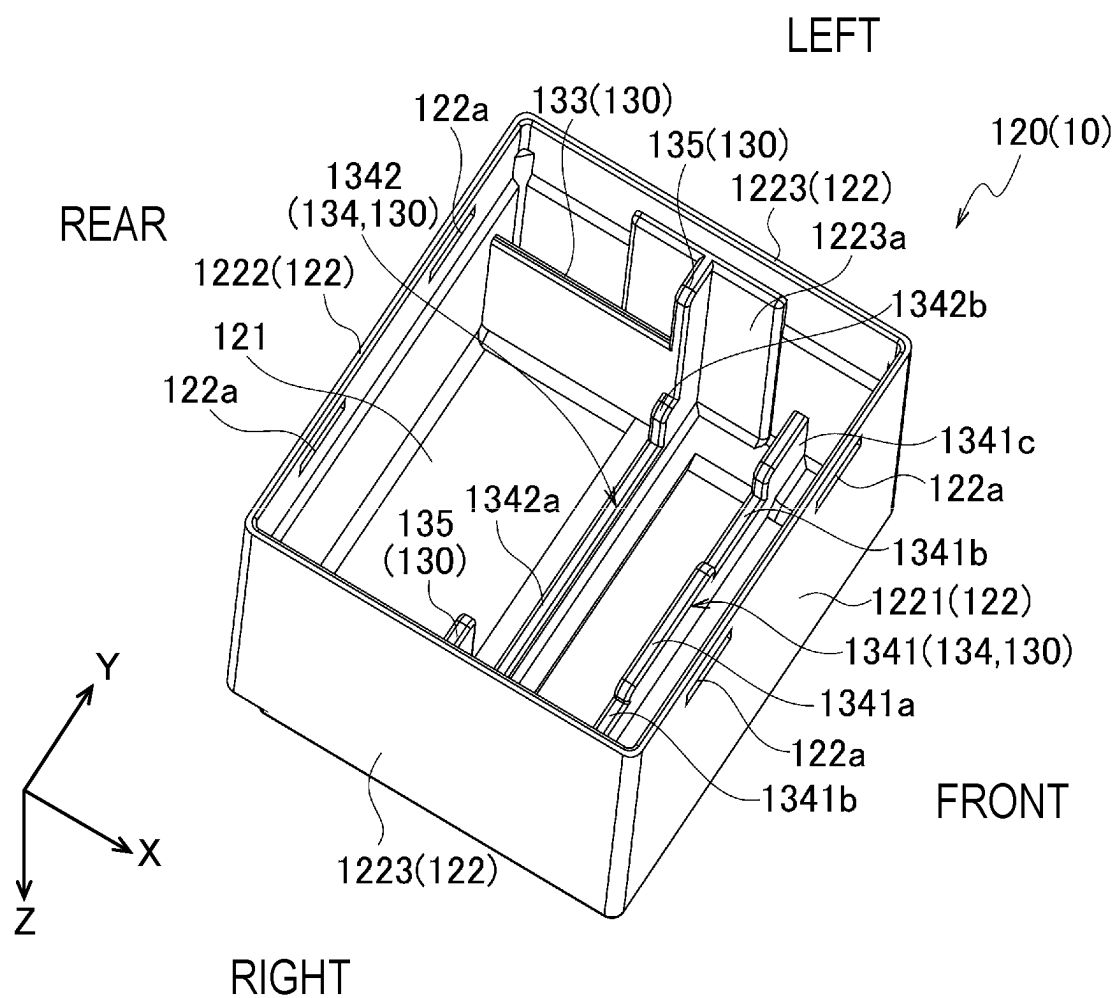


FIG. 17

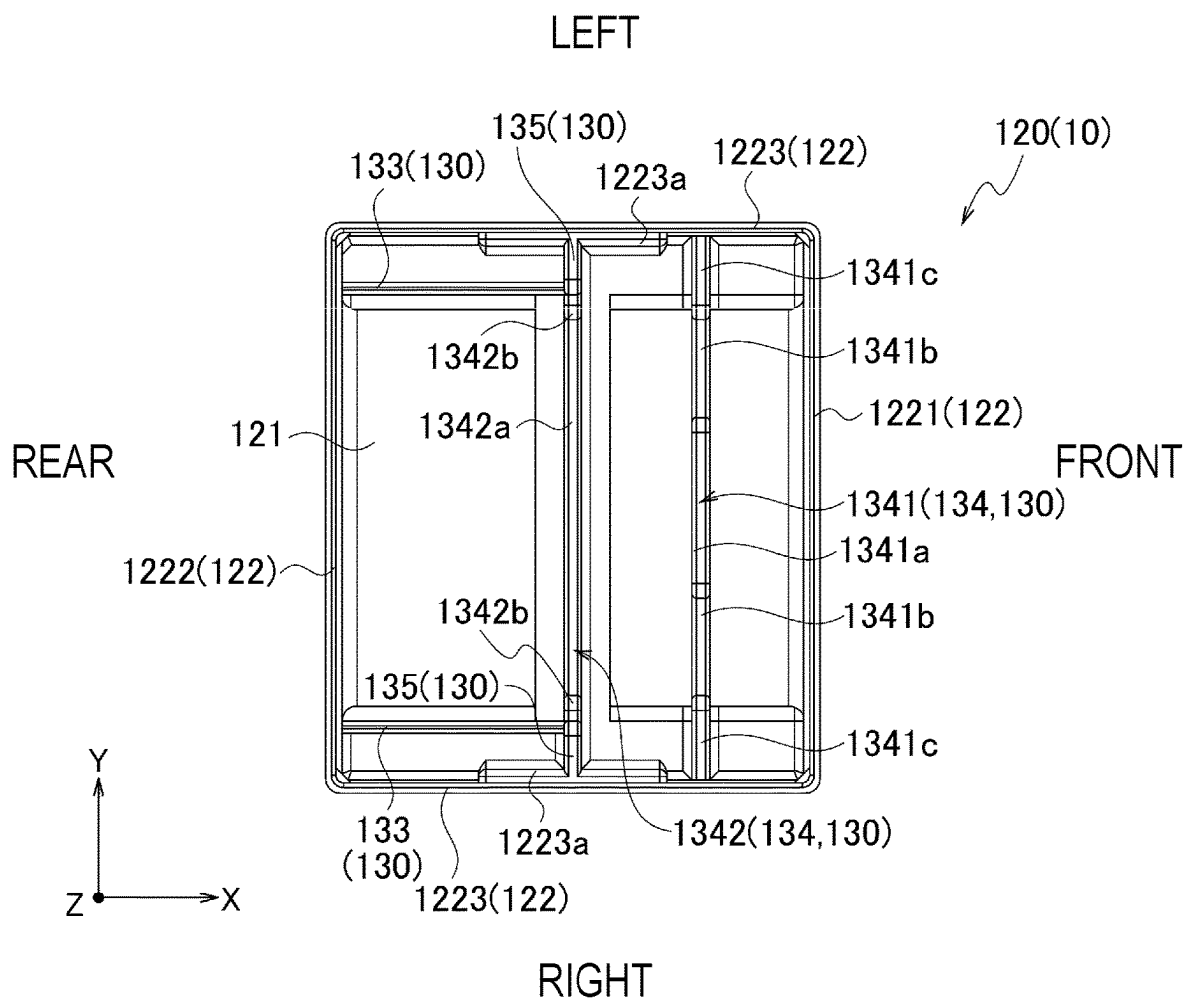


FIG. 18

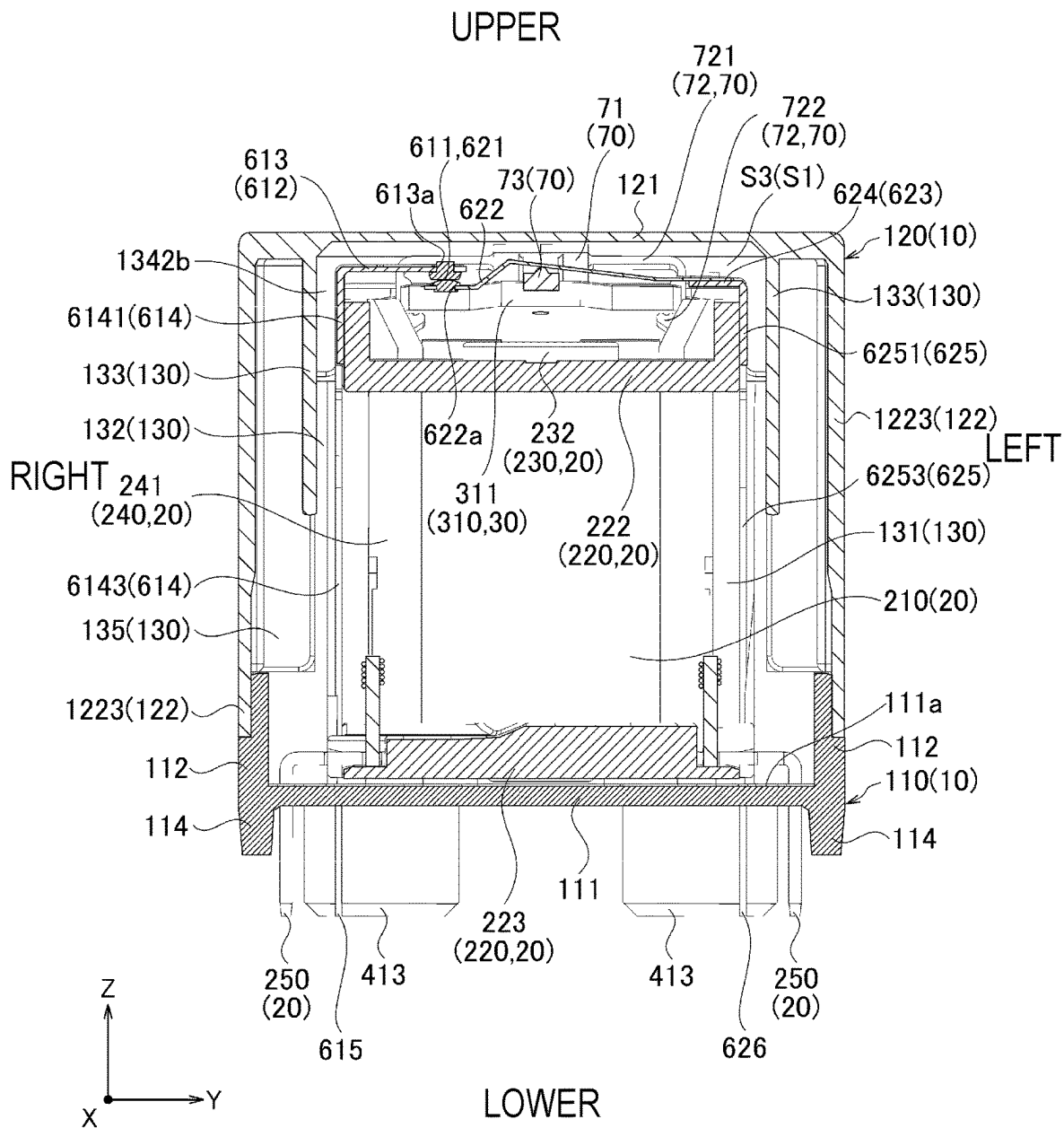


FIG. 19

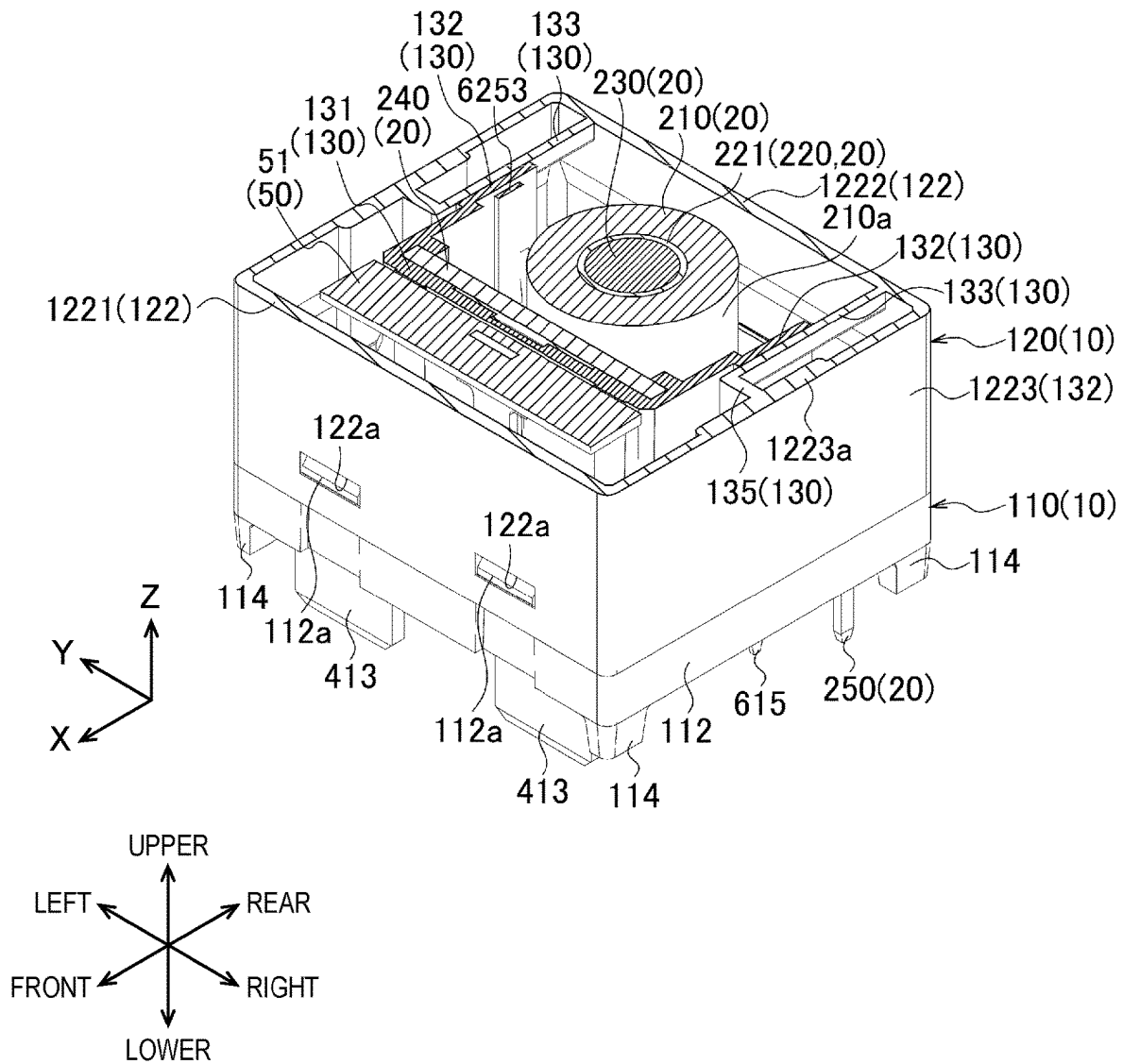
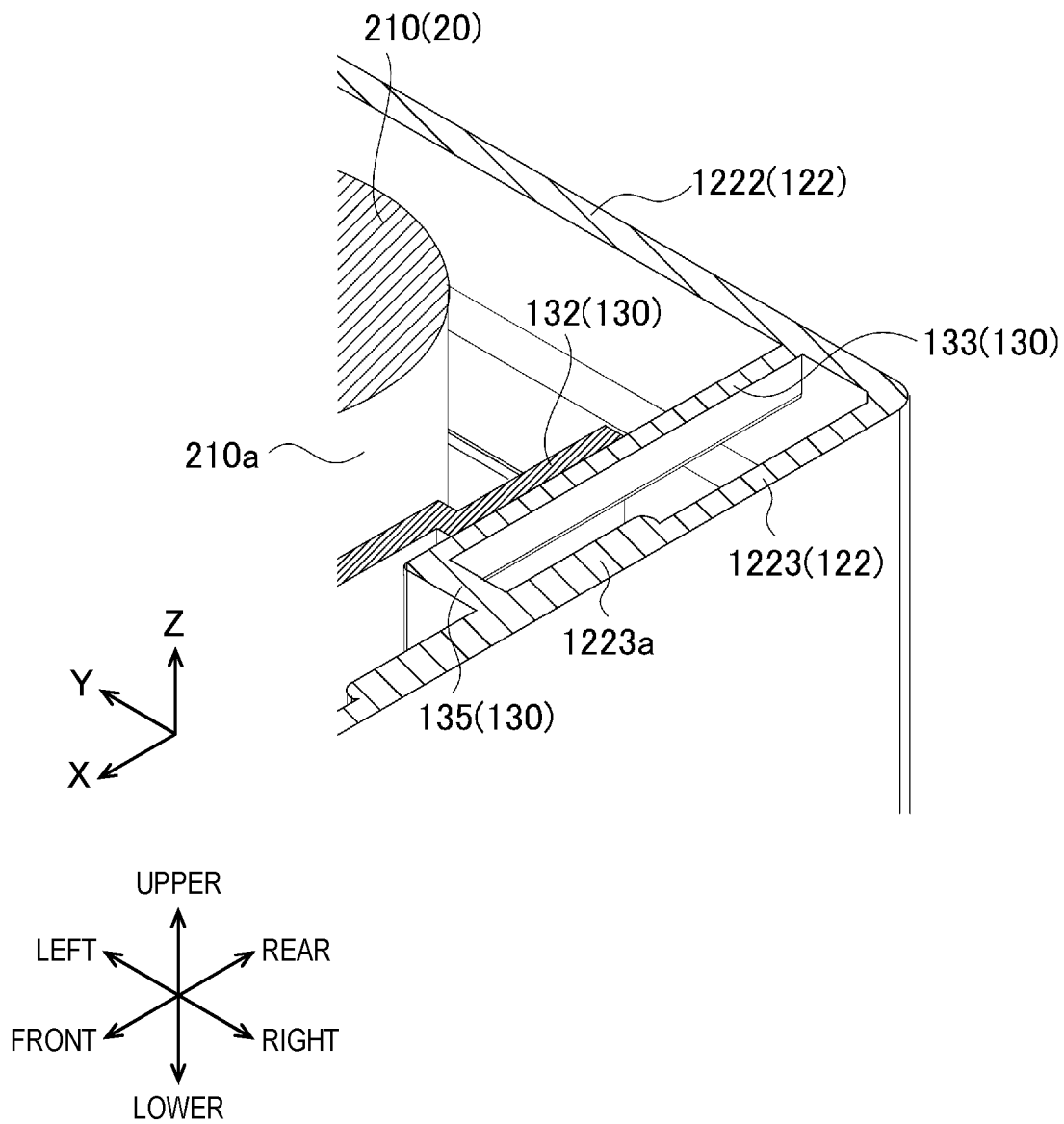
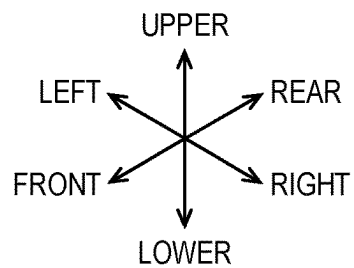


FIG. 20





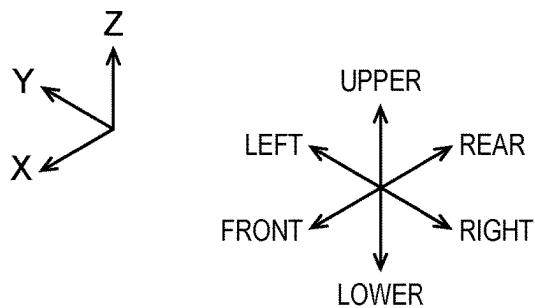


FIG. 23

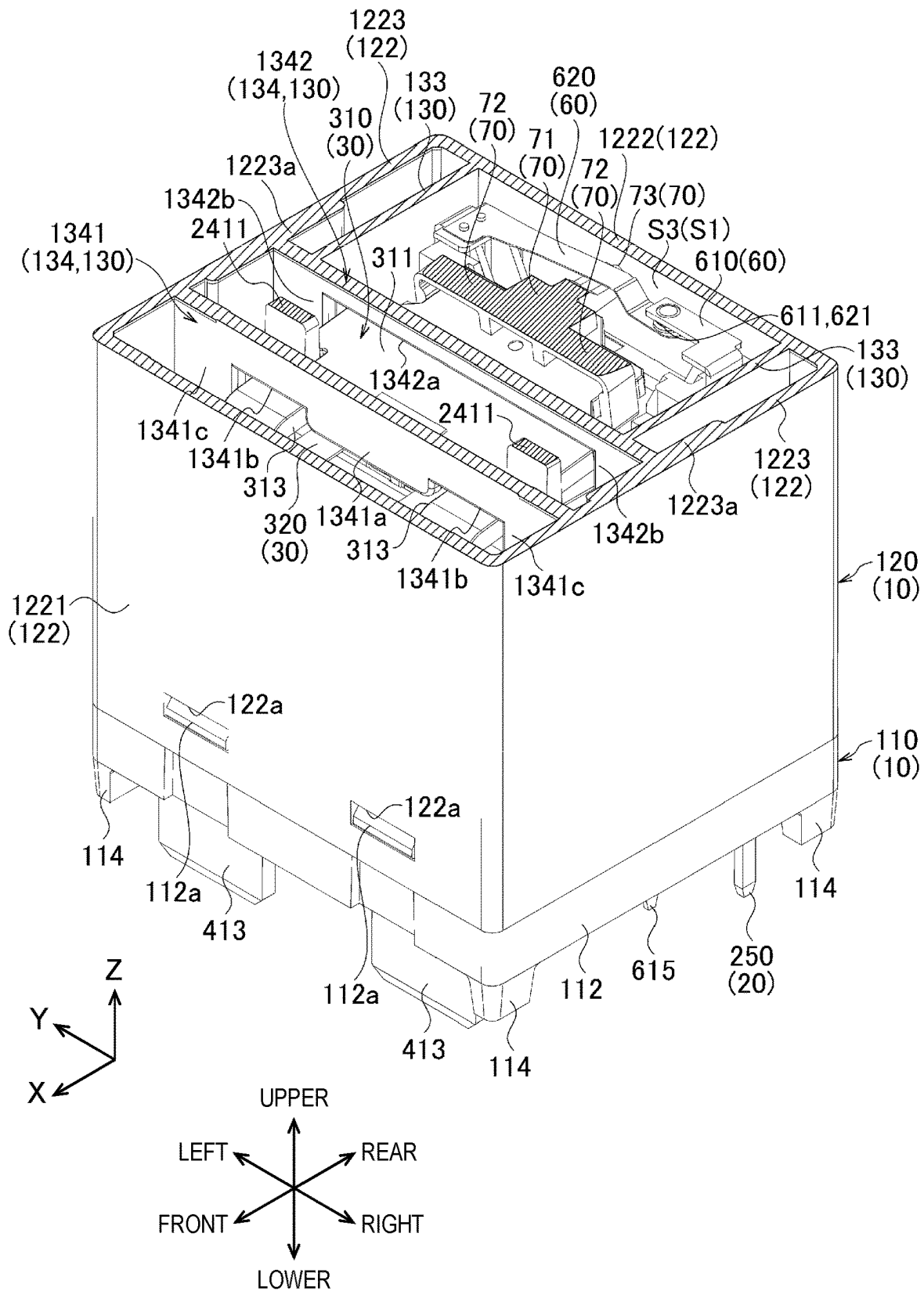


FIG. 24

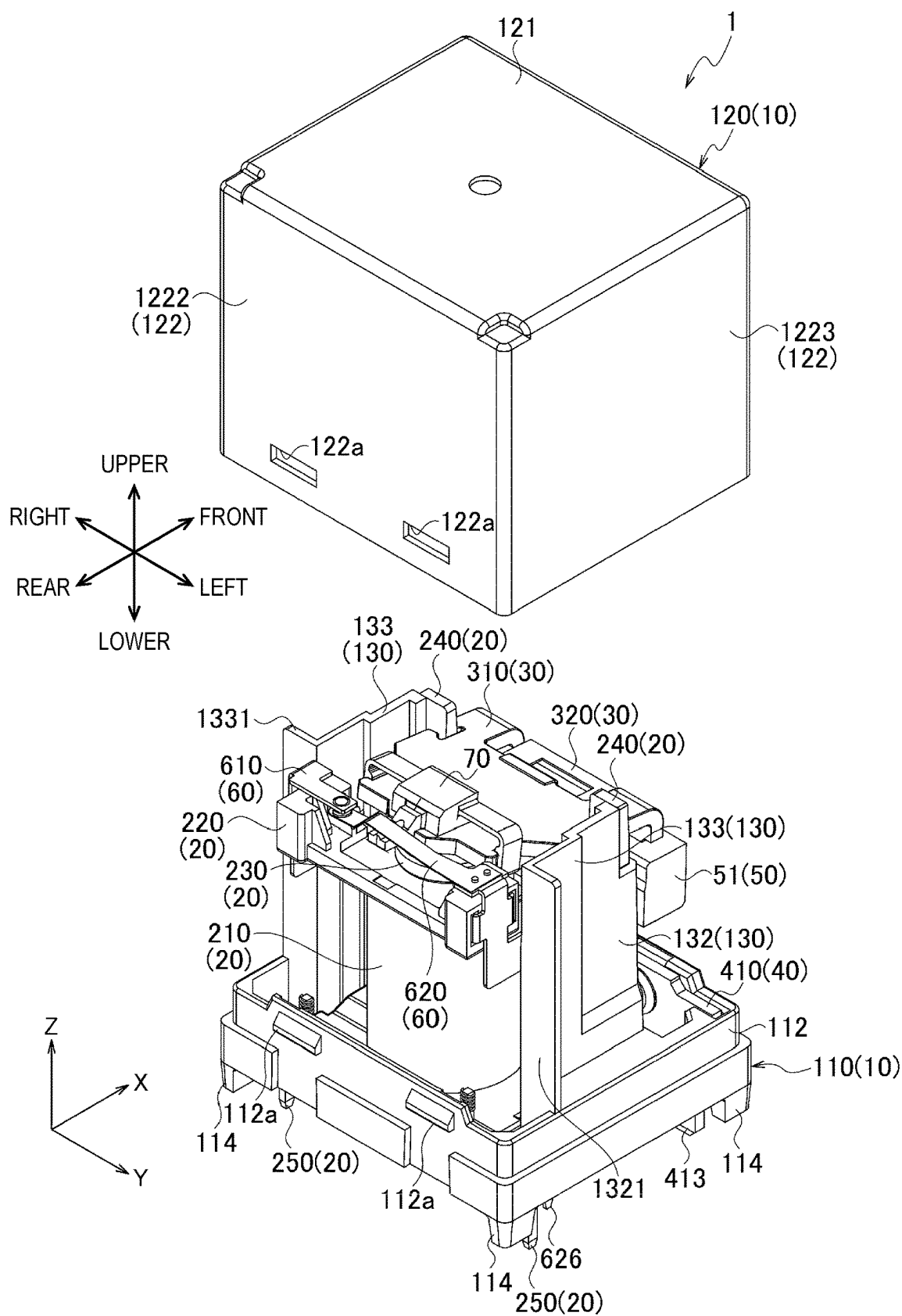


FIG. 26

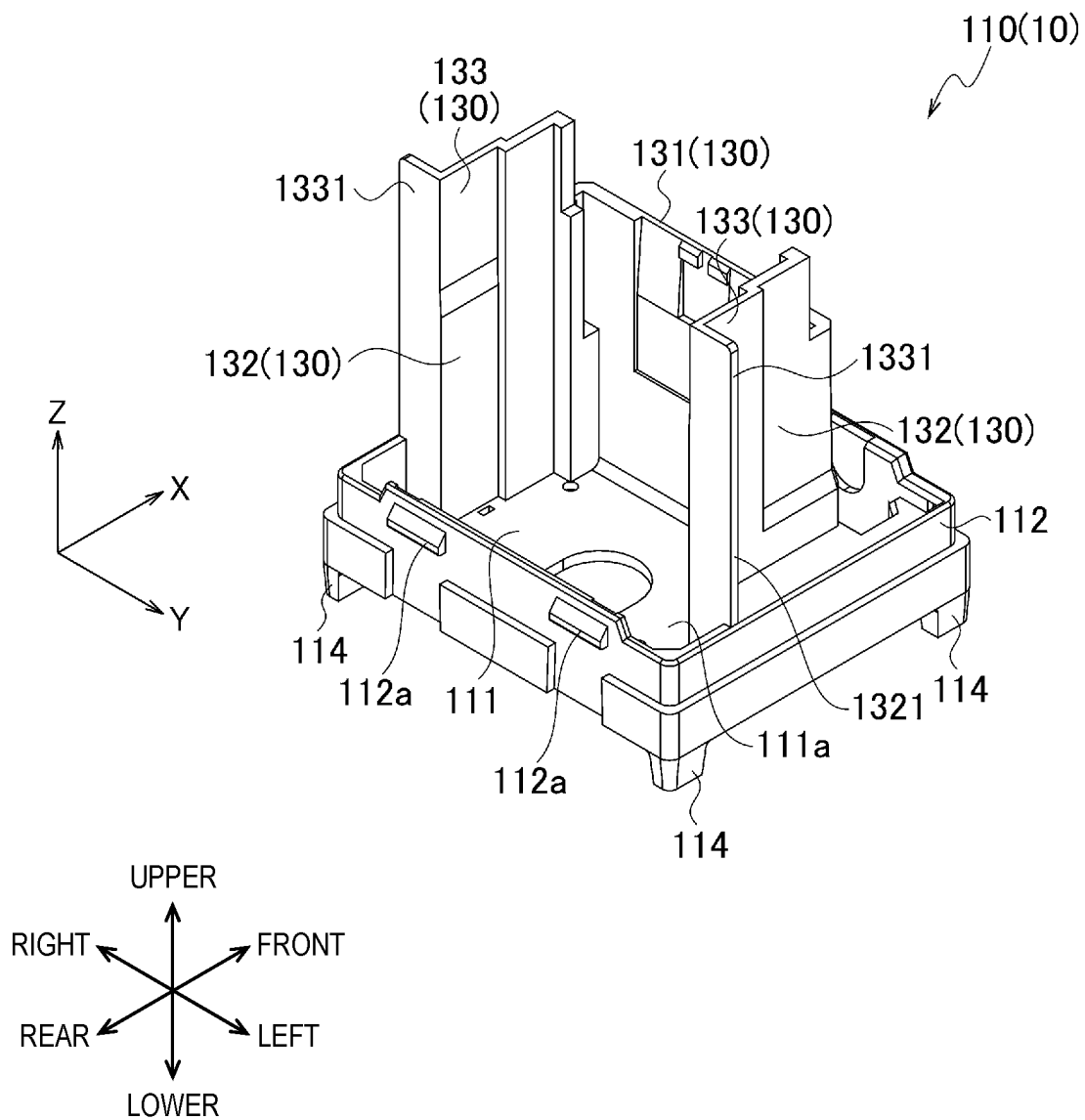


FIG. 27

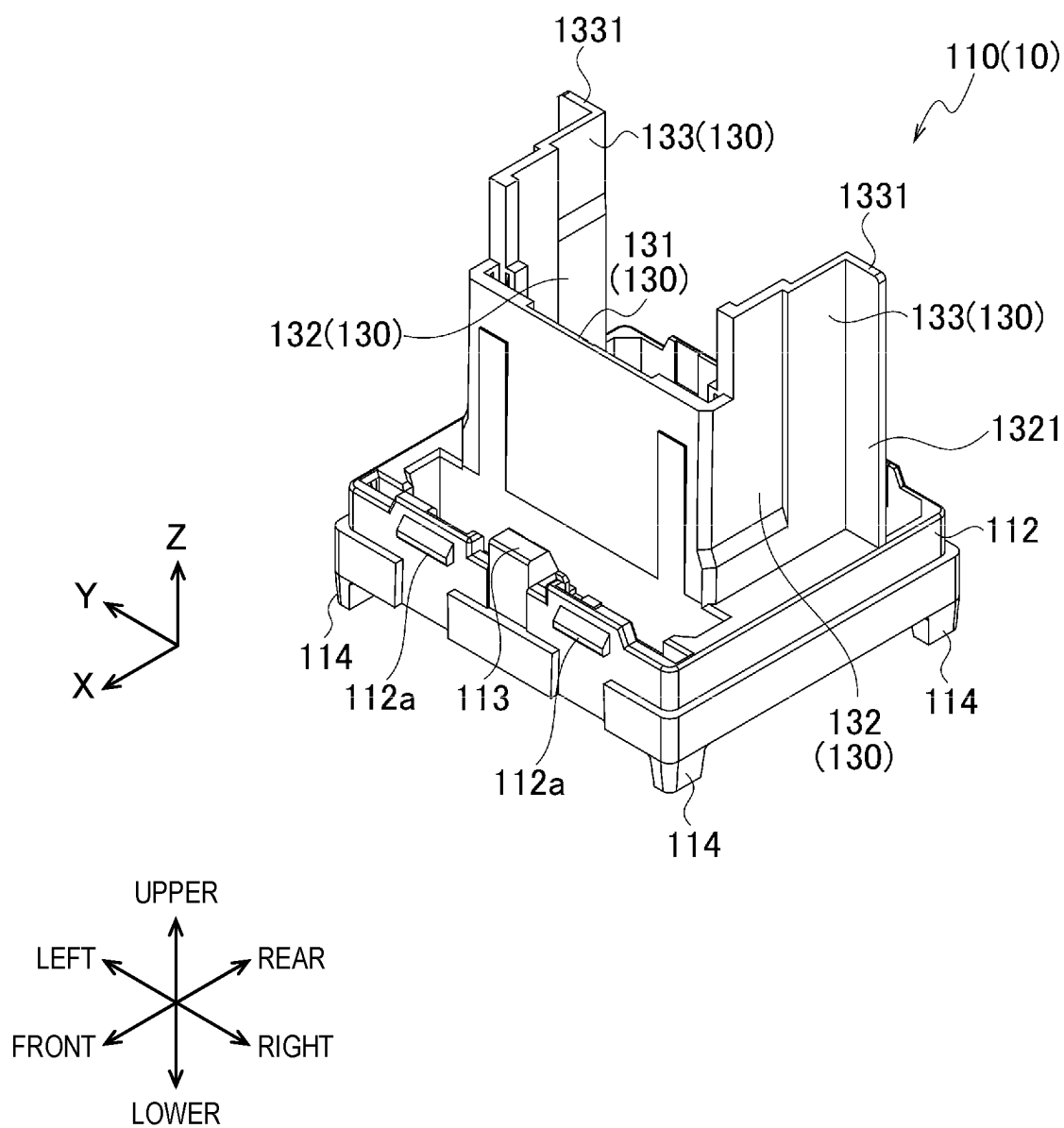


FIG. 28

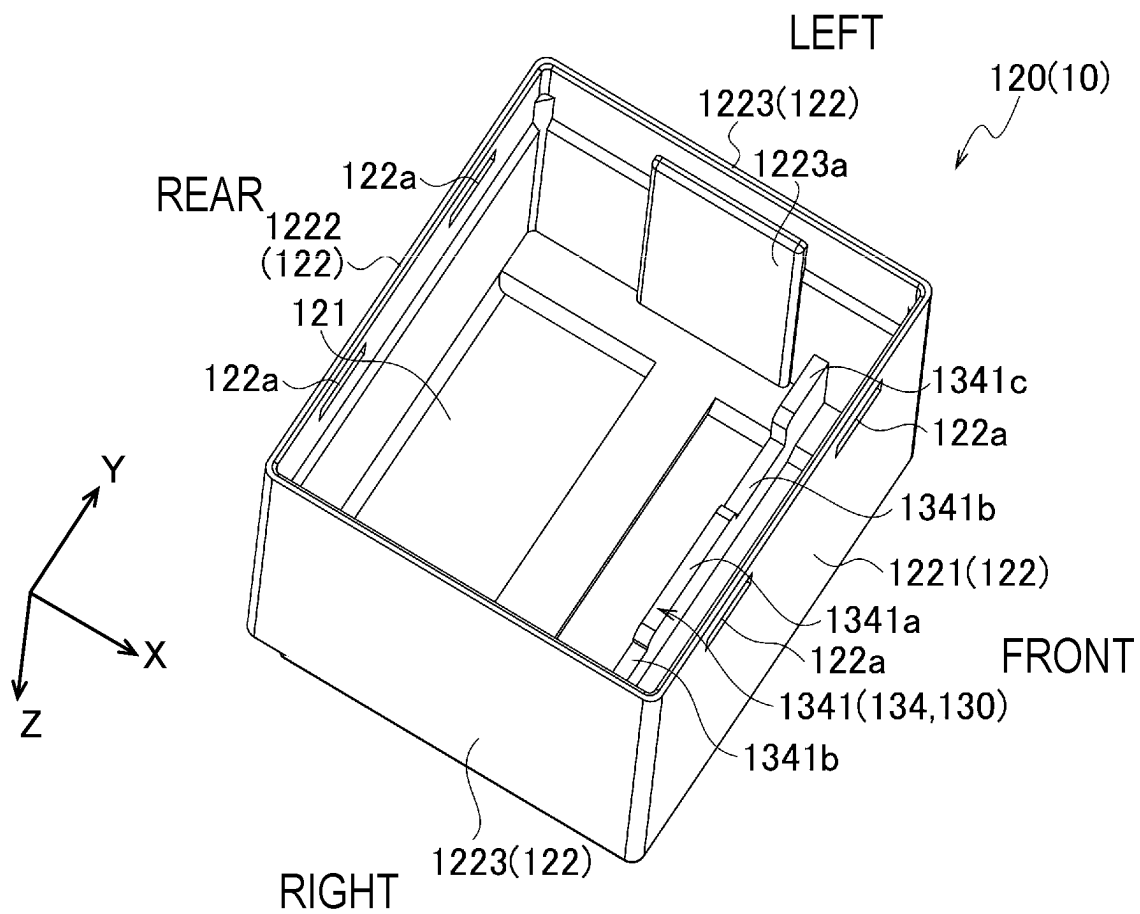


FIG. 29

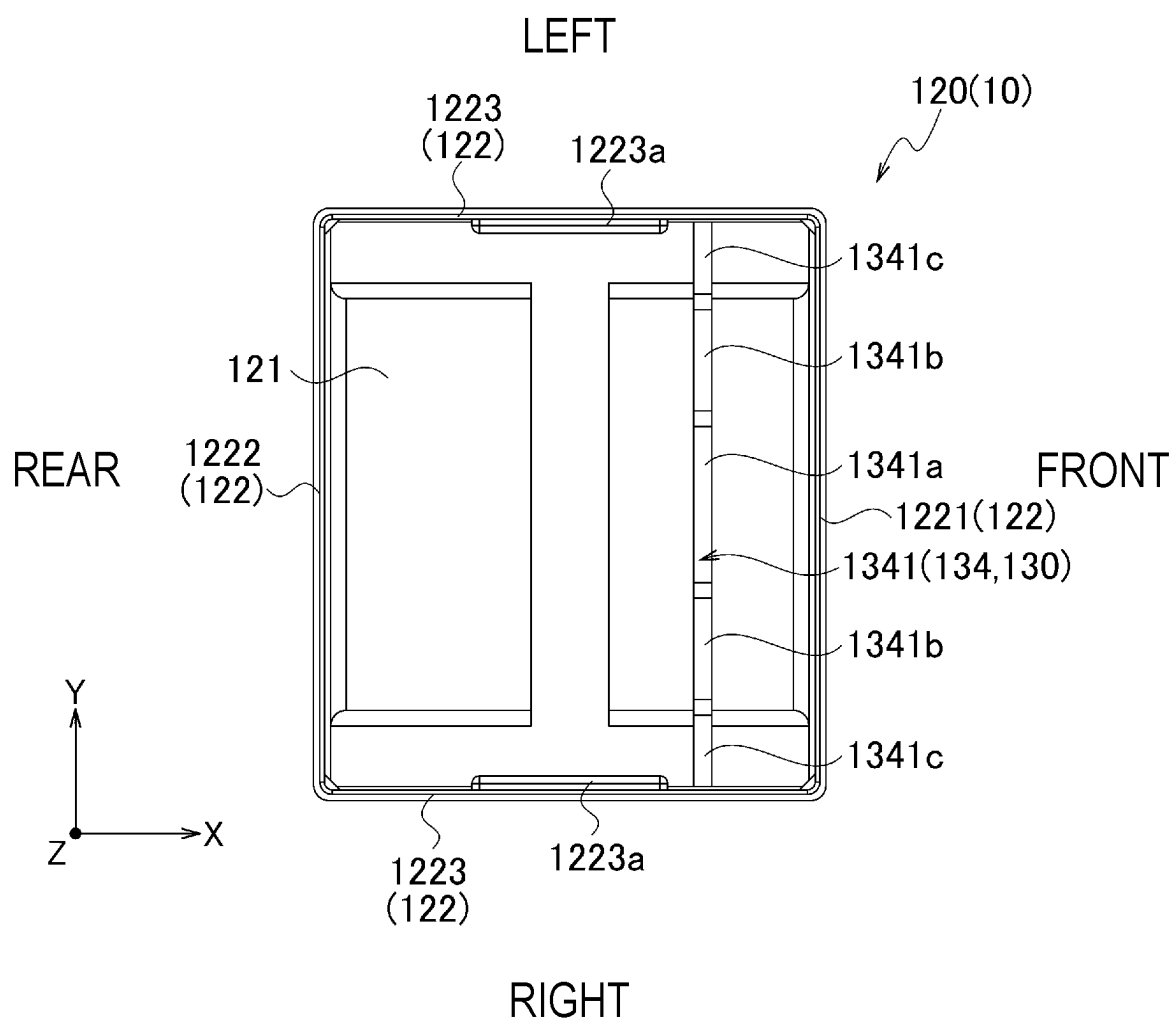


FIG. 30

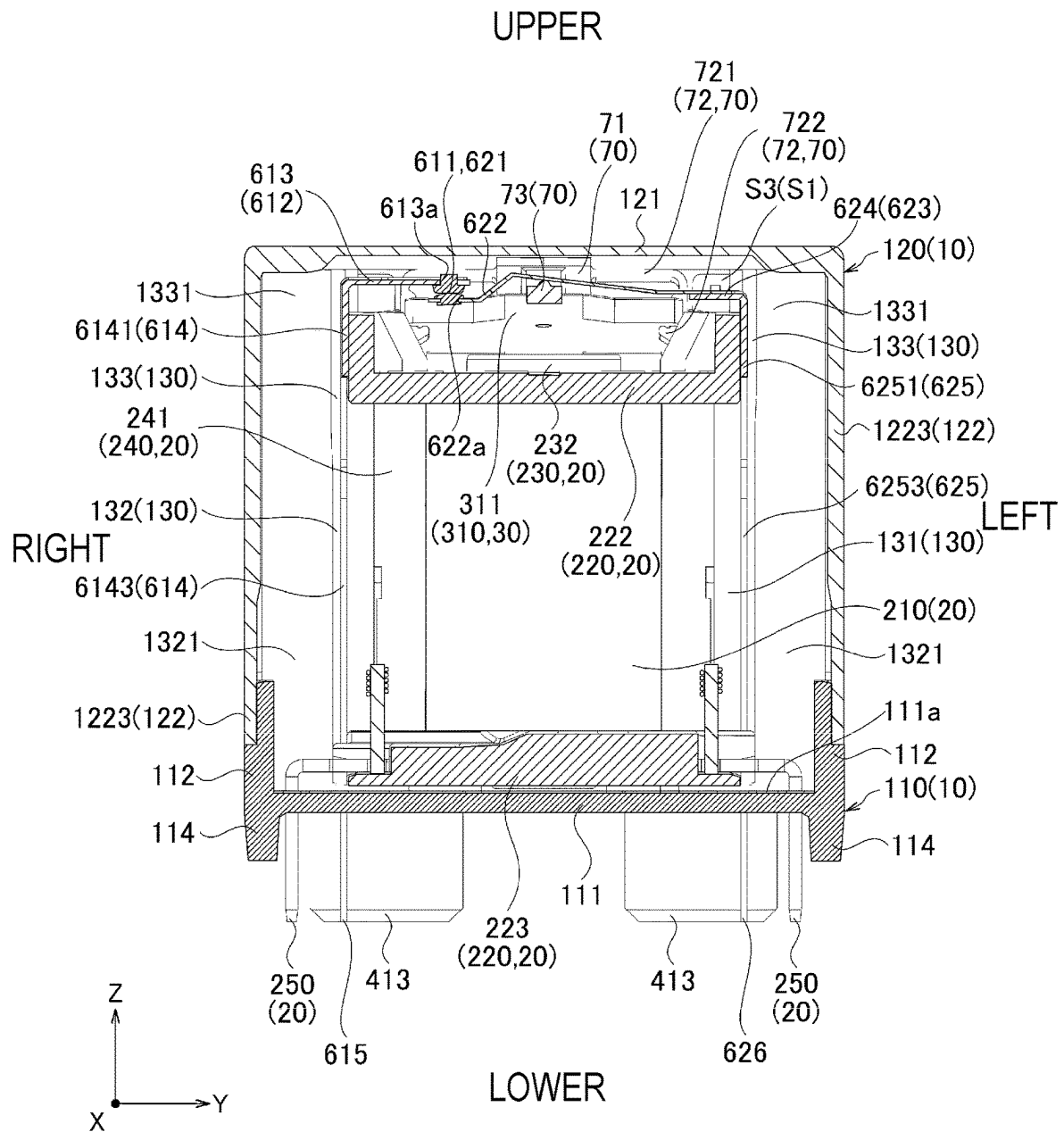


FIG. 31

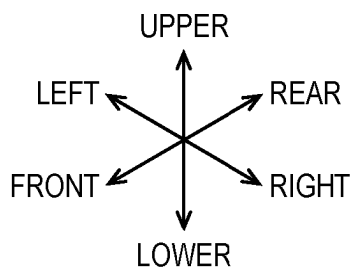
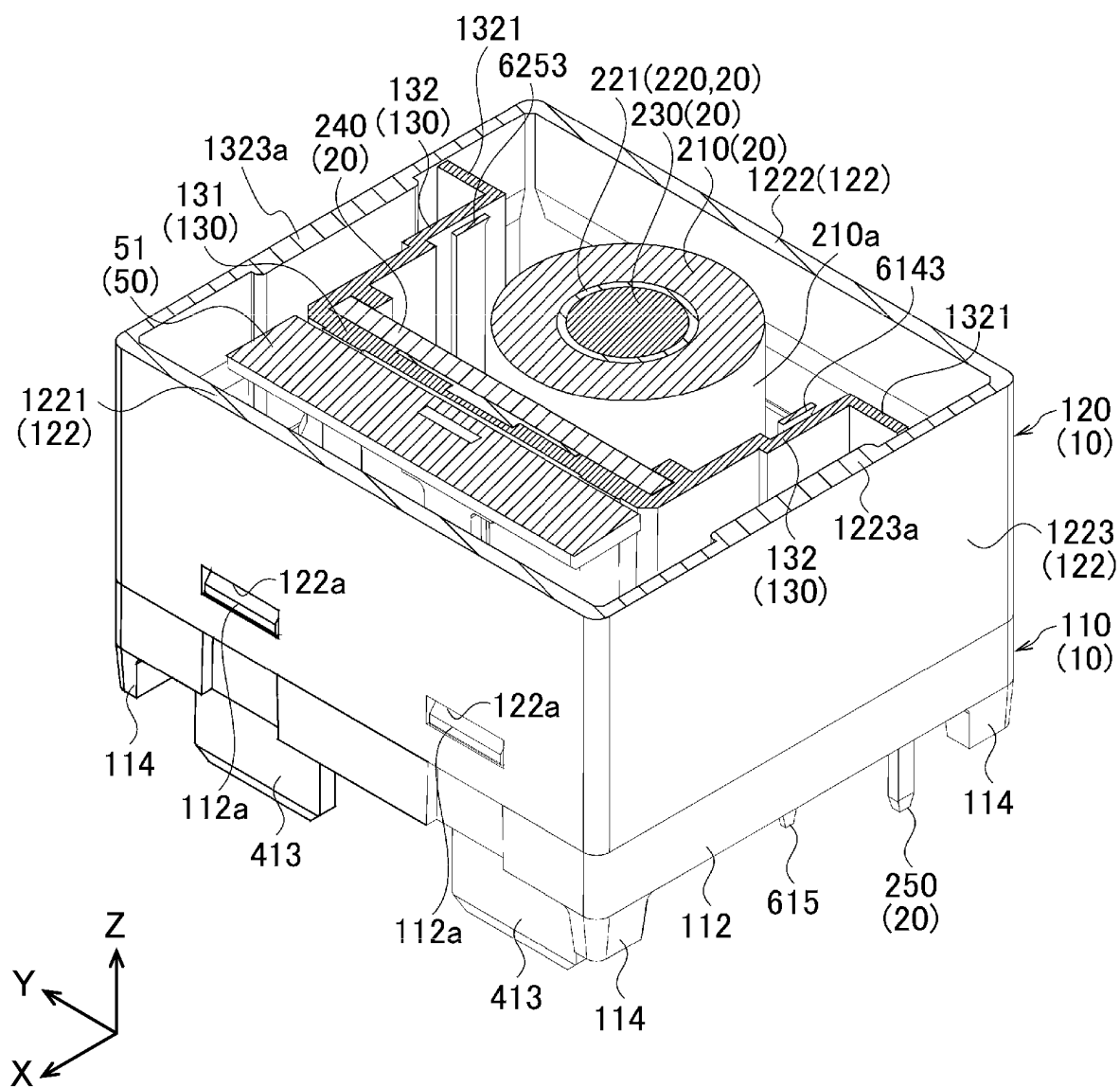
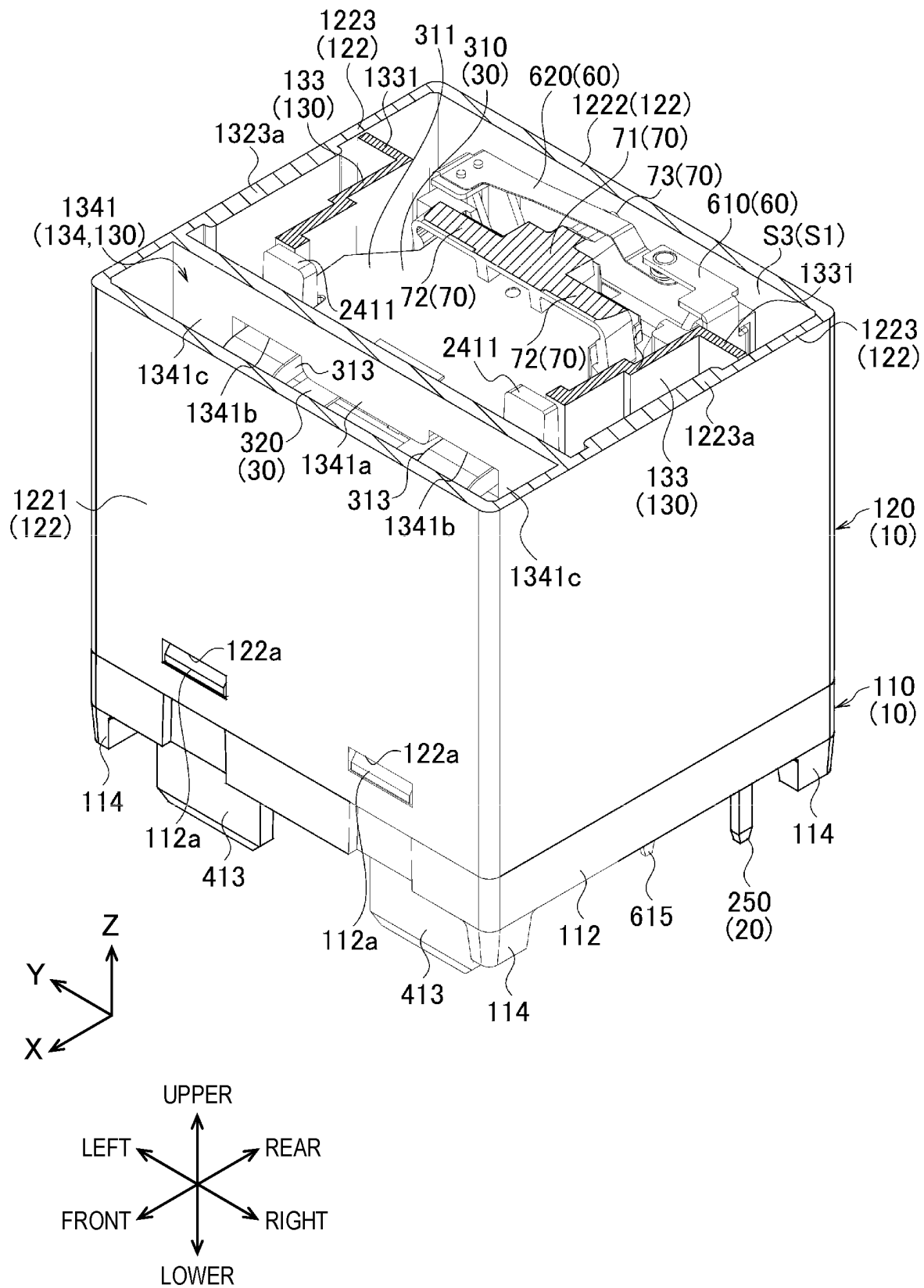


FIG. 32



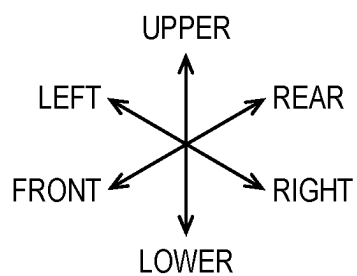


FIG. 34

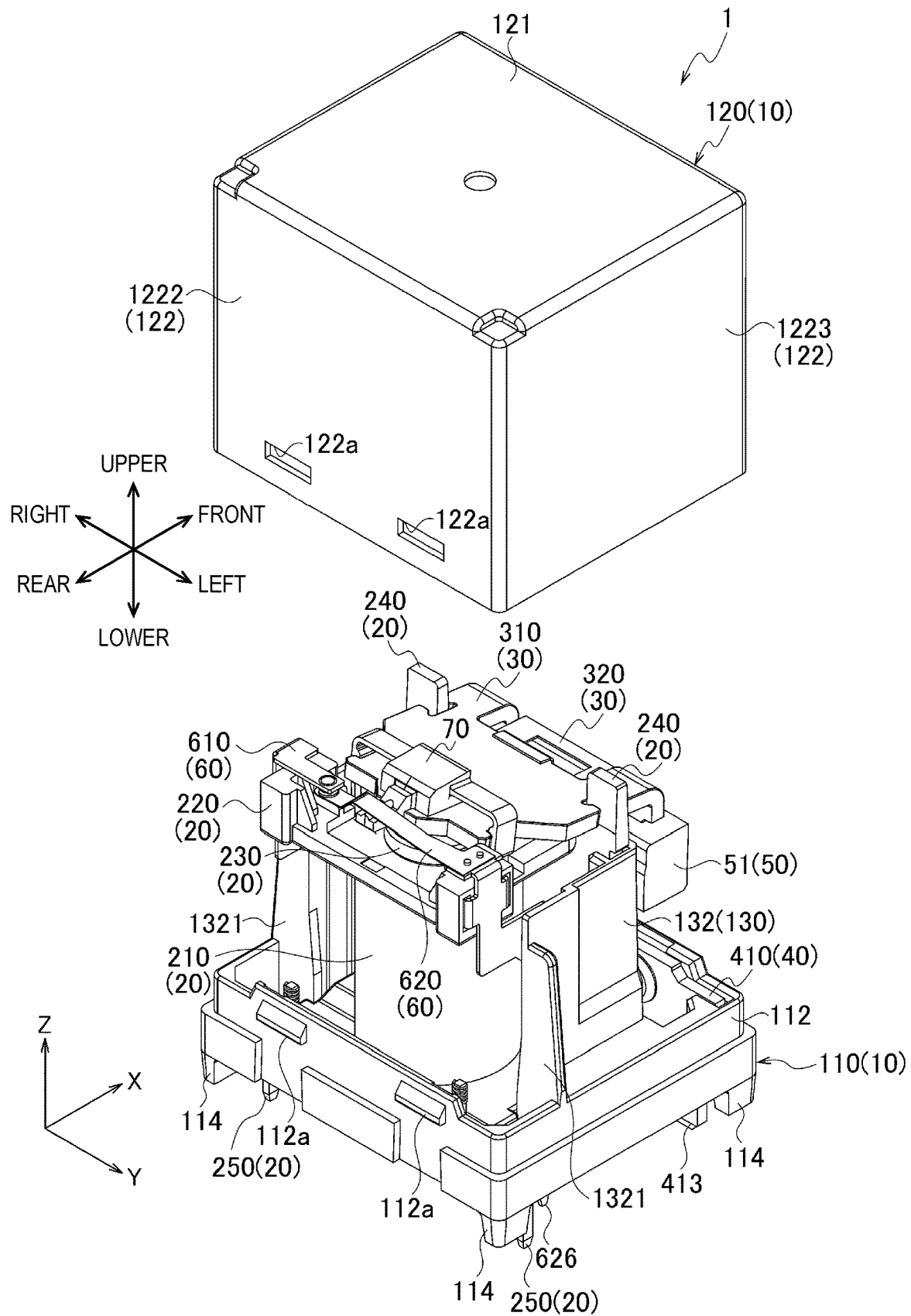


FIG. 35

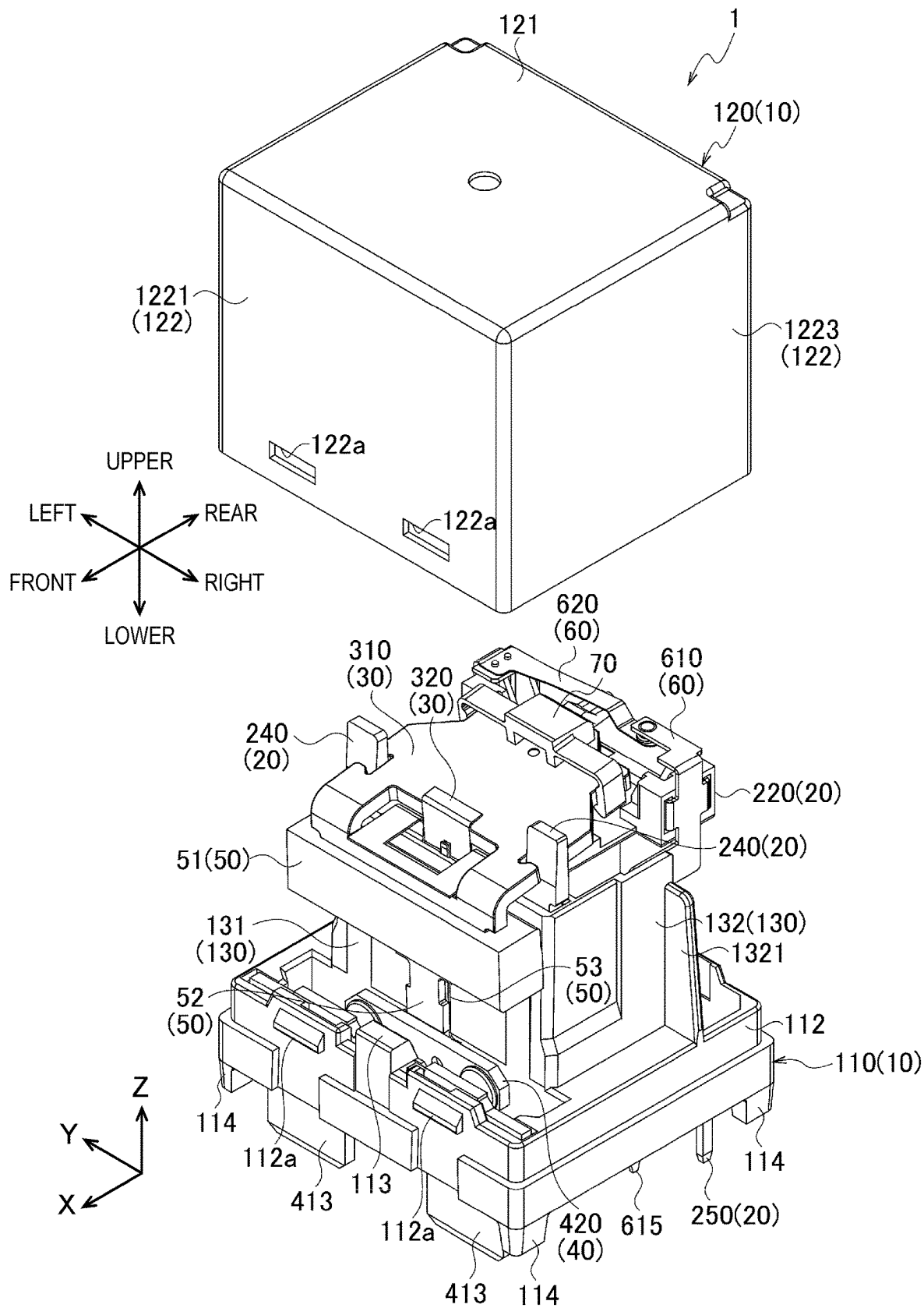


FIG. 36

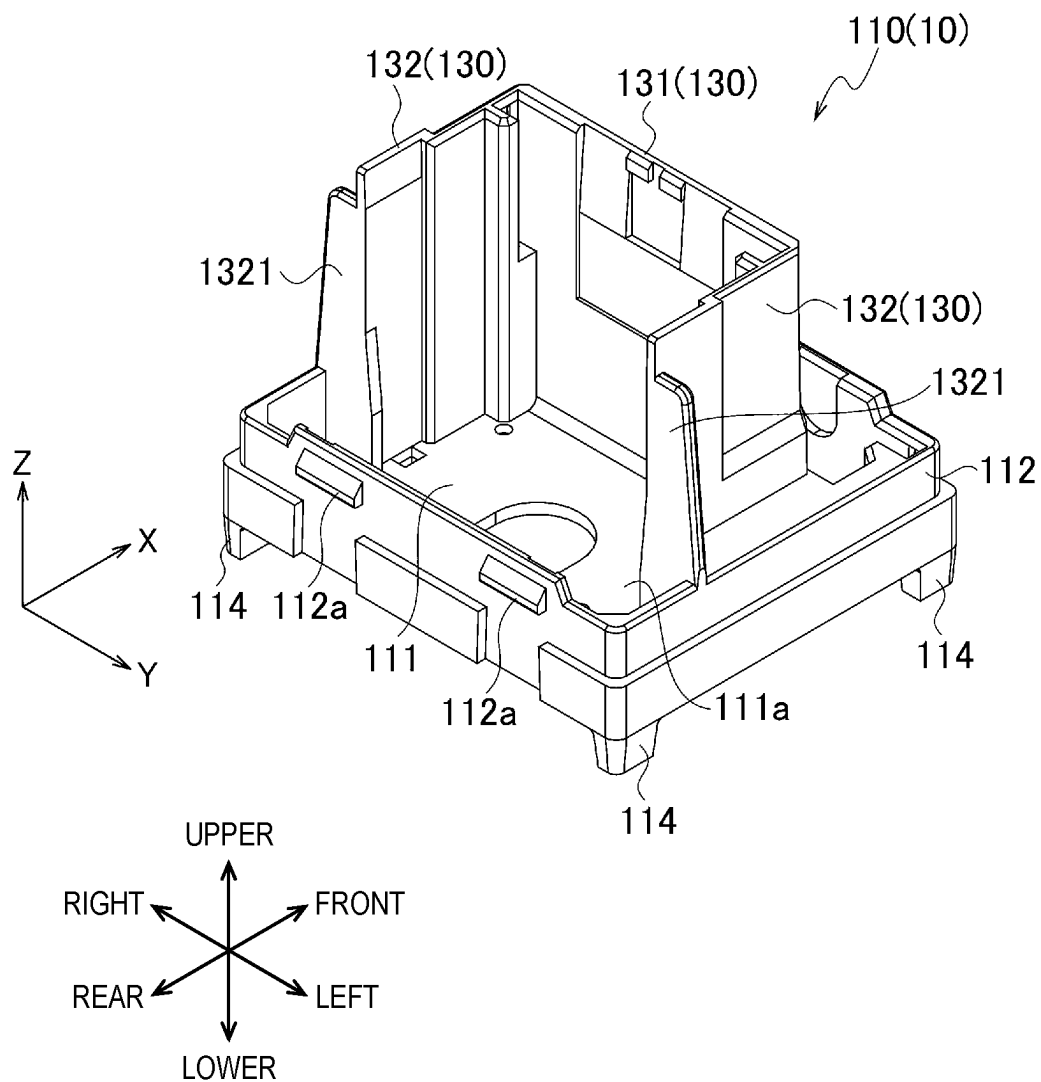


FIG. 37

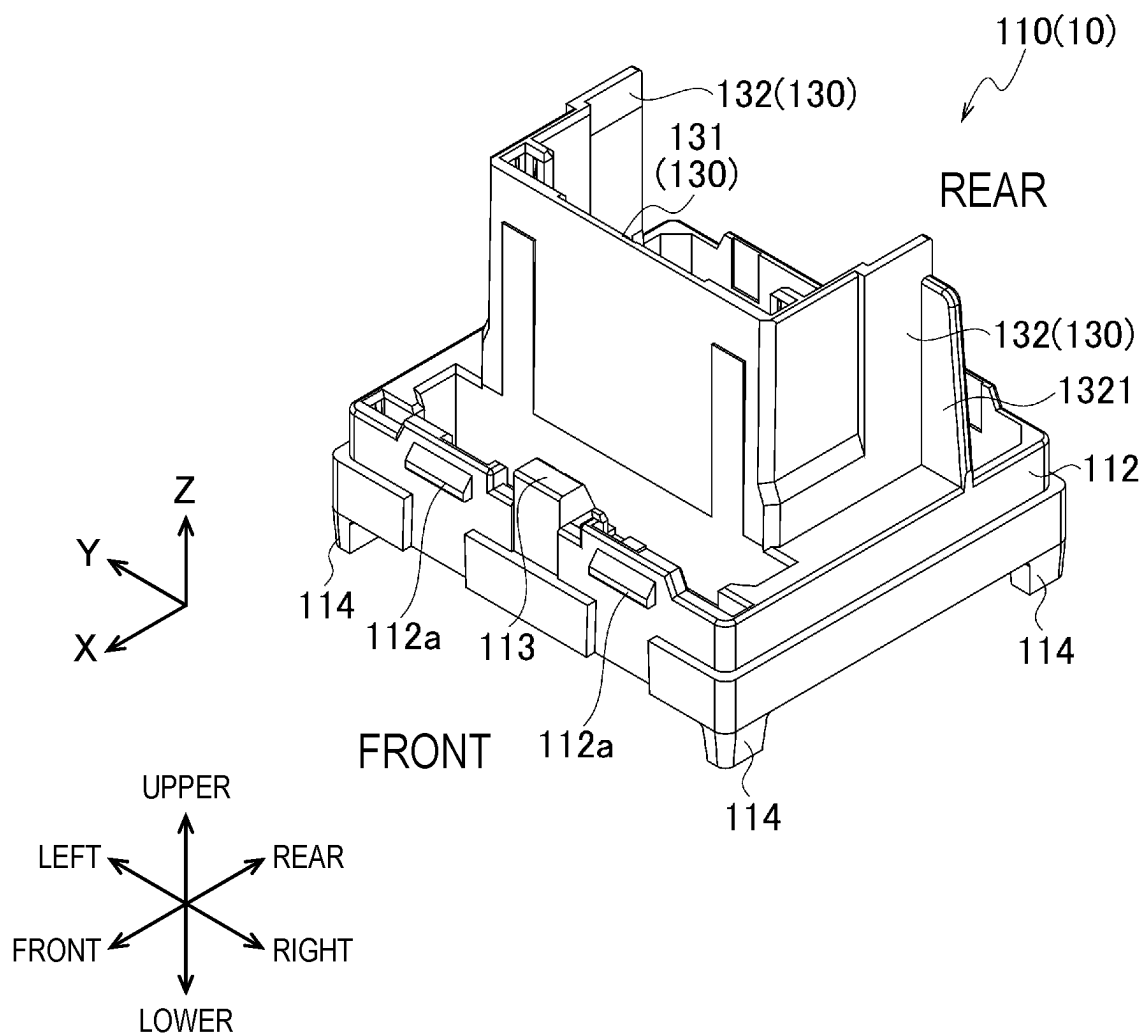


FIG. 38

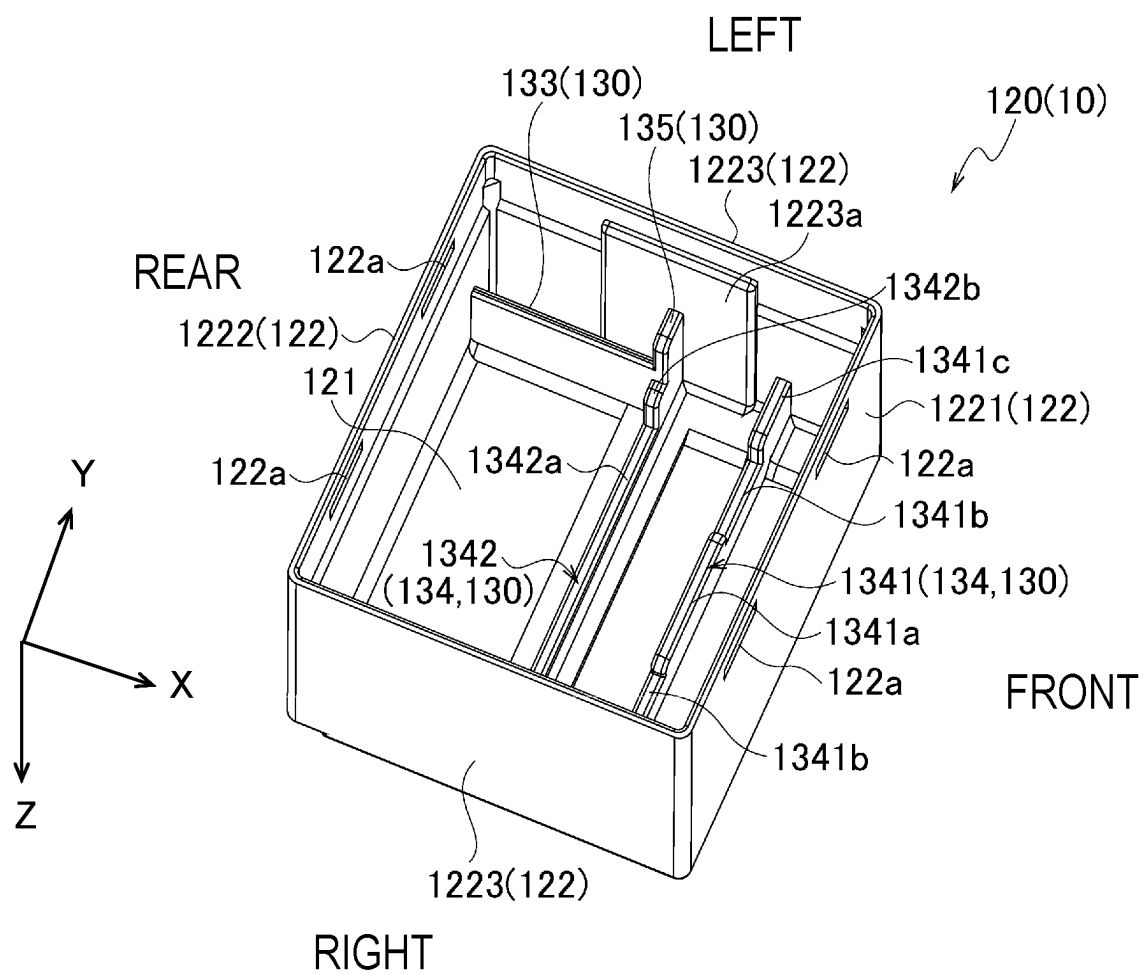


FIG. 39

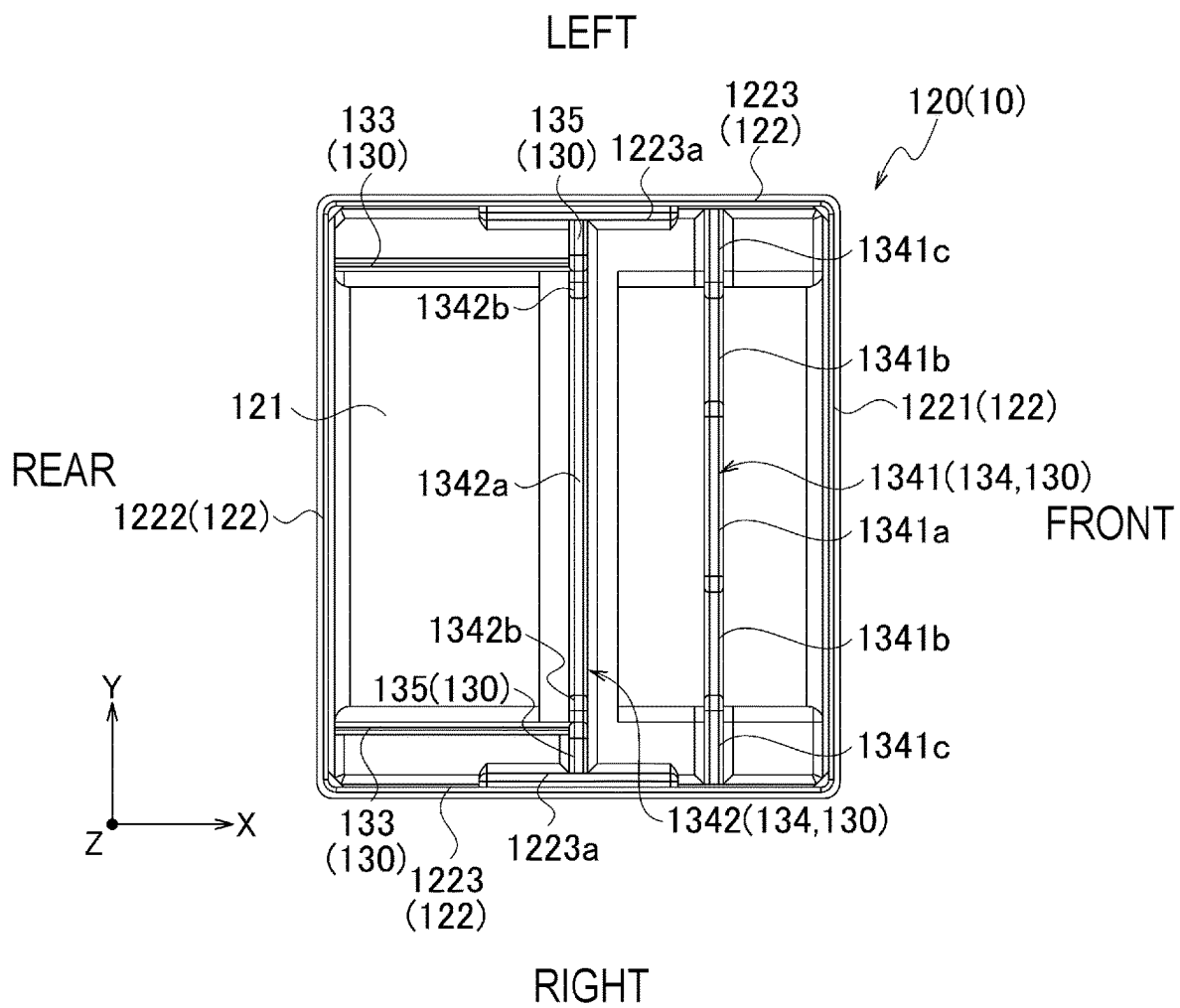


FIG. 40

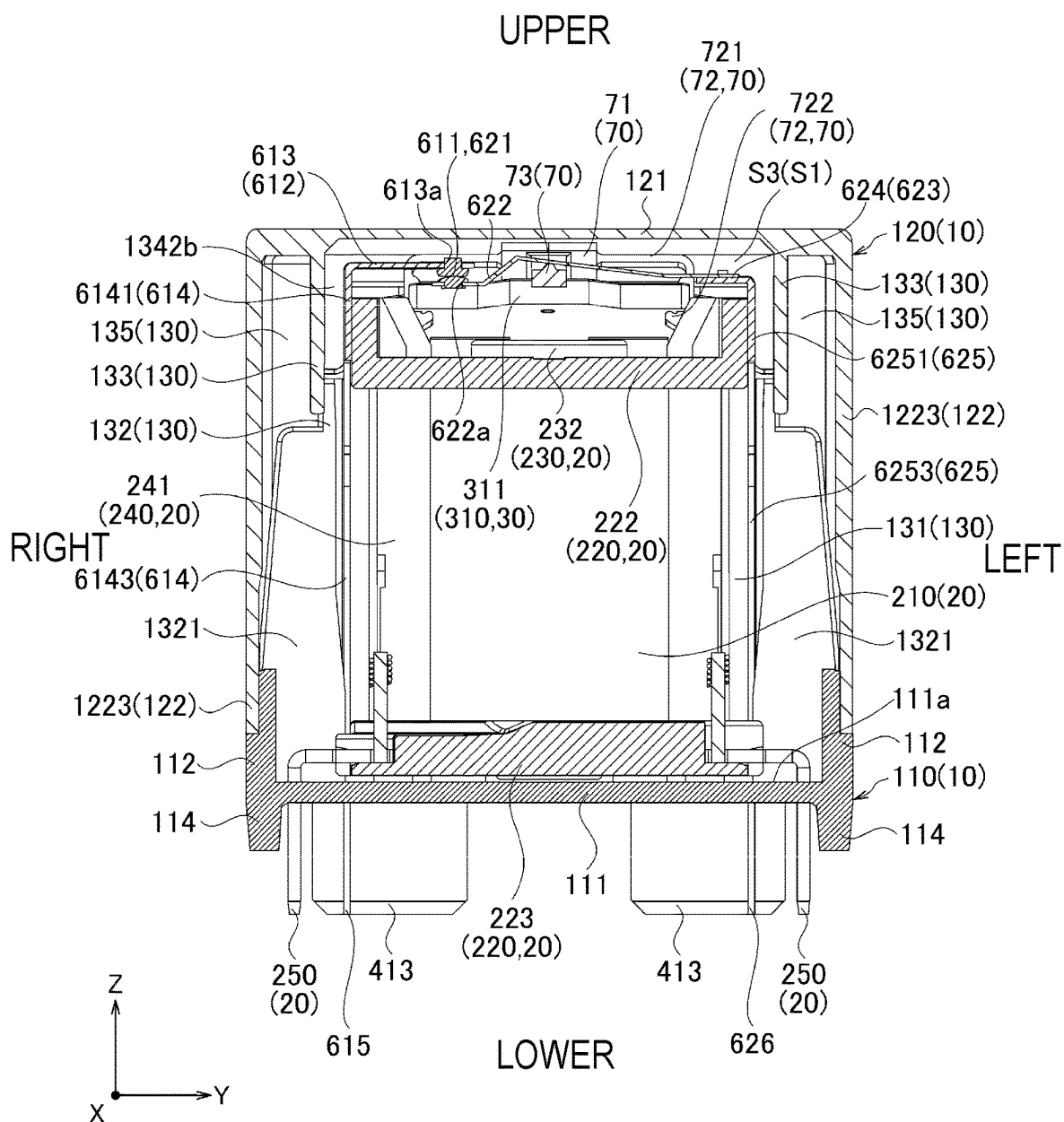


FIG. 41

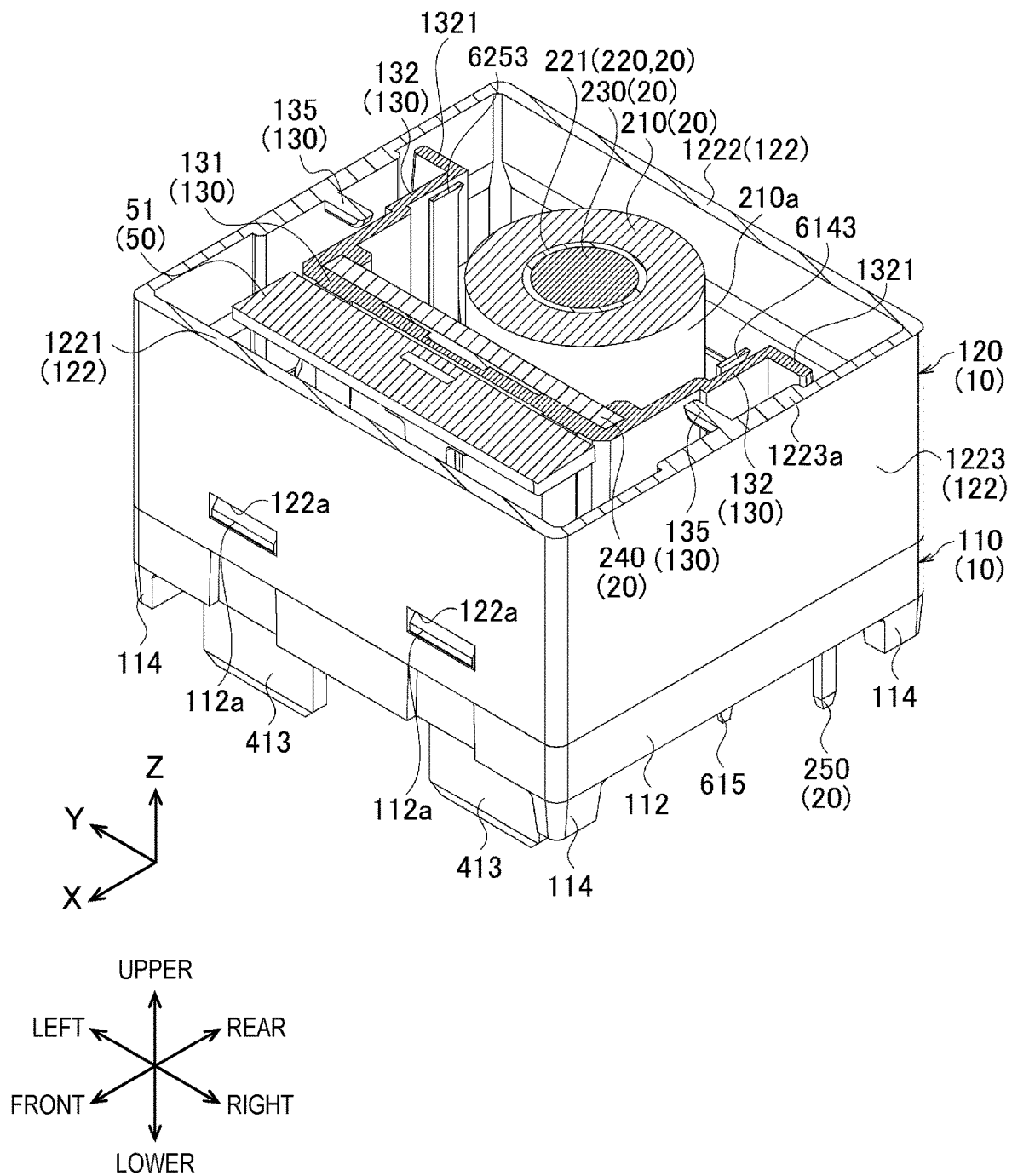


FIG. 42

RIGHT

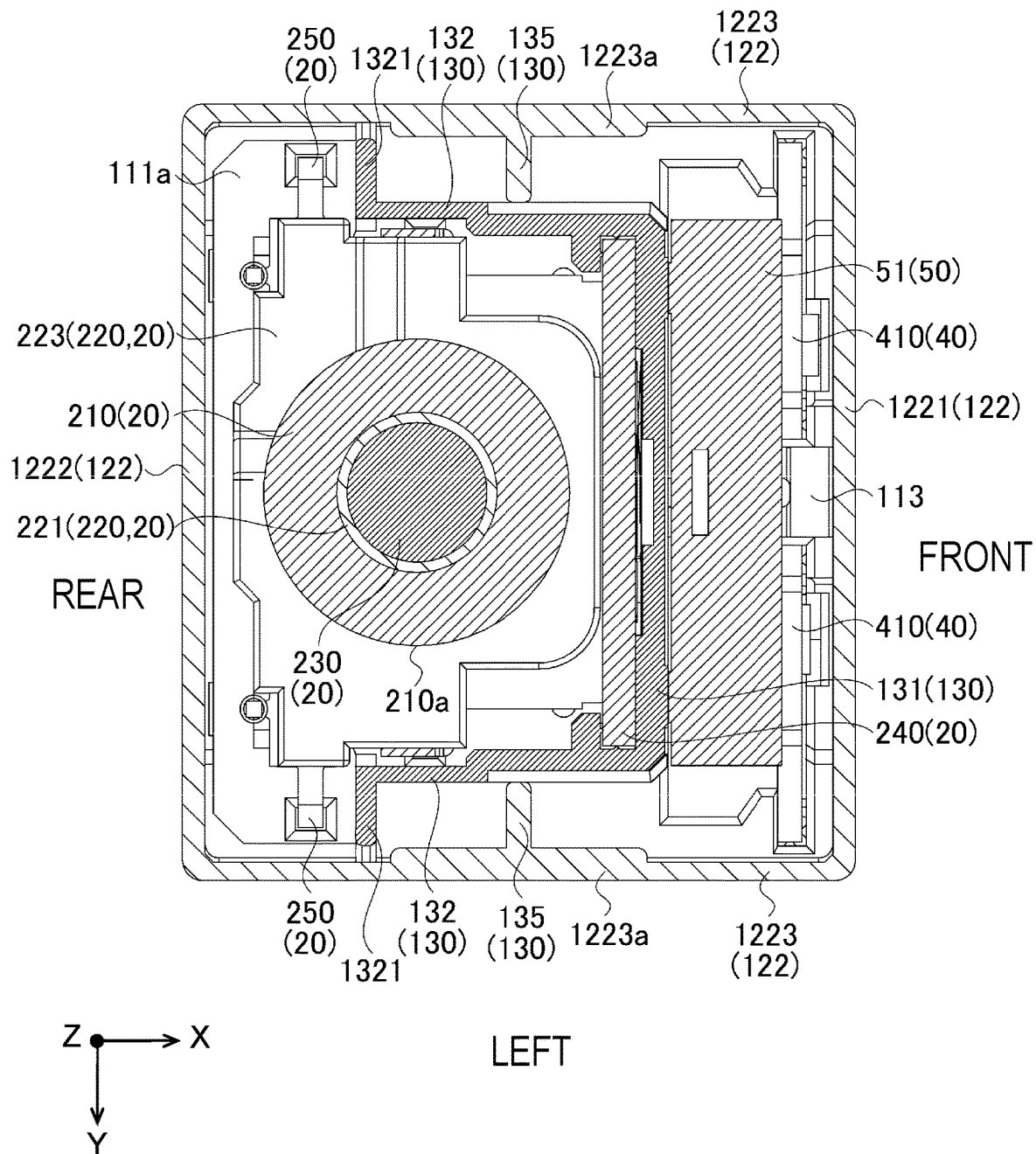
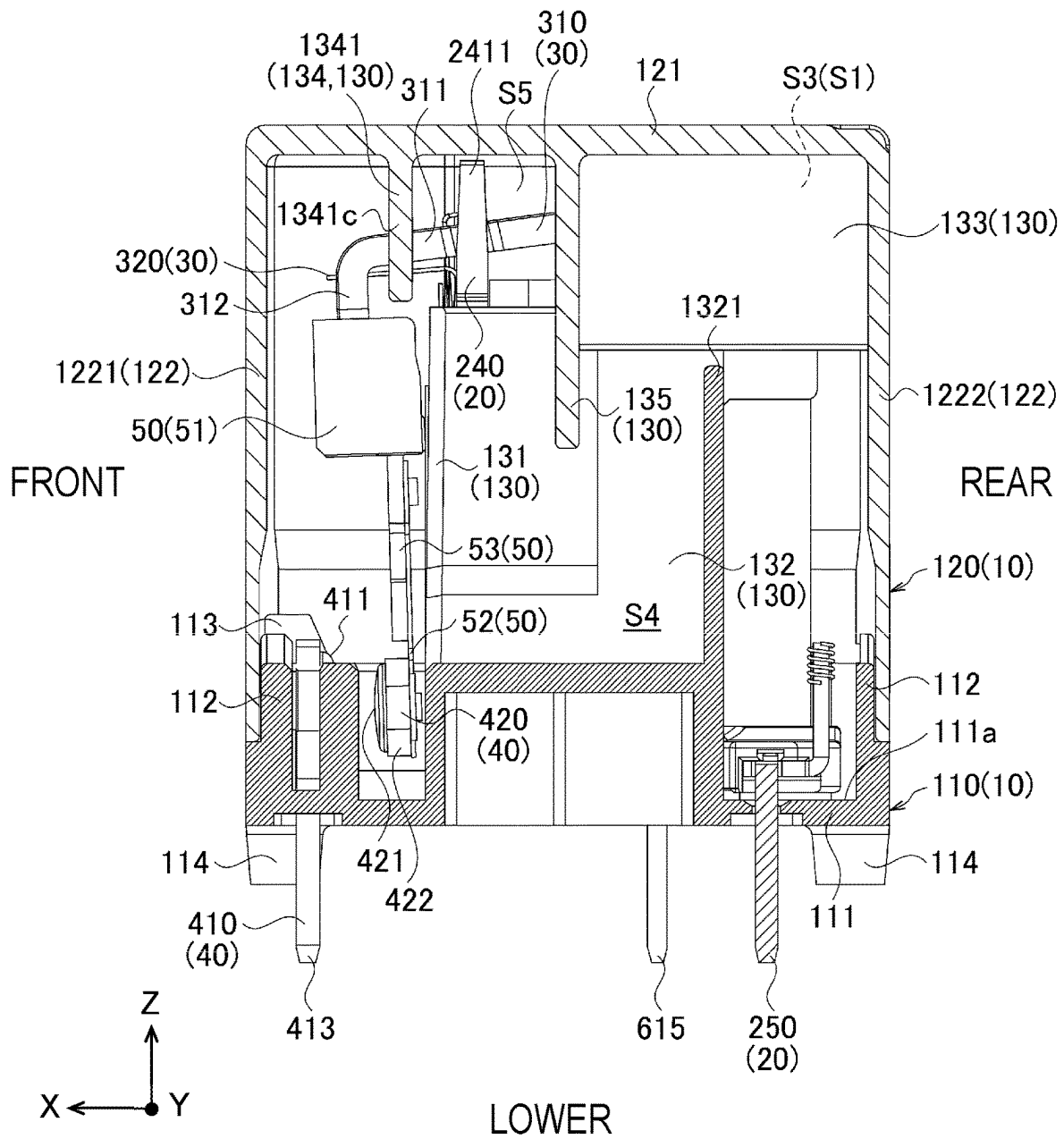


FIG. 44

UPPER



1

ELECTROMAGNETIC RELAY**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a U.S. national stage application of the PCT International Application No. PCT/JP2021/045405 filed on Dec. 9, 2021, which claims the benefit of foreign priority of Japanese patent application No. 2021-000828 filed on Jan. 6, 2021, the contents all of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to an electromagnetic relay.

BACKGROUND ART

Conventionally, as disclosed in the following PTL 1, an electromagnetic relay including a housing having a base and a case covering the base and having an internal space is known.

In PTL 1, the electromagnetic relay includes a main contact portion having a main fixed contact and a main moving contact that comes into contact with or separates from the main fixed contact, and an auxiliary contact portion having an auxiliary fixed contact and an auxiliary moving contact that comes into contact with or separates from the auxiliary fixed contact. The main contact portion and the auxiliary contact portion are disposed in an internal space formed in the housing. By bringing the main fixed contact and the main moving contact into contact with or separates from each other, conduction and non-conduction between the main fixed contact portion and the main moving contact portion can be switched. On the other hand, by bringing the auxiliary fixed contact and the auxiliary moving contact into contact with or separated from each other, conduction and non-conduction between the auxiliary fixed contact portion and the auxiliary moving contact portion can be switched.

CITATION LIST**Patent Literature**

PTL 1: Unexamined Japanese Patent Publication No. 2015-115248

SUMMARY OF THE INVENTION

As an electromagnetic relay having a main contact portion and an auxiliary contact portion, there is an electromagnetic relay in which a current flowing through the main contact portion is larger than that in the electromagnetic relay disclosed in PTL 1.

In the electromagnetic relay in which the current flowing through the main contact portion is large, there is a high possibility that the auxiliary contact portion is affected by the consumable powder scattered when the main contact portion comes into contact with or separates from the auxiliary contact portion.

Therefore, in the electromagnetic relay in which the current flowing through the main contact portion is large, it is preferable to more reliably prevent the auxiliary contact portion from being affected by the consumable powder or

2

the like scattered when the main contact portion is brought into contact with or separated from the auxiliary contact portion.

That is, in the electromagnetic relay including the main contact portion and the auxiliary contact portion, it is preferable to more reliably prevent the contact reliability of the auxiliary contact portion from deteriorating even when the current flowing through the main contact portion is large.

Therefore, an object of the present disclosure is to provide an electromagnetic relay capable of more reliably preventing the contact reliability of the auxiliary contact portion from deteriorating even when the current flowing through the main contact portion is large.

An electromagnetic relay according to the present disclosure includes: a housing including a base and a case that opens downward and is provided on the base, the housing having an internal space surrounded by the base and the case; a main fixed contact; a main moving contact that comes into contact with and separates from the main fixed contact; an auxiliary fixed contact; an auxiliary moving contact that comes into contact with and separates from the auxiliary fixed contact; a coil provided in the housing such that a central axis extends in a vertical direction; an iron core disposed so as to be surrounded by the coil; a yoke disposed around the coil; an armature that is disposed to face an upper end of the iron core and swings in response to excitation/non-excitation of the coil; a movable part provided with the main moving contact and configured to move in response to swinging of the armature; and a partition wall that divides the internal space of the housing into a first space and a second space. The main fixed contact and the main moving contact are disposed in front of the coil. The main fixed contact, the main moving contact, the auxiliary fixed contact, the auxiliary moving contact, the coil, the iron core, the yoke, the armature, the movable part, and the partition wall are provided in the internal space of the housing. The main fixed contact and the main moving contact are disposed in the first space. The auxiliary fixed contact and the auxiliary moving contact are disposed in the second space. The partition wall includes a first side wall connected to the base, a pair of second side walls connected to the base, and a pair of third side walls. When viewed from above, the first side wall is disposed between: (1) the main fixed contact and the main moving contact; and (2) the coil, the iron core, the yoke, the auxiliary fixed contact, and the auxiliary moving contact. One of the pair of second side walls is connected to one end of the first side wall. An other of the pair of second side walls is connected to an other end of the first side wall. One of the pair of third side walls is disposed so as to be adjacent to one of the pair of second side walls. An other of the pair of third side walls is disposed so as to be adjacent to an other of the pair of second side walls. The coil is surrounded by the first side wall and the pair of second side walls. (1) Each of the pair of third side walls is located on a side of the auxiliary fixed contact and the auxiliary moving contact, or (2) each of the pair of third side walls is located above the upper end of the iron core.

According to the present disclosure, it is possible to suppress a decrease in contact reliability of the auxiliary contact portion (the auxiliary fixed contact and the auxiliary moving contact) even when a current flowing through the main contact portion (configured by the main fixed contact and the main moving contact) is large.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an electromagnetic relay according to a first exemplary embodiment as viewed from one direction.

3

FIG. 2 is a perspective view of the electromagnetic relay according to the first exemplary embodiment as viewed from another direction.

FIG. 3 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is an exploded perspective view of a state in which a cover is removed as viewed from one direction.

FIG. 4 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is an exploded perspective view of a state in which the cover is removed as viewed from another direction.

FIG. 5 is a plan view illustrating members other than the cover of the electromagnetic relay according to the first exemplary embodiment as viewed from above.

FIG. 6 is an exploded perspective view of members other than the cover of the electromagnetic relay according to the first exemplary embodiment as viewed from one direction.

FIG. 7 is an exploded perspective view of members other than the cover of the electromagnetic relay according to the first exemplary embodiment as viewed from another direction.

FIG. 8 is an exploded perspective view of the electromagnetic device included in the electromagnetic relay according to the first exemplary embodiment as viewed from one direction.

FIG. 9 is an exploded perspective view of a moving member, a movable part, and a main moving contact portion included in the electromagnetic relay according to the first exemplary embodiment as viewed from one direction.

FIG. 10 is an exploded perspective view of the moving member, the movable part, and the main moving contact portion included in the electromagnetic relay according to the first exemplary embodiment as viewed from another direction.

FIG. 11 is an exploded perspective view of an auxiliary contact portion included in the electromagnetic relay according to the first exemplary embodiment as viewed from one direction.

FIG. 12 is a view illustrating contact or separation between the main contact portion and the auxiliary contact portion according to the first exemplary embodiment, and is a perspective view illustrating a state in which the main contact portion and the auxiliary contact portion are at a second position.

FIG. 13 is a view illustrating contact or separation between the main contact portion and the auxiliary contact portion according to the first exemplary embodiment, and is a perspective view illustrating a state in which the main contact portion and the auxiliary contact portion are at a first position.

FIG. 14 is a view illustrating the contact or separation between the main contact portion and the auxiliary contact portion according to the first exemplary embodiment, and is a vertical cross-sectional view illustrating a state in which the main contact portion and the auxiliary contact portion are at a second position.

FIG. 15 is a view illustrating the contact or separation between the main contact portion and the auxiliary contact portion according to the first exemplary embodiment, and is a vertical cross-sectional view illustrating a state in which the main contact portion and the auxiliary contact portion are at the first position.

FIG. 16 is a perspective view illustrating the inside of the cover according to the first exemplary embodiment.

FIG. 17 is a rear view illustrating the cover according to the first exemplary embodiment.

4

FIG. 18 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is a cross-sectional view illustrating a state in which a position where the auxiliary contact of the electromagnetic relay exists is cut by a vertical plane extending in the horizontal direction.

FIG. 19 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is a perspective view illustrating a state in which a center in a vertical direction of the electromagnetic relay is cut by a horizontal plane.

FIG. 20 is an enlarged perspective view of a part of FIG. 19.

FIG. 21 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is a perspective view illustrating a state in which a position where a third side wall of the electromagnetic relay exists is cut by a vertical plane extending in the front-rear direction.

FIG. 22 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is a perspective view illustrating a state in which the position where the pressing wall of the electromagnetic relay exists is cut by a vertical plane extending in the left-right direction.

FIG. 23 is a view illustrating the electromagnetic relay according to the first exemplary embodiment, and is a perspective view illustrating a state in which a position where the auxiliary drive unit of the electromagnetic relay exists is cut by a horizontal plane.

FIG. 24 is an exploded perspective view of a state in which a cover of the electromagnetic relay according to a second exemplary embodiment is removed as viewed from one direction.

FIG. 25 is an exploded perspective view of a state in which the cover of the electromagnetic relay according to the second exemplary embodiment is removed as viewed from another direction.

FIG. 26 is a perspective view of a base according to the second exemplary embodiment as viewed from one direction.

FIG. 27 is a perspective view of a base according to the second exemplary embodiment as viewed from another direction.

FIG. 28 is a perspective view illustrating the inside of a cover according to the second exemplary embodiment.

FIG. 29 is a rear view illustrating the cover according to the second exemplary embodiment.

FIG. 30 is a view illustrating an electromagnetic relay according to the second exemplary embodiment, and is a cross-sectional view illustrating a state in which a position where an auxiliary contact of the electromagnetic relay exists is cut by a vertical plane extending in the left-right direction.

FIG. 31 is a view illustrating an electromagnetic relay according to the second exemplary embodiment, and is a perspective view illustrating a state in which a center in a vertical direction of the electromagnetic relay is cut by a horizontal plane.

FIG. 32 is a view illustrating an electromagnetic relay according to the second exemplary embodiment, and is a perspective view illustrating a state in which a position where an auxiliary drive unit of the electromagnetic relay exists is cut by a horizontal plane.

FIG. 33 is a view illustrating the electromagnetic relay according to the second exemplary embodiment, and is a perspective view illustrating a state in which the position where the extension part of the second side wall of the

5

electromagnetic relay exists is cut by a vertical plane extending in the front-rear direction.

FIG. 34 is an exploded perspective view of a state in which a cover of the electromagnetic relay according to a third exemplary embodiment is removed as viewed from one direction.

FIG. 35 is an exploded perspective view of a state in which the cover of the electromagnetic relay according to the third exemplary embodiment is removed as viewed from another direction.

FIG. 36 is a perspective view of a base according to the third exemplary embodiment as viewed from one direction.

FIG. 37 is a perspective view of a base according to the third exemplary embodiment as viewed from another direction.

FIG. 38 is a perspective view illustrating the inside of a cover according to the third exemplary embodiment.

FIG. 39 is a rear view illustrating the cover according to the third exemplary embodiment.

FIG. 40 is a view illustrating the electromagnetic relay according to the third exemplary embodiment, and is a cross-sectional view illustrating a state in which a position where the auxiliary contact of the electromagnetic relay exists is cut by a vertical plane extending in the horizontal direction.

FIG. 41 is a view illustrating an electromagnetic relay according to the third exemplary embodiment, and is a perspective view illustrating a state in which a center in a vertical direction of the electromagnetic relay is cut by a horizontal plane.

FIG. 42 is a plan view of the electromagnetic relay illustrated in FIG. 41.

FIG. 43 is a view illustrating the electromagnetic relay according to the third exemplary embodiment, and is a perspective view illustrating a state in which the position where the extension part of the second side wall of the electromagnetic relay exists is cut by a vertical plane extending in the front-rear direction.

FIG. 44 is a side view of the electromagnetic relay illustrated in FIG. 43.

DESCRIPTION OF EMBODIMENT

An exemplary embodiment of the present disclosure will now be described in detail with reference to the attached drawings. In the following description, the vertical direction in a state where a base is positioned below a case and coils are disposed such that the axial direction extends in the vertical direction is referred to as a Z direction (axial direction). Specifically, a description will be given assuming that the vertical direction in a state where the base is positioned below the case in a state where the base portion extends along a horizontal plane and the coil is disposed such that the axial direction extends in the vertical direction is a Z direction (axial direction).

In addition, a direction intersecting the Z direction (axial direction) will be described as an X direction (first direction: front-rear direction: a direction in which the main fixed contact and the main moving contact face each other). A direction intersecting the X direction and the Z direction will be described as a Y direction (second direction: width direction: longitudinal direction of moving contactor). Specifically, the Z direction is orthogonal to the X direction, and the Y direction is orthogonal to the X direction and the Z direction.

In addition, description will be given by defining an upper side and a lower side in a state where the base is positioned

6

below the case, defining a side on which the main fixed contact is disposed as a front side in the front-rear direction, and defining a side on which the main moving contact is disposed as a rear side in the front-rear direction.

Substantially identical constitutional elements are included in the plurality of exemplary embodiments described hereinafter respectively. Accordingly, in the following description, common symbols are given to the substantially identical constitutional elements, and the repeated explanation of these constitutional elements is omitted.

Further, in the present disclosure, the description will be given by using terms indicating directions such as "upper", "lower", "left", "right", "front", and "rear". However, these terms merely indicate a relative positional relationship, and the present disclosure is not limited thereto. For example, when electromagnetic relay 1 of the present disclosure is installed in an inclined manner, the direction of electromagnetic relay 1 in an actual use state may be different from the direction described in the present disclosure.

First Exemplary Embodiment

As illustrated in FIGS. 1 and 2, electromagnetic relay 1 according to the present exemplary embodiment includes housing 10 formed of a resin material in a hollow box shape. In the present exemplary embodiment, housing 10 includes base 110 and case 120 covering base 110, and has a substantially rectangular parallelepiped outer surface. In addition, internal space S1 is formed in housing 10 in a state where case 120 is attached to base 110. Note that the shape of the outer surface of housing 10 is not limited to a rectangular parallelepiped shape, and may have any shape.

Electromagnetic device (drive unit) 20 is disposed on the rear side in the X direction (front-rear direction: first direction) in internal space S1 of housing 10, and main contact portion 40 is disposed on the front side in the X direction (front-rear direction: first direction). Further, auxiliary contact portion 60 is disposed behind housing 10 in the X direction (front-rear direction: first direction) and above housing 10 in the Z direction (vertical direction: axial direction) in internal space S1.

Here, in the present exemplary embodiment, main contact portion 40 is a so-called normally closed contact that is turned on in the initial state, and auxiliary contact portion 60 is a so-called normally open contact that is turned off in the initial state. Note that main contact portion 40 can be a so-called normally open contact that is turned off in the initial state, and auxiliary contact portion 60 can be a so-called normally closed contact that is turned on in the initial state.

Base 110 includes base portion 111 in a substantially rectangular plate-shape extending along a substantially horizontal plane (a direction intersecting the Z direction: an XY plane), and a peripheral wall 112 continuously connected to a peripheral edge of base portion 111 and extending in the Z direction (vertical direction) (see FIGS. 3 to 7). A stepped portion is formed on the opening peripheral edge on the upper end side of peripheral wall 112, and the outer periphery is smaller than that on the lower end side. A pair of protrusions 112a is juxtaposed in the horizontal direction on each of the front surface and the rear surface above the stepped portion of peripheral wall 112.

On the other hand, case 120 has a substantially box shape that opens downward, and case 120 is attached to base 110 from above.

Case 120 includes top wall 121 in a substantially rectangular plate-shape extending along a substantially horizontal

plane (a direction intersecting the Z direction: an XY plane), and peripheral wall **122** extending downward in the Z direction (vertical direction) from a peripheral edge of top wall **121** (see FIGS. 3 to 5).

Peripheral wall **122** includes front wall **1221** that is located on the front side in the X direction (front-rear direction) and extends in the Y direction (width direction) and the Z direction (vertical direction), and a rear wall **1222** that is located backward in the X direction (front-rear direction) and extends in the Y direction (width direction) and the Z direction (vertical direction). Peripheral wall **122** includes a pair of side walls **1223** that is continuously connected to front wall **1221** and rear wall **1222** on both sides in the Y direction (width direction) and extends in the X direction (front-rear direction) and the Z direction (vertical direction).

A pair of insertion holes **122a** into which protrusion **112a** of base **110** are inserted when case **120** is attached to base **110** are juxtaposed in the horizontal direction in lower portions of the front wall **1221** and rear wall **1222**.

In the present exemplary embodiment, base **110** includes first side wall **131** that is continuously provided so as to rise upward from bottom surface **111a** of base portion **111** and extends in the Y direction (width direction). Further, base **110** includes a pair of second side walls **132** that is continuously connected from both ends in the Y direction (width direction) of first side wall **131** backward in the X direction (front-rear direction). The pair of second side walls **132** also extends upward from bottom surface **111a** of base portion **111**. Yoke **240** of electromagnetic device **20** is held by first side wall **131** and the pair of second side walls **132** extending upward from bottom surface **111a** of base portion **111**, so that three sides (at least a part) of side surface **210a** of coil **210** are surrounded.

As described above, in the present exemplary embodiment, electromagnetic device **20** is disposed on the rear side of first side wall **131**. Main contact portion **40** is disposed on the front side of first side wall **131** (see FIGS. 2 to 4). That is, in the present exemplary embodiment, electromagnetic device **20** and main contact portion **40** are disposed in internal space **S1** in a state partitioned in the X direction (front-rear direction) by first side wall **131**.

In addition, partition wall **113** is formed on the front side of first side wall **131** of base **110**, and a creepage distance between a pair of fixed contact portions **310**, **310** described later is secured by partition wall **113**.

In addition, base **110** is provided with raising member **114** for providing a gap between base **110** and a printed circuit board (not illustrated) when electromagnetic relay **1** is disposed on the printed circuit board.

Electromagnetic device (drive unit) **20** is a device that generates an electromagnetic force, and includes coil **210** that generates a magnetic flux by being energized, and hollow cylindrical coil bobbin **220** around which coil **210** is wound (see FIG. 8).

As coil **210**, for example, a conductive wire can be used. In the present exemplary embodiment, coil **210** is disposed in internal space **S1** of housing **10** such that the axial direction extends in the Z direction (vertical direction) in a state where base **110** is positioned on the lower side of case **120**.

coil bobbin **220** is formed of resin which is an insulating material, and a cylindrical portion **221** having a cylindrical shape and extending in the Z direction (vertical direction: axial direction of the coil) is formed at a central portion of coil bobbin **220**. Through-hole **2211** penetrating in the Z

direction (vertical direction: axial direction of coil) is formed inside cylindrical portion **221**.

In addition, coil bobbin **220** includes upper flange portion **222** in a substantially rectangular shape which is continuously connected to the upper end of cylindrical portion **221** around which coil **210** is wound on the outer surface and protrudes radially outward of cylindrical portion **221**. Coil bobbin **220** includes lower flange portion **223** in a substantially rectangular shape which is continuously connected to the lower end of cylindrical portion **221** and protrudes radially outward of cylindrical portion **221**.

Further, in the present exemplary embodiment, upper flange portion **222** is formed with upper auxiliary contact retainer **2221** that holds auxiliary contact portion **60**. Upper auxiliary contact retainer **2221** is formed at each of both ends in the Y direction (width direction) at the rear end in the X direction (front-rear direction) of upper flange portion **222**. Each of upper auxiliary contact retainers **2221** is provided with press-fitting opening **2221a** that opens outward in the Y direction (width direction) and into which press-fitting piece **6141a**, **6251a** of auxiliary contact portion **60** to be described later are press-fitted (see FIGS. 6 to 8). Restriction wall **2221b** that regulates coming-off and rotation of press-fitting piece **6141a**, **6251a** is formed around press-fitting opening **2221a** in upper auxiliary contact retainer **2221** (see FIGS. 6 to 8).

On the other hand, lower auxiliary contact retainer **2231** that holds auxiliary contact portion **60** is formed in lower flange portion **223**. In the present exemplary embodiment, lower flange portion **223** is formed such that the rear end side in the X direction (front-rear direction) is wider than the front end side, and lower auxiliary contact retainers **2231** are respectively formed at both ends in the Y direction (width direction) on the front end side of the wide portion. Each of lower auxiliary contact retainers **2231** is provided with press-fitting opening **2231a** that opens outward in the Y direction (width direction) and into which press-fitting piece **6143a**, **6253a** of auxiliary contact portion **60** to be described later are press-fitted (see FIGS. 6 to 8). Restriction wall **2231b** that restricts coming off and rotation of press-fitting piece **6143a**, **6253a** is formed around press-fitting opening **2231a** in lower auxiliary contact retainer **2231** (see FIGS. 6 to 8).

Electromagnetic device **20** also includes iron core **230** that is inserted into through-hole **2211** formed in cylindrical portion **221** of coil bobbin **220** and is magnetized (through which magnetic flux passes) by energized coil **210**. Iron core **230** is disposed inside coil **210**.

Iron core **230** includes shaft portion **231** in a substantially columnar shape extending in the Z direction (vertical direction) and head **232** in a substantially columnar shape formed to have a larger diameter than shaft portion **231** and continuously connected to an upper end of shaft portion **231** (see FIG. 8).

Electromagnetic device **20** includes yoke **240** disposed around coil **210** wound around cylindrical portion **221**. In the present exemplary embodiment, yoke **240** is a substantially plate-shaped member made of a magnetic material, and has a substantially L shape in a side view (as viewed along the Y direction). That is, yoke **240** includes vertical wall (upright portion) **241** disposed on the front side of coil **210** wound around cylindrical portion **221** so as to extend substantially along the vertical plane, and horizontal wall **242** extending to the rear side from the lower end of vertical wall **241** (see FIG. 8). Such yoke **240** can be formed, for example, by bending one plate.

As described above, yoke **240** is supported by first side wall **131** and the pair of second side walls **132** extending upward from bottom surface **111a** of base portion **111** (see FIGS. **3** and **4**). A pair of protruding portions (extension parts) **2411** projecting upward is formed at both ends in the Y direction (width direction) of vertical wall (upright portion) **241** of yoke **240**, and armature **310** is disposed between the pair of protruding portions (extension parts) **2411**.

Further, electromagnetic device **20** includes a pair of coil terminals **250** to which both ends of coil **210** are connected, and electromagnetic device **20** is driven by energizing coil **210** via the pair of coil terminals **250**.

Then, moving member **30** is moved by switching the driving state of electromagnetic device **20**.

In the present exemplary embodiment, moving member **30** includes armature **310** disposed so as to face head **232** of iron core **230** in the vertical direction (Z direction), and hinge spring **320** attached across armature **310** and yoke **240**.

Armature **310** is formed of a conductive metal, and is disposed so as to be swingable in the vertical direction (Z direction) with respect to head **232** of iron core **230** according to the excitation/non-excitation of coil **210**.

In the present exemplary embodiment, armature **310** includes horizontal wall **311** facing head **232** of iron core **230** in the vertical direction (Z direction), and vertical wall **312** extending downward from the front end of horizontal wall **311** in the X direction (front-rear direction) (see FIGS. **9** and **10**).

Horizontal wall **311** of armature **310** is attached to the upper end of vertical wall **241** so as to be swingable in the vertical direction (Z direction), and armature **310** can rotate in the Z direction (vertical direction) about a portion supported by yoke **240**.

Specifically, cutouts **3111** are formed at both ends in the Y direction (width direction) at the front end in the X direction (front-rear direction) of horizontal wall **311**. Protruding portion (extension portion) **2411** of yoke **240** is inserted into cutout **3111**, whereby armature **310** is supported by yoke **240**. As described above, in the present exemplary embodiment, cutout **3111** is a portion supported by yoke **240** of armature **310**.

Further, in the present exemplary embodiment, through-hole **313** penetrating in the Z direction (vertical direction) is formed at the front end of armature **310** in the X direction (front-rear direction). Then, in a state of being inserted into hinge spring **320** and through-hole **313**, armature **310** and yoke **240** are attached. At this time, armature **310** is biased by hinge spring **320** in a direction in which horizontal wall **311** is separated from head **232** of iron core **230**.

When coil **210** is energized, armature **310** is rotated so that horizontal wall **311** approaches head **232** of iron core **230**. Specifically, horizontal wall **311** of armature **310** is attracted to head **232** of iron core **230** by energizing coil **210**, and armature **310** is rotated so that horizontal wall **311** approaches head **232** of iron core **230**. That is, by energizing coil **210** via the pair of coil terminals **250**, horizontal wall **311** of armature **310** rotates downward in the Z direction (vertical direction). At this time, vertical wall **312** continuously connected to horizontal wall **311** rotates forward in the X direction (front-rear direction).

The swing range of armature **310** is set between a position where horizontal wall **311** is farthest from head **232** of iron core **230** and a position where horizontal wall **311** is closest to head **232** of iron core **230**.

In the present exemplary embodiment, the swing range of armature **310** is set between an initial position where hori-

zontal wall **311** is disposed to be separated upward from head **232** of iron core **230** by a predetermined gap and an abutment position where horizontal wall **311** abuts on head **232** of iron core **230**.

Therefore, in the present exemplary embodiment, when coil **210** is energized, armature **310** moves to the abutment position where horizontal wall **311** abuts on head **232** of iron core **230**, and when the energization to coil **210** is stopped, the armature returns to the initial position by the biasing force of hinge spring **320**.

As described above, armature **310** according to the present exemplary embodiment is disposed to face head **232** of iron core **230** via a predetermined gap when coil **210** is not energized, and swings so as to be attracted to head **232** side of iron core **230** when coil **210** is energized.

Then, by switching the driving state of electromagnetic device **20** and swinging armature **310**, it is possible to switch conduction and non-conduction between main fixed contact portion **410** and main moving contact portion **420** paired with each other (having main contacts that come into contact with or separate from each other).

In the present exemplary embodiment, main contact portion **40** that opens and closes the main contact according to turning on and off of energization of coil **210** is provided on the front side of electromagnetic device **20**.

Main contact portion **40** includes main fixed contact portion **410** and main moving contact portion **420**, and main fixed contact portion **410** includes main fixed contact **411** and main body **412** having main fixed contact **411**. On the other hand, main moving contact portion **420** includes main moving contact **421** that moves relative to main fixed contact **411** and can come into contact with or separate from main fixed contact **411**, and moving contactor **422** having main moving contact **421**.

In the present exemplary embodiment, main contact portion **40** includes only one set of main fixed contact portion **410** and the main moving contact portion **420** (having main contacts that come into contact with or separate from each other) forming a pair with each other (see FIGS. **6** and **7**).

In the present exemplary embodiment, the set of main fixed contact portion **410** and main moving contact portion **420** having the main contacts that come into contact with or separate from each other includes the pair of main fixed contact portions **410** and one main moving contact portion **420**.

Specifically, two main fixed contact portions **410** having a symmetrical shape with respect to the XZ plane are a pair of main fixed contact portions **410**. Two main fixed contact portions **410** forming a pair are fixed to base **110** (housing **10**) in a state of being separated in the Y direction (width direction: axial direction and direction intersecting the first direction: second direction).

Each of main fixed contact portions **410** includes main body **412** having one main fixed contact **411** (see FIGS. **6** and **7**). In the present exemplary embodiment, a member to be a main fixed contact is inserted into insertion hole **412a** formed in main body **412** so as to penetrate in the plate thickness direction, and riveting is performed, so that main body **412** has main fixed contact **411** (see FIGS. **14** and **15**). As described above, in the present exemplary embodiment, main body **412** has a function as a fixed-side main contact holder that holds main fixed contact **411**.

Main fixed contact **411** is not necessarily formed on main body **412** by rivet joining, and can be formed by various methods. For example, it is also possible to cause a portion protruding by performing dowel processing on main body **412** to function as a main fixed contact. Further, by bringing

11

main moving contact **421** into contact with a part of the flat surface of main body **412**, it is also possible to cause a part of the flat surface of main body **412** to function as the main fixed contact.

Further, main fixed contact portion **410** includes terminal portion **413** which is connected to a lower end of main body **412** and is fixed to base **110** (housing **10**) in a state where a distal end (connection part) protrudes outward (downward) from base **110** (housing **10**).

In the present exemplary embodiment, insertion hole **115** penetrating in the Z direction (vertical direction) is formed in base **110**. Then, the distal end (connection part: lower end) of terminal portion **413** is inserted into insertion hole **115** from above. With such a configuration, main fixed contact portion **410** is fixed to base **110** (housing **10**) in a state where the distal end (connection part: lower end) of terminal portion **413** protrudes outward (downward) from base **110** (see FIGS. **14** and **15**). Main fixed contact portion **410** is fixed to base **110** (housing **10**) with an adhesive or the like.

At this time, main fixed contact portion **410** is fixed to base **110** (housing **10**) with main fixed contact **411** facing backward in the X direction (front-rear direction). That is, main fixed contact portion **410** is fixed to base **110** (housing **10**) in a state where a surface of main body **412** on a side where main fixed contact **411** is formed (rear surface: surface on a side facing main moving contact **421**) faces backward.

Note that main fixed contact **411**, main body **412**, and terminal portion **413** can be formed of, for example, a conductive material such as a silver-based material or a copper-based material.

As described above, in the present exemplary embodiment, two main fixed contacts **411** are disposed side by side in the Y direction that is the direction orthogonal (intersecting) to the direction in which main fixed contact **411** and main moving contact **421** relatively move. One of two main bodies **412** has one main fixed contact **411**, and the other main body has other main fixed contact **411**.

On the other hand, one main moving contact portion **420** includes one moving contactor **422**, and one moving contactor **422** includes a pair of main moving contacts **421** arranged side by side in the Y direction (width direction) (see FIGS. **9** and **10**).

In the present exemplary embodiment, a member to be a main moving contact is inserted into insertion hole **422a** formed on both sides in the longitudinal direction of moving contactor **422** having a substantially rectangular plate shape so as to penetrate in the plate thickness direction, and riveting is performed. Thus, main moving contactor **422** includes main moving contact **421** (see FIGS. **14** and **15**). As described above, in the present exemplary embodiment, moving contactor **422** has a function as a movable-side main contact holder that holds main moving contact **421**.

Main moving contact **421** is not necessarily formed on moving contactor **422** by riveting, and can be formed by various methods. For example, it is also possible to cause a portion protruding by performing dowel processing on moving contactor **422** to function as a main moving contact. Further, by bringing a part of the flat surface of moving contactor **422** into contact with main fixed contact **411**, it is also possible to cause a part of the flat surface of moving contactor **422** to function as the main moving contact.

One of main moving contact portions **420** is disposed so as to be located behind two main fixed contact portions **410** forming a pair in the X direction (front-rear direction) with the plate thickness direction substantially coinciding with the X direction (front-rear direction) and the longitudinal

12

direction substantially coinciding with the Y direction (width direction). At this time, main moving contact portion **420** is disposed in a state where main moving contact **421** faces main fixed contact **411** in the X direction (front-rear direction). Specifically, moving contactor **422** is disposed such that main moving contact **421** formed on one side in the Y direction (width direction) faces main fixed contact **411** of main fixed contact portion **410** disposed on one side in the Y direction (width direction) in the X direction (front-rear direction). Similarly, moving contactor **422** is disposed such that main moving contact **421** formed on the other side in the Y direction (width direction) faces main fixed contact **411** of main fixed contact portion **410** disposed on the other side in the Y direction (width direction) in the X direction (front-rear direction). Thus, one main moving contact **421** comes into contact with or separates from one main fixed contact **411** of two main fixed contacts **411**, and other main moving contact **421** comes into contact with or separates from other main fixed contact **411**. One moving contactor **422** includes two main moving contacts **421**.

Note that main moving contact **421** and moving contactor **422** can also be formed of, for example, a conductive material such as a silver-based material or a copper-based material.

A set including the pair of main fixed contact portions **410** and one main moving contact portion **420** having such a configuration is accommodated on the front side of first side wall **131** in the X direction (front-rear direction: first direction) in internal space **S1** (see FIGS. **12** to **15**).

Here, main moving contact portion **420** is disposed so as to be relatively swingable in the X direction (front-rear direction) with respect to the pair of main fixed contact portions **410**.

In the present exemplary embodiment, main contact portion **40** is continuously connected to armature **310** via movable part **50**. By swinging movable part **50** in the X direction (front-rear direction) along with the swinging of armature **310**, main moving contact portion **420** swings in the X direction (front-rear direction) in conjunction with the operation of movable part **50**. That is, by causing movable part **50** to hold main moving contact portion **420**, main moving contact portion **420** swings relative to the pair of main fixed contact portions **410** in the X direction (front-rear direction).

As illustrated in FIGS. **9** and **10**, in the present exemplary embodiment, movable part **50** includes holder part **51** which is formed of an insulating resin material and has an upper portion in which insertion hole **511** into which vertical wall **312** of armature **310** is inserted and held is formed. Movable part **50** includes movable plate **52** continuously connected to a lower portion of holder part **51**, and movable spring **53** connecting movable plate **52** and moving contactor **422**.

In the present exemplary embodiment, through-hole **521** penetrating in the plate thickness direction is formed in an upper portion of movable plate **52** in the Z direction (vertical direction). Then, in a state where the upper end of movable plate **52** is inserted into an insertion hole (not illustrated) formed at the lower end of holder part **51**, a projection formed in the insertion hole (not illustrated) of holder part **51** is inserted into through-hole **521**, so that movable plate **52** is held by holder part **51**.

In addition, projection **522** protruding backward is formed at a central portion of movable plate **52** in the Z direction (vertical direction), and upper through-hole **531** penetrating in the plate thickness direction is formed at an upper portion of movable spring **53** in the Z direction (vertical direction). Projection **522** of movable plate **52** is inserted into upper

13

through-hole **531** of movable spring **53**, so that movable spring **53** is held by movable plate **52**.

Further, lower through-hole **532** penetrating in the plate thickness direction is formed at a lower portion of movable spring **53** in the Z direction (vertical direction), and projection **422b** protruding backward is formed at a central portion of moving contactor **422** in the Y direction (width direction). Projection **422b** of moving contactor **422** is inserted into lower through-hole **532** of movable spring **53** so that movable spring **53** holds moving contactor **422**.

In this way, main contact portion **40** is continuously connected to armature **310** via movable part **50**.

With such a configuration, main moving contact portion **420** swings relative to the pair of main fixed contact portions **410** in the X direction (front-rear direction) as armature **310** swings. Therefore, main moving contact **421** swings so as to draw an arc around the upper end of vertical wall **312**.

Further, in the present exemplary embodiment, auxiliary contact portion **60** is disposed in internal space **S1** of housing **10** separately from main contact portion **40**. Auxiliary contact portion **60** is disposed in internal space **S1** in a state in which the auxiliary contacts (auxiliary fixed contact **611** and auxiliary moving contact **621**) exist at positions on the rear side in the X direction (front-rear direction) and on the upper end side of coil **210**.

Auxiliary contact portion **60** includes auxiliary fixed contact portion **610** and auxiliary moving contact portion **620**, and auxiliary fixed contact portion **610** includes auxiliary fixed contact **611** and terminal portion **612** having auxiliary fixed contact **611**. On the other hand, auxiliary moving contact portion **620** includes auxiliary moving contact **621** that moves relative to auxiliary fixed contact **611** and can come into contact with or separate from auxiliary fixed contact **611**, auxiliary moving terminal portion **622** having auxiliary moving contact **621**, and terminal portion **623** connected to auxiliary moving terminal portion **622**. The movable member of auxiliary moving terminal portion **622** is formed of a leaf spring.

In the present exemplary embodiment, auxiliary contact portion **60** includes only one set of auxiliary fixed contact portion **610** and auxiliary moving contact portion **620** (having auxiliary contacts that come into contact with or separate from each other) forming a pair with each other (see FIGS. 6 and 7).

In the present exemplary embodiment, the set of auxiliary fixed contact portion **610** and auxiliary moving contact portion **620** having the auxiliary contacts that come into contact with or separate from each other includes one auxiliary fixed contact portion **610** and one auxiliary moving contact portion **620**. One auxiliary fixed contact **611** is formed in one auxiliary fixed contact portion **610**, and only one auxiliary moving contact **621** that comes into contact with or separates from one auxiliary fixed contact **611** is formed in one auxiliary moving contact portion **620**.

In the present exemplary embodiment, as described above, auxiliary fixed contact portion **610** includes terminal portion **612** having one auxiliary fixed contact **611**, and terminal portion **612** includes horizontal piece **613** extending in the horizontal direction and elongated in the Y direction (width direction). Auxiliary fixed contact **611** is formed on horizontal piece **613**. In the present exemplary embodiment, a member to be an auxiliary fixed contact is inserted into insertion hole **613a** formed in horizontal piece **613** so as to penetrate in the plate thickness direction, and riveting is performed, so that horizontal piece **613** has auxiliary fixed contact **611** (see FIG. 11). As described above, in the present exemplary embodiment, horizontal

14

piece **613** has a function as a fixed-side auxiliary contact holder that holds auxiliary fixed contact **611**.

Auxiliary fixed contact **611** is not necessarily formed on horizontal piece **613** by riveting, and can be formed by various methods. For example, it is also possible to cause a portion protruding by performing dowel processing on horizontal piece **613** to function as an auxiliary fixed contact. In addition, by bringing auxiliary fixed contact **611** into contact with a part of the flat surface of horizontal piece **613**, it is also possible to cause a part of the flat surface of horizontal piece **613** to function as an auxiliary fixed contact. A plurality of auxiliary fixed contacts **611** may be provided on horizontal piece **613** (terminal portion **612**).

In addition, terminal portion **612** includes side piece portion **614** that is continuously connected to an outer end of horizontal piece **613** in the Y direction (width direction), extends along the XZ plane, and is elongated in the Z direction (vertical direction).

In the present exemplary embodiment, side piece portion **614** includes first side piece portion **6141** continuously connected to an outer end of horizontal piece **613** in the Y direction (width direction), and coupling part **6142** extending forward in the X direction (front-rear direction) from a lower end of first side piece portion **6141**. Side piece portion **614** further includes second side piece portion **6143** extending downward in the Z direction (vertical direction) from a front end lower portion of coupling part **6142**. As described above, in the present exemplary embodiment, side piece portion **614** has a shape bent into a crank shape when viewed from the Y direction (width direction).

In the present exemplary embodiment, auxiliary fixed contact portion **610** is held by coil bobbin **220**.

Specifically, press-fitting pieces **6141a** protruding inward in the Y direction (width direction) are provided at both ends in the X direction (front-rear direction) of first side piece portion **6141**. The pair of press-fitting pieces **6141a** is press-fitted into the pair of press-fitting openings **2221a** of upper auxiliary contact retainer **2221** formed in upper flange portion **222**.

Press-fitting piece **6143a** protruding inward in the Y direction (width direction) is provided at the lower end of second side piece portion **6143** on the front end side in the X direction (front-rear direction). Then, press-fitting piece **6143a** is press-fitted into press-fitting opening **2231a** of lower auxiliary contact retainer **2231** formed in lower flange portion **223**.

In this manner, the pair of press-fitting pieces **6141a** is press-fitted into the pair of press-fitting openings **2221a**, and press-fitting piece **6143a** is press-fitted into press-fitting opening **2231a**, so that auxiliary fixed contact portion **610** is held by coil bobbin **220**.

Auxiliary fixed contact portion **610** includes connection part **615** extending downward from the lower end of second side piece portion **6143**. Connection part **615** is formed so as to protrude outward (downward) from base **110** in a state where side piece portion **614** is held by coil bobbin **220** arranged on base **110**.

In the present exemplary embodiment, auxiliary fixed contact portion **610** is held by coil bobbin **220** with auxiliary fixed contact **611** facing downward in the Z direction (vertical direction). That is, auxiliary fixed contact portion **610** is held by coil bobbin **220** in a state where the surface of horizontal piece **613** on the side where auxiliary fixed contact **611** is formed (lower surface: surface facing auxiliary moving contact **621**) faces downward.

15

In addition, auxiliary fixed contact **611**, terminal portion **612**, and connection part **615** can be formed of, for example, a conductive material such as a silver-based material or a copper-based material.

On the other hand, auxiliary moving contact portion **620** includes auxiliary moving terminal portion **622**, and auxiliary moving terminal portion **622** includes a leaf spring extending in the horizontal direction and elongated in the Y direction (width direction). Auxiliary moving terminal portion **622** includes auxiliary moving contact **621**.

In the present exemplary embodiment, auxiliary moving terminal portion **622** has a shape bent in a crank shape such that the distal end (end on the inner side in the Y direction) is located downward. A member to serve as the auxiliary moving contact is inserted into insertion hole **622a** formed at the distal end of auxiliary moving terminal portion **622** so as to penetrate in the plate thickness direction, and riveting is performed to allow auxiliary moving terminal portion **622** to include auxiliary moving contact **621** (see FIG. 11). As described above, in the present exemplary embodiment, auxiliary moving terminal portion **622** has a function as a movable-side auxiliary contact holder that holds auxiliary moving contact **621**.

The formation of auxiliary moving contact **621** on auxiliary moving terminal portion **622** does not need to be performed by riveting, and can be performed by various methods. For example, it is also possible to cause a portion protruding by performing dowel processing on auxiliary moving terminal portion **622** to function as an auxiliary moving contact. Further, by bringing auxiliary moving contact **621** into contact with a part of the flat surface of auxiliary moving terminal portion **622**, a part of the flat surface of auxiliary moving terminal portion **622** can function as the auxiliary moving contact. A plurality of auxiliary moving contacts **621** may be provided in auxiliary moving terminal portion **622**.

Further, terminal portion **623** is continuously provided at an outer end of auxiliary moving terminal portion **622** in the Y direction (width direction). Terminal portion **623** includes horizontal piece **624** extending in the horizontal direction and elongated in the Y direction (width direction), and auxiliary moving terminal portion **622** is connected to horizontal piece **624**. Specifically, a pair of insertion holes **622b** is formed at an outer end of auxiliary moving terminal portion **622** in the Y direction (width direction) so as to be aligned in the X direction (front-rear direction), and a pair of projections **624a** is formed on horizontal piece **624** so as to be aligned in the X direction (front-rear direction). Auxiliary moving terminal portion **622** is connected to horizontal piece **624** by riveting in a state where the pair of projections **624a** are respectively inserted into the pair of insertion holes **622b**.

In addition, terminal portion **623** includes side piece portion **625** which is connected to an outer end of horizontal piece **624** in the Y direction (width direction), extends along the XZ plane, and is elongated in the Z direction (vertical direction).

In the present exemplary embodiment, side piece portion **625** includes first side piece portion **6251** connected to an outer end of horizontal piece **624** in the Y direction (width direction), and coupling part **6252** extending forward in the X direction (front-rear direction) from a lower end of first side piece portion **6251**. Side piece portion **625** further includes second side piece portion **6253** extending downward in the Z direction (vertical direction) from a front end lower portion of coupling part **6252**. As described above, in

16

the present exemplary embodiment, side piece portion **625** also has a shape bent into a crank shape as viewed in the Y direction (width direction).

Further, in the present exemplary embodiment, auxiliary moving contact portion **620** is held by coil bobbin **220**.

Specifically, press-fitting pieces **6251a** protruding inward in the Y direction (width direction) are provided at both ends in the X direction (front-rear direction) of first side piece portion **6251**. The pair of press-fitting pieces **6251a** are press-fitted into the pair of press-fitting openings **2221a** of upper auxiliary contact retainer **2221** formed in upper flange portion **222**.

Press-fitting piece **6253a** protruding inward in the Y direction (width direction) is provided at the lower end of second side piece portion **6253** on the front end side in the X direction (front-rear direction). Then, press-fitting piece **6253a** is press-fitted into press-fitting opening **2231a** of lower auxiliary contact retainer **2231** formed in lower flange portion **223**.

In this manner, the pair of press-fitting pieces **6251a** is press-fitted into the pair of press-fitting openings **2221a**, and press-fitting piece **6253a** is press-fitted into press-fitting opening **2231a**, so that auxiliary moving contact portion **620** is held by coil bobbin **220**.

Auxiliary moving contact portion **620** includes connection part **626** extending downward from the lower end of second side piece portion **6253**. Connection part **626** is formed so as to protrude outward (downward) from base **110** in a state where side piece portion **625** is held by coil bobbin **220** arranged on base **110**.

In the present exemplary embodiment, auxiliary moving contact portion **620** is held by coil bobbin **220** with auxiliary moving contact **621** facing upward in the Z direction (vertical direction). That is, auxiliary moving contact portion **620** is held by coil bobbin **220** in a state where a surface of auxiliary moving terminal portion **622** on a side where auxiliary moving contact **621** is formed (upper surface: surface facing auxiliary fixed contact **611**) faces upward.

In addition, auxiliary moving contact **621**, auxiliary moving terminal portion **622**, terminal portion **623**, and the connection part **626** can be formed of, for example, a conductive material such as a silver-based material or a copper-based material.

A set including one auxiliary fixed contact portion **610** and one auxiliary moving contact portion **620** having such a configuration is accommodated behind first side wall **131** in the X direction (front-rear direction: first direction) and on the upper end side of coil **210** in internal space S1 (see FIGS. 12 to 15). Auxiliary fixed contact **611** and auxiliary moving contact **621** are disposed above head **232** of iron core **230** in the Z direction (vertical direction).

Here, auxiliary moving contact portion **620** is disposed so that auxiliary moving terminal portion **622** can swing relative to auxiliary fixed contact portion **610** in the Z direction (vertical direction). In the present exemplary embodiment, auxiliary moving terminal portion **622** can be relatively swung in the Z direction (vertical direction) with respect to auxiliary fixed contact portion **610** by auxiliary drive unit **70**. That is, by switching the driving state of electromagnetic device **20** and swinging auxiliary drive unit **70**, conduction and non-conduction between auxiliary fixed contact portion **610** and auxiliary moving contact portion **620** paired with each other (having auxiliary contacts that come into contact with or separate from each other) can be switched.

In the present exemplary embodiment, auxiliary drive unit **70** is made of an insulating resin material and is held by horizontal wall **311** of armature **310**. Auxiliary drive unit **70**

17

is swung in the Z direction (vertical direction) along with the swinging of armature 310. In this way, auxiliary moving terminal portion 622 is swung in the Z direction (vertical direction) along with the swinging of auxiliary drive unit 70 in the Z direction (vertical direction).

Auxiliary drive unit 70 includes main body 71 and fixed part 72 which is continuously provided so as to protrude outward in the Y direction (width direction) from main body 71 and is held by horizontal wall 311 of armature 310. Further, auxiliary drive unit 70 includes push-up part 73 that is continuously provided so as to protrude from main body 71 toward the rear side in the X direction (front-rear direction) and pushes up auxiliary moving terminal portion 622.

In the present exemplary embodiment, fixed part 72 includes arm portion 721 that protrudes outward in the Y direction (width direction), and hook portion 722 that is continuously provided downward in the Z direction (vertical direction) from an outer end in the Y direction (width direction) of arm portion 721.

Held part 3112 which holds auxiliary drive unit 70 is formed on a rear portion of horizontal wall 311 of armature 310 in the X direction (front-rear direction). By hooking a pair of hook portions 72 on the held part 3112, auxiliary drive unit 70 is held by horizontal wall 311 of armature 310.

As described above, in the present exemplary embodiment, by swinging auxiliary drive unit 70 in conjunction with the swinging of armature 310, conduction and non-conduction between auxiliary fixed contact portion 610 and auxiliary moving contact portion 620 having the auxiliary contact that comes into contact with or separates from each other can be switched. That is, main contact portion 40 is brought into and out of contact with one end of armature 310, and auxiliary contact portion 60 comes into contact with or separates from the other end of armature 310.

With such a configuration, auxiliary moving contact portion 620 swings relative to auxiliary fixed contact portion 610 in the Z direction (vertical direction) as armature 310 swings. At this time, auxiliary moving contact 621 swings so as to draw an arc around the outer end in the Y direction (width direction) of auxiliary moving terminal portion 622.

In the present exemplary embodiment, auxiliary moving terminal portion 622 is connected to terminal portion 623 held by coil bobbin 220 in a state where auxiliary moving contact 621 is separated from auxiliary fixed contact 611 in the natural state. In the state where the energization to coil 210 is stopped, push-up part 73 of auxiliary drive unit 70 is pushed up by coming into contact with auxiliary moving terminal portion 622, and auxiliary moving contact 621 comes into contact with auxiliary fixed contact 611.

On the other hand, in a state where coil 210 is energized, the rear end side of horizontal wall 311 of armature 310 rotates downward, and auxiliary drive unit 70 moves downward as the rear end side of horizontal wall 311 rotates downward. When auxiliary drive unit 70 moves downward, auxiliary moving terminal portion 622 moves downward by the elastic restoring force, and auxiliary moving contact 621 is separated from auxiliary fixed contact 611.

Alternatively, auxiliary drive unit 70 may drive auxiliary moving terminal 622 using another method. As another method of driving the auxiliary moving terminal 622 by auxiliary drive unit 70, for example, there is a method in which, when auxiliary drive unit 70 is separated from auxiliary moving terminal 622, auxiliary moving contact 621 is brought into contact with auxiliary fixed contact 611 by the elastic restoring force of auxiliary moving terminal portion 622 to be in the conductive state, and auxiliary moving contact 621 is separated from auxiliary fixed contact

18

611 to be in the non-conductive state by auxiliary drive unit 70 pushing down auxiliary moving terminal 622.

As described above, in the present exemplary embodiment, auxiliary contact portion 60 is provided such that the on state and the off state are opposite to main contact portion 40.

Next, an example of the operation of electromagnetic relay 1 having the above-described configuration will be described.

First, in a state where coil 210 is not energized, horizontal wall 311 of armature 310 is moved in a direction away from head 232 of iron core 230 by the elastic force of hinge spring 320. At this time, since vertical wall 312 of armature 310 is located behind in the X direction (front-rear direction), movable part 50 is also located behind in the X direction (front-rear direction). That is, main moving contact portion 420 held by movable part 50 is separated from main fixed contact portion 410, and main moving contact 421 is separated from main fixed contact 411 (see FIGS. 12 and 14).

On the other hand, since auxiliary drive unit 70 also moves in the direction away from head 232 of iron core 230, auxiliary moving terminal portion 622 is pushed up by push-up part 73 of auxiliary drive unit 70, and auxiliary moving contact 621 comes into contact with auxiliary fixed contact 611 (see FIGS. 12 and 14).

When coil 210 is energized from the off state, horizontal wall 311 of armature 310 is attracted downward (toward iron core 230) by the electromagnetic force, and approaches head 232 of iron core 230 against the elastic force of hinge spring 320. Then, vertical wall 312 rotates forward as horizontal wall 311 rotates downward (toward iron core 230), and movable part 50 rotates forward as vertical wall 312 rotates forward. Consequently, moving contactor 422 held by movable part 50 rotates forward toward main fixed contact portion 410, and main moving contact 421 of moving contactor 422 comes into contact with main fixed contact 411 of main fixed contact portion 410. Thus, the pair of main fixed contact portions 410 is electrically connected by main moving contact portion 420 (see FIGS. 13 and 15).

On the other hand, since auxiliary drive unit 70 also moves in the direction approaching head 232 of iron core 230, push-up part 73 of auxiliary drive unit 70 is lowered, and auxiliary moving contact 621 is separated from auxiliary fixed contact 611. Accordingly, the electrical connection between auxiliary fixed contact portion 610 and auxiliary moving contact portion 620 is released (see FIGS. 13 and 15).

Then, when the energization to coil 210 is stopped in this state, horizontal wall 311 of armature 310 rotates upward (the side away from iron core 230) by the biasing force of hinge spring 320 and returns to the initial position. In addition, vertical wall 312 rotates backward as horizontal wall 311 rotates upward, and movable part 50 rotates backward as vertical wall 312 rotates backward. As a result, moving contactor 422 held by movable part 50 rotates backward so as to be separated from main fixed contact portion 410, and main moving contact 421 of moving contactor 422 is separated from main fixed contact 411 of main fixed contact portion 410. In this way, the electrical connection between the pair of main fixed contact portions 410 is released.

On the other hand, since auxiliary drive unit 70 also moves in a direction away from head 232 of iron core 230, auxiliary moving terminal portion 622 is pushed up by push-up part 73 of auxiliary drive unit 70 and returns to the initial position. As a result, auxiliary moving contact 621 comes into contact with auxiliary fixed contact 611, and

19

auxiliary fixed contact portion **610** and auxiliary moving contact portion **620** are electrically connected.

As described above, in the present exemplary embodiment, when armature **310** is at the initial position, main moving contact **421** and main fixed contact **411** are separated from each other, and auxiliary moving contact **621** and auxiliary fixed contact **611** are at the second position in contact with each other (see FIGS. **12** and **14**). On the other hand, when armature **310** is at the abutment position, main moving contact **421** and main fixed contact **411** are in contact with each other, and auxiliary moving contact **621** and auxiliary fixed contact **611** are at the first position to be separated from each other (see FIGS. **13** and **15**).

Therefore, while coil **210** is not energized, the pair of main fixed contact portions **410** are insulated from each other, and while coil **210** is energized, the pair of main fixed contact portions **410** are electrically connected to each other. As described above, in the present exemplary embodiment, main moving contact **421** is configured to be able to relatively reciprocate (turn) in the first direction (X direction: front-rear direction) with respect to main fixed contact **411** between the first position and the second position.

On the other hand, auxiliary fixed contact portion **610** and auxiliary moving contact portion **620** are insulated while coil **210** is not energized, and auxiliary fixed contact portion **610** and auxiliary moving contact portion **620** are electrically connected while coil **210** is energized. As described above, in the present exemplary embodiment, auxiliary moving contact **621** is configured to be able to reciprocate (turn) relatively in the axial direction (Z direction: vertical direction) with respect to auxiliary fixed contact **611** between the first position and the second position.

Here, when energization to coil **210** is stopped in a state where main moving contact **421** and main fixed contact **411** are located at the first position where the main moving contact and the main fixed contact are in contact with each other, opening of main moving contact **421** to separate from main fixed contact **411** is started.

When the opening is started, an arc is generated between main moving contact **421** and main fixed contact **411** in the initial stage of the opening, and the current conduction state is continued by the arc.

Therefore, by extending the arc generated between main moving contact **421** and main fixed contact **411** backward in the X direction (front-rear direction), the arc generated between main moving contact **421** and main fixed contact **411** is extinguished more reliably and more quickly.

At this time, the arc generated between main moving contact **421** and main fixed contact **411** is extended in the space formed between second side wall **132** and side wall **1223** of case **120**. Therefore, in the present exemplary embodiment, a space formed between second side wall **132** and side wall **1223** of case **120** is arc extension space **S4** for extending an arc (see FIG. **5**).

If the arc generated between main moving contact **421** and main fixed contact **411** is extended backward in the X direction (front-rear direction), electromagnetic relay **1** can be downsized. However, when the arc is extended backward in the X direction (front-rear direction), consumable powder or the like may be scattered backward in the X direction (front-rear direction) together with the extended arc. At this time, if the main contact side space (first space) **S2** in which main contact portion **40** is disposed and auxiliary contact side space (second space) **S3** in which auxiliary contact portion **60** is disposed in internal space **S1** of housing **10**

20

communicate with each other through a large passage, auxiliary contact portion **60** may be affected by the consumable powder or the like.

As described above, when main contact side space **S2** in which main contact portion **40** is disposed and auxiliary contact side space **S3** in which auxiliary contact portion **60** is disposed are completely communicated with each other, auxiliary contact portion **60** may be affected by consumable powder or the like. In particular, in an electromagnetic relay through which a large current flows, auxiliary contact portion **60** is greatly affected by consumable powder or the like.

Therefore, in the present exemplary embodiment, even when the current flowing through main contact portion **40** is large, it is possible to more reliably prevent the contact reliability of auxiliary contact portion **60** from deteriorating. Specifically, partition wall **130** is formed which divides internal space **S1** into main contact side space **S2** where main contact portion **40** exists and auxiliary contact side space **S3** where auxiliary contact portion **60** exists. That is, main contact side space **S2** and auxiliary contact side space **S3** of internal space **S1** can be defined by the continuous portion of partition wall **130**.

By providing partition wall **130**, main contact side space **S2** in which main contact portion **40** is disposed and auxiliary contact side space **S3** in which auxiliary contact portion **60** is disposed can communicate with each other through a narrower gap. That is, by providing partition wall **130**, it is possible to prevent main contact side space **S2** and auxiliary contact side space **S3** from communicating with each other through a relatively wide gap as much as possible. In this way, it is possible to more reliably prevent the consumable powder or the like generated in main contact portion **40** from entering auxiliary contact side space **S3** where auxiliary contact portion **60** exists.

In the present exemplary embodiment, partition wall **130** includes first side wall **131** that is continuously connected to base **110** so as to extend along the Z direction (vertical direction: axial direction) on the front side of coil **210** in the X direction (front-rear direction) (one side in the first direction intersecting the axial direction).

Partition wall **130** includes a pair of second side walls **132** disposed on both sides in the Y direction (width direction: a second direction intersecting with the axial direction of the coil and the first direction) and continuously connected to base **110** so as to extend along the Z direction (vertical direction: axial direction).

As described above, in the present exemplary embodiment, first side wall **131** and the pair of second side walls **132** for supporting yoke **240** also function as partition wall **130** that divides into main contact side space **S2** where main contact portion **40** exists and auxiliary contact side space **S3** where auxiliary contact portion **60** exists.

First side wall **131** is continuously connected to base **110** such that main contact portion **40** and movable part **50** are located on the front side in the X direction (front-rear direction) and auxiliary contact portion **60** and coil **210** are located on the rear side in the X direction (front-rear direction) (the other side in the first direction intersecting the axial direction).

At least a part of side surface **210a** of coil **210** is surrounded by first side wall **131** and the pair of second side walls **132**.

Further, in the present exemplary embodiment, partition wall **130** includes a pair of third side walls **133** disposed side by side in the Y direction (width direction: second direction)

21

and provided along the Z direction (vertical direction: axial direction) and along second side wall **132** (see FIGS. **16** and **17**).

In the present exemplary embodiment, third side wall **133** is formed inside case **120**. Specifically, third side wall **133** extends from the inner surface of top wall **121** along the X direction (front-rear direction: first direction) and the Z direction (vertical direction: axial direction). Third side wall **133** is formed such that the rear end in the X direction (front-rear direction: first direction) is in contact with the inner surface of rear wall **1222**.

With case **120** attached to base **110**, the lower end of third side wall **133** is disposed outside second side wall **132** in the Y direction (width direction: second direction). At this time, a gap is hardly formed between the lower end of third side wall **133** and second side wall **132** (see FIGS. **18** to **20**).

As described above, in the present exemplary embodiment, third side wall **133** is provided on case **120** so as to overlap second side wall **132** as viewed in the Y direction (width direction: second direction).

Here, main contact side space **S2** includes internal space **S1** facing a front side surface of first side wall **131** in the X direction, internal space **S1** facing an outer side surface of the pair of second side walls **132** in the Y direction, and internal space **S1** facing an outer side surface of third side wall **133** in the Y direction. On the other hand, auxiliary contact side space **S3** includes internal space **S1** facing the side surface on the rear side in the X direction of first side wall **131**, internal space **S1** facing the inner side surface in the Y direction of the pair of second side walls **132**, and internal space **S1** facing the inner side surface in the Y direction of third side wall **133**. In the portion where second side wall **132** and third side wall **133** overlap, internal space **S1** facing the outer side surface in the Y direction of one of second side wall **132** and third side wall **133** is main contact side space **S2**, and internal space **S1** facing the inner side surface in the Y direction of the other is auxiliary contact side space **S3**.

Further, the pair of third side walls **133** is provided at the same height as the contact peripheral portion of auxiliary contact portion **60** in the Z direction (vertical direction: axial direction) with the bottom surface **111a** of base **110** as a reference in a state where base **110** is located below case **120** (see FIGS. **18** and **21**). That is, the pair of third side walls **133** overlaps at least a part of the contact peripheral portion of auxiliary contact portion **60** when viewed from the direction orthogonal to the Z direction. In the present exemplary embodiment, the contact peripheral portion includes auxiliary fixed contact **611**, terminal portion **612** (upper end of horizontal piece **613** and side piece portion **614**) around auxiliary fixed contact **611**, auxiliary moving contact **621** and auxiliary moving terminal portion **622** movable within the movable range of auxiliary moving terminal portion **622**, horizontal piece **624**, and the upper end of side piece portion **625**.

The contact peripheral portion of auxiliary contact portion **60** is located at the upper end of auxiliary contact portion **60**. The pair of third side walls **133** is provided at the same height as at least one of auxiliary fixed contact **611** and auxiliary moving contact **621** in the Z direction with reference to bottom surface **111a** of base **110**. Note that the pair of third side walls **133** may be provided higher than head **232** of iron core **230** in the Z direction with respect to bottom surface **111a** of base **110**.

In the present exemplary embodiment, when viewed from the Y direction (width direction: second direction), the entire auxiliary moving terminal portion **622** provided with aux-

22

iliary moving contact **621** and entire horizontal piece **613** provided with auxiliary fixed contact **611** (terminal portion **612** around auxiliary fixed contact **611**) overlap third side wall **133**. That is, the contact peripheral portion (entire auxiliary moving terminal portion **622** provided with auxiliary moving contact **621** and entire horizontal piece **613** provided with auxiliary fixed contact **611**) of auxiliary contact portion **60** is disposed between a pair of third side walls **133** in the Y direction (width direction: second direction).

In this way, arc extension space **S4** and auxiliary contact side space **S3** of main contact side space **S2** are divided by third side wall **133**. Third side wall **133** more reliably prevents the consumable powder or the like generated in main contact portion **40** from entering auxiliary contact side space **S3** through arc extension space **S4**.

In the present exemplary embodiment, main contact side space **S2** is located above main contact portion **40** and has a substantially L-shaped space when viewed from the Y direction (width direction: second direction). This space is armature arrangement space **S5** in which armature **310** is disposed (see FIGS. **14** and **15**). Since armature **310** is disposed in armature arrangement space **S5** in a state of being allowed to swing, a relatively large gap is formed between armature **310** and case **120** in armature arrangement space **S5**. Therefore, there is a possibility that armature **310** is displaced at the time of swinging, or consumable powder or the like generated in main contact portion **40** enters auxiliary contact side space **S3** through a relatively large gap formed between armature **310** and case **120**.

Therefore, in the present exemplary embodiment, partition wall **130** includes fourth side wall **134** provided on case **120** so as to protrude downward in the Z direction (vertical direction: axial direction) in a state where base **110** is positioned below case **120**. Fourth side wall **134** extends in the Y direction (width direction: second direction) and faces armature **310** in the Z direction (vertical direction: axial direction) (see FIGS. **14** and **15**).

Further, in the present exemplary embodiment, fourth side wall **134** includes pressing wall **1341** which is disposed on the front side (main contact portion **40** side) in the X direction (front-rear direction: first direction) with respect to vertical wall (upright portion) **241** of yoke **240**, and is capable of pressing armature **310**.

Pressing wall **1341** includes protrusion **1341a** formed at the central portion in the Y direction (width direction: second direction) and a pair of recesses **1341b** formed at both ends in the Y direction (width direction: second direction) of protrusion **1341a** (see FIGS. **16** and **17**). Further, pressing wall **1341** includes a pair of pressing wall extended portions **1341c** formed on the outer sides of the pair of recesses **1341b** in the Y direction (width direction: second direction). The pair of pressing wall extended portions **1341c** is formed such that the outside in the Y direction (width direction: second direction) is in contact with the inner surface of side wall **1223**.

With case **120** attached to base **110**, the lower end of protrusion **1341a** is inserted into through-hole **313** formed in armature **310** (see FIGS. **22** and **23**). In addition, in a state where case **120** is attached to base **110**, the pair of recesses **1341b** faces both sides of through-hole **313** of armature **310** in the Y direction (width direction: second direction) in the Z direction (vertical direction: axial direction). Further, in a state where case **120** is attached to base **110**, armature **310** is disposed between the pair of pressing wall extended portions **1341c** in the Y direction (width direction: second direction).

23

By providing pressing wall **1341** having such a shape in case **120**, position deviation of armature **310** at the time of swinging is suppressed. In addition, since armature arrangement space **S5** is divided in the X direction (front-rear direction: first direction) by pressing wall **1341**, the consumable powder and the like generated in main contact portion **40** can be more reliably suppressed from entering auxiliary contact side space **S3** by pressing wall **1341**.

Further, fourth side wall **134** is disposed behind (on auxiliary contact portion **60** side) the vertical wall (upright portion) **241** of yoke **240** in the X direction (front-rear direction: first direction), and has partition wall **1342** capable of dividing space **S5** above armature **310**.

Main contact side space **S2** includes internal space **S1** facing the side surface on the front side in the X direction of fourth side wall **134** (pressing wall **1341**, partition wall **1342**). On the other hand, auxiliary contact side space **S3** includes internal space **S1** facing the side surface on the rear side in the X direction of fourth side wall **134** (pressing wall **1341**, partition wall **1342**).

Partition wall **1342** includes recess **1342a** formed at the central portion in the Y direction (width direction: second direction) and a pair of partition wall extended portions **1342b** formed at both ends in the Y direction (width direction: second direction) of recess **1342a** (see FIGS. **16** and **17**). The pair of partition wall extended portions **1342b** is formed such that the outside in the Y direction (width direction: second direction) is continuously connected to the front end of third side wall **133**.

In a state where case **120** is attached to base **110**, the lower end of recess **1342a** faces armature **310** (the upper surface of water wall **311**) in the Z direction (vertical direction: axial direction) (see FIGS. **14** and **15**). Further, in a state where case **120** is attached to base **110**, armature **310** is disposed between the pair of partition wall extended portions **1342b** in the Y direction (width direction: second direction).

As described above, in the present exemplary embodiment, partition wall **1342** has partition wall extended portions **1342b** disposed on both sides of armature **310** in the Y direction (width direction: second direction). In the present exemplary embodiment, partition wall extended portion **1342b** has a portion located outside second side wall **132** in the Y direction (width direction: second direction) (see FIG. **21**).

By providing partition wall **1342** having such a shape in case **120**, it is possible to suppress the consumable powder or the like that has not been blocked by pressing wall **1341** from entering auxiliary contact side space **S3** by partition wall **1342**.

Further, in the present exemplary embodiment, partition wall **130** includes fifth side wall **135** that extends in the Z direction (vertical direction: axial direction) and is provided on case **120** so as to be disposed outside second side wall **132** in the Y direction (width direction: second direction).

In the present exemplary embodiment, thick portion **1223a** is formed at the central portion of side wall **1223** of case **120** in the X direction (front-rear direction: first direction). Fifth side wall **135** extends so as to protrude inward in the Y direction (width direction: second direction) from a center in the X direction (front-rear direction: first direction) in thick portion **1223a**. Fifth side wall **135** extends downward in the Z direction (vertical direction: axial direction) from top wall **121** of case **120**. Then, in a state where case **120** is attached to base **110**, the lower end of fifth side wall **135** is extended to a position substantially in contact with base portion **111** of base **110**.

24

Further, in the present exemplary embodiment, fifth side wall **135** is formed integrally with third side wall **133** and partition wall extended portion **1342b**. Specifically, fifth side wall **135**, third side wall **133**, and partition wall extended portion **1342b** are formed by integral molding.

Since arc extension space **S4** is divided in the X direction (front-rear direction: first direction) by providing fifth side wall **135** having such a shape, the consumable powder or the like generated in main contact portion **40** can be more reliably suppressed from entering auxiliary contact side space **S3** by fifth side wall **135**.

In the present exemplary embodiment, main contact side space **S2** includes internal space **S1** facing the side surface on the front side in the X direction of fifth side wall **135**. On the other hand, auxiliary contact side space **S3** includes internal space **S1** facing the side surface on the rear side in the X direction of fifth side wall **135**.

Second Exemplary Embodiment

As illustrated in FIGS. **24** to **27**, electromagnetic relay **1** according to the present exemplary embodiment has substantially the same configuration as electromagnetic relay **1** shown in the first exemplary embodiment.

That is, electromagnetic relay **1** according to the present exemplary embodiment includes housing **10** having base **110** and case **120** covering base **110**.

Housing **10** further includes main contact portion **40** that has main fixed contact **411** and main moving contact **421** that comes into contact with and separates from main fixed contact **411** and is disposed in internal space **S1** formed in the housing.

Auxiliary fixed contact **611** and auxiliary moving contact **621** that comes into contact with or separates from auxiliary fixed contact **611** are provided, and auxiliary contact portion **60** disposed in internal space **S1** is provided.

In a state where base **110** is located below case **120**, coil **210** disposed in internal space **S1** such that the axial direction extends in the Z direction (vertical direction), and yoke **240** disposed around coil **210** are provided.

Armature **310** that swings in response to the excitation/non-excitation of coil **210**, and movable part **50** that moves main moving contact **421** in response to the swing of armature **310** are provided.

Electromagnetic relay **1** according to the present exemplary embodiment also includes partition wall **130** that divides internal space **S1** into main contact side space **S2** where main contact portion **40** exists and auxiliary contact side space **S3** where auxiliary contact portion **60** exists.

Partition wall **130** includes first side wall **131** continuously connected to base **110** so as to extend along the Z direction (vertical direction: axial direction) on the front side of coil **210** in the X direction (front-rear direction) (one side in the first direction intersecting the axial direction).

Partition wall **130** includes a pair of second side walls **132** disposed on both sides in the Y direction (width direction: a second direction intersecting with the axial direction of the coil and the first direction) and continuously connected to base **110** so as to extend along the Z direction (vertical direction: axial direction).

First side wall **131** is continuously connected to base **110** such that main contact portion **40** and movable part **50** are located on the front side in the X direction (front-rear direction) and auxiliary contact portion **60** and coil **210** are located on the rear side in the X direction (front-rear direction) (the other side in the first direction intersecting the axial direction).

25

At least a part of side surface **210a** of coil **210** is surrounded by first side wall **131** and the pair of second side walls **132**.

Further, also in the present exemplary embodiment, partition wall **130** includes a pair of third side walls **133** disposed side by side in the Y direction (width direction: second direction) and provided along the Z direction (vertical direction: axial direction) and along second side wall **132**.

In a state where base **110** is located below case **120**, the pair of third side walls **133** is provided at the same height as the contact peripheral portion of auxiliary contact portion **60** in the Z direction (vertical direction: axial direction) with bottom surface **111a** of base **110** as a reference (see FIGS. **30** and **32**). The pair of third side walls **133** is provided at the same height as at least one of auxiliary fixed contact **611** and auxiliary moving contact **621** in the Z direction with reference to bottom surface **111a** of base **110**. Also in the present exemplary embodiment, the pair of third side walls **133** may be provided higher than head **232** of iron core **230** in the Z direction with reference to bottom surface **111a** of base **110**.

Further, also in the present exemplary embodiment, partition wall **130** includes fourth side wall **134** provided on case **120** so as to protrude downward in the Z direction (vertical direction: axial direction) in a state where base **110** is located below case **120** (see FIGS. **28** and **29**). Fourth side wall **134** extends in the Y direction (width direction: second direction) and faces armature **310** in the Z direction (vertical direction: axial direction) (see FIGS. **32** and **33**).

Fourth side wall **134** is disposed on the front side (main contact portion **40** side) in the X direction (front-rear direction: first direction) with respect to vertical wall (upright portion) **241** of yoke **240**, and includes pressing wall **1341** capable of pressing armature **310**.

Pressing wall **1341** includes protrusion **1341a** formed at the center in the Y direction (width direction: second direction) and a pair of recesses **1341b** formed at both ends in the Y direction (width direction: second direction) of protrusion **1341a** (see FIGS. **28** and **29**). Further, pressing wall **1341** includes a pair of pressing wall extended portions **1341c** formed on the outer sides of the pair of recesses **1341b** in the Y direction (width direction: second direction). The pair of pressing wall extended portions **1341c** is formed such that the outside in the Y direction (width direction: second direction) is in contact with the inner surface of side wall **1223**.

With case **120** attached to base **110**, the lower end of protrusion **1341a** is inserted into through-hole **313** formed in armature **310** (see FIG. **32**). In addition, in a state where case **120** is attached to base **110**, the pair of recesses **1341b** faces both sides of through-hole **313** of armature **310** in the Y direction (width direction: second direction) in the Z direction (vertical direction: axial direction). Further, in a state where case **120** is attached to base **110**, armature **310** is disposed between the pair of pressing wall extended portions **1341c** in the Y direction (width direction: second direction).

Here, in the present exemplary embodiment, third side wall **133** is formed on base **110**, and third side wall **133** and second side wall **132** are integrally formed (see FIGS. **26** and **27**). Specifically, third side wall **133** and second side wall **132** are formed by integral molding. In addition, a wall portion of partition wall **130** extending in the X direction (front-rear direction: first direction) and the Z direction (vertical direction: axial direction) is formed such that an upper end protrudes upward in the Z direction (vertical direction: axial direction) from an upper end of first side wall **131** (see FIGS. **26** and **27**). In the wall portion described

26

above, a portion continuously provided to first side wall **131** (a portion located below the upper end of first side wall **131** in the wall described above) is defined as second side wall **132**, and a portion protruding upward from the portion defined as the second side wall (a portion not continuously provided to first side wall **131** in the wall described above) is defined as third side wall **133**.

In the present exemplary embodiment, second side wall **132** has extension part **1321** extending outward in the Y direction (width direction: second direction). Extension part **1321** extends in the Y direction (width direction: second direction) and the Z direction (vertical direction: axial direction) from a rear end of second side wall **132** in the X direction (front-rear direction: first direction). Further, at the upper end of extension part **1321**, extension part **1331** is extended so as to extend in the Y direction (width direction: second direction) and the Z direction (vertical direction: axial direction) from the rear end of third side wall **133** in the X direction (front-rear direction: first direction), and extension part **1321** and extension part **1331** are integrally formed. As described above, in the present exemplary embodiment, the wall portion constituted by extension part **1321** and extension part **1331** is formed from the lower end of second side wall **132** to the upper end of third side wall **133** (see FIGS. **26** and **27**).

Further, in the present exemplary embodiment, in a state where case **120** is attached to base **110**, extension part **1321** is configured such that a gap is hardly formed between the end surface on the outer side in the Y direction (width direction: second direction) and the inner surface of side wall **1223** of case **120** (see FIGS. **30** to **32**).

As described above, in the present exemplary embodiment, arc extension space **S4** is divided in the X direction (front-rear direction: first direction) by providing extension part **1321**. Then, extension part **1321** more reliably prevents the consumable powder or the like generated in main contact portion **40** from entering auxiliary contact side space **S3** (see FIGS. **32** and **33**).

In the present exemplary embodiment, fourth side wall **134** that is disposed on the front side (main contact portion **40** side) in the X direction (front-rear direction: first direction) with respect to vertical wall (upright portion) **241** of yoke **240** and does not include pressing wall **1341** capable of pressing armature **310** is exemplified.

Such electromagnetic relay **1** can also achieve substantially the same operations and effects as those of electromagnetic relay **1** described in the first exemplary embodiment.

Third Exemplary Embodiment

As illustrated in FIGS. **34** to **37**, electromagnetic relay **1** according to the present exemplary embodiment has substantially the same configuration as electromagnetic relay **1** shown in the first exemplary embodiment.

That is, electromagnetic relay **1** according to the present exemplary embodiment includes housing **10** having base **110** and case **120** covering base **110**.

Housing **10** further includes main contact portion **40** that has main fixed contact **411** and main moving contact **421** that comes into contact with and separates from main fixed contact **411** and is disposed in internal space **S1** formed in the housing.

Auxiliary fixed contact **611** and auxiliary moving contact **621** that comes into contact with or separates from auxiliary fixed contact **611** are provided, and auxiliary contact portion **60** disposed in internal space **S1** is provided.

In a state where base **110** is located below case **120**, coil **210** disposed in internal space **S1** such that the axial direction extends in the Z direction (vertical direction), and yoke **240** disposed around coil **210** are provided.

Armature **310** that swings in response to the excitation/non-excitation of coil **210**, and movable part **50** that moves main moving contact **421** in response to the swing of armature **310** are provided.

Electromagnetic relay **1** according to the present exemplary embodiment also includes partition wall **130** that divides internal space **S1** into main contact side space **S2** where main contact portion **40** exists and auxiliary contact side space **S3** where auxiliary contact portion **60** exists.

Partition wall **130** includes first side wall **131** continuously connected to base **110** so as to extend along the Z direction (vertical direction: axial direction) on the front side of coil **210** in the X direction (front-rear direction) (one side in the first direction intersecting the axial direction).

Partition wall **130** includes a pair of second side walls **132** disposed on both sides in the Y direction (width direction: a second direction intersecting with the axial direction of the coil and the first direction) and continuously connected to base **110** so as to extend along the Z direction (vertical direction: axial direction).

First side wall **131** is continuously connected to base **110** such that main contact portion **40** and movable part **50** are located on the front side in the X direction (front-rear direction) and auxiliary contact portion **60** and coil **210** are located on the rear side in the X direction (front-rear direction) (the other side in the first direction intersecting the axial direction).

At least a part of side surface **210a** of coil **210** is surrounded by first side wall **131** and the pair of second side walls **132**.

Further, also in the present exemplary embodiment, partition wall **130** includes a pair of third side walls **133** disposed side by side in the Y direction (width direction: second direction) and provided along the Z direction (vertical direction: axial direction) and along second side wall **132**.

Also in the present exemplary embodiment, third side wall **133** is formed inside case **120**. Specifically, third side wall **133** extends from the inner surface of top wall **121** along the X direction (front-rear direction: first direction) and the Z direction (vertical direction: axial direction). Third side wall **133** is formed such that the rear end in the X direction (front-rear direction: first direction) is in contact with the inner surface of rear wall **1222**.

With case **120** attached to base **110**, the lower end of third side wall **133** is disposed outside second side wall **132** in the Y direction (width direction: second direction). At this time, a gap is hardly formed between the lower end of third side wall **133** and second side wall **132** (see FIG. **40**).

As described above, in the present exemplary embodiment, third side wall **133** is provided on case **120** so as to overlap second side wall **132** as viewed in the Y direction (width direction: second direction).

In a state where base **110** is located below case **120**, the pair of third side walls **133** is provided at the same height as the contact peripheral portion of auxiliary contact portion **60** in the Z direction (vertical direction: axial direction) with respect to bottom surface **111a** of base **110** (see FIGS. **32**, **40**, and **43**). The pair of third side walls **133** is provided at the same height as at least one of auxiliary fixed contact **611** and auxiliary moving contact **621** in the Z direction with reference to bottom surface **111a** of base **110**. Also in the present exemplary embodiment, the pair of third side walls **133** may

be provided higher than head **232** of iron core **230** in the Z direction with reference to bottom surface **111a** of base **110**.

Further, also in the present exemplary embodiment, partition wall **130** includes fourth side wall **134** provided on case **120** so as to protrude downward in the Z direction (vertical direction: axial direction) in a state where base **110** is located below case **120** (see FIGS. **38** and **39**). Fourth side wall **134** extends in the Y direction (width direction: second direction) and faces armature **310** in the Z direction (vertical direction: axial direction) (see FIGS. **40**, **43**, and **44**).

Fourth side wall **134** is disposed on the front side (main contact portion **40** side) in the X direction (front-rear direction: first direction) with respect to vertical wall (upright portion) **241** of yoke **240**, and includes pressing wall **1341** capable of pressing armature **310**.

Pressing wall **1341** includes protrusion **1341a** formed at the center in the Y direction (width direction: second direction) and a pair of recesses **1341b** formed at both ends in the Y direction (width direction: second direction) of protrusion **1341a** (see FIGS. **38** and **39**). Further, pressing wall **1341** includes a pair of pressing wall extended portions **1341c** formed on the outer sides of the pair of recesses **1341b** in the Y direction (width direction: second direction). The pair of pressing wall extended portions **1341c** is formed such that the outside in the Y direction (width direction: second direction) is in contact with the inner surface of side wall **1223**.

With case **120** attached to base **110**, the lower end of protrusion **1341a** is inserted into through-hole **313** formed in armature **310**. In addition, in a state where case **120** is attached to base **110**, the pair of recesses **1341b** faces both sides of through-hole **313** of armature **310** in the Y direction (width direction: second direction) in the Z direction (vertical direction: axial direction). Further, in a state where case **120** is attached to base **110**, armature **310** is disposed between the pair of pressing wall extended portions **1341c** in the Y direction (width direction: second direction).

Fourth side wall **134** includes partition wall **1342** which is disposed on the rear side (auxiliary contact portion **60** side) in the X direction (front-rear direction: first direction) with respect to vertical wall (upright portion) **241** of yoke **240**, and is capable of dividing space **S5** above armature **310**.

Partition wall **1342** includes recess **1342a** formed at the center in the Y direction (width direction: second direction) and a pair of partition wall extended portions **1342b** formed at both ends in the Y direction (width direction: second direction) of recess **1342a** (see FIGS. **38** and **39**). The pair of partition wall extended portions **1342b** is formed such that the outside in the Y direction (width direction: second direction) is continuously connected to the front end of third side wall **133**.

With case **120** attached to base **110**, the lower end of recess **1342a** faces armature **310** (the upper surface of water wall **311**) in the Z direction (the vertical direction: the axial direction). Further, in a state where case **120** is attached to base **110**, armature **310** is disposed between the pair of partition wall extended portions **1342b** in the Y direction (width direction: second direction).

Further, also in the present exemplary embodiment, partition wall **130** includes fifth side wall **135** that extends in the Z direction (vertical direction: axial direction) and is provided on case **120** so as to be disposed outside second side wall **132** in the Y direction (width direction: second direction).

Also in the present exemplary embodiment, thick portion **1223a** is formed at the center of side wall **1223** of case **120**

in the X direction (front-rear direction: first direction). Fifth side wall 135 extends so as to protrude inward in the Y direction (width direction: second direction) from a center in the X direction (front-rear direction: first direction) in thick portion 1223a.

Fifth side wall 135 extends downward in the Z direction (vertical direction: axial direction) from top wall 121 of case 120.

Here, in the present exemplary embodiment, in a state where case 120 is attached to base 110, the lower end of fifth side wall 135 is located at an intermediate portion between base portion 111 of base 110 and top wall 121 of case 120. The lower end of fifth side wall 135 only needs to be positioned between base portion 111 of base 110 and top wall 121 of case 120, and the amount of protrusion of fifth side wall 135 from top wall 121 can be appropriately set. However, the amount of protrusion of fifth side wall 135 from top wall 121 is preferably such an amount that a space linearly communicating from main contact portion 40 to auxiliary contact side space S3 is not formed in arc extension space S4. That is, the shortest route from main contact portion 40 to auxiliary contact side space S3 through arc extension space S4 is preferably a polygonal line when viewed from the Y direction (width direction: second direction). Accordingly, in order for the consumable powder or the like generated in main contact portion 40 to enter auxiliary contact side space S3, it is necessary to change the direction at least once. Therefore, it is possible to more reliably prevent the consumable powder or the like generated in main contact portion 40 from entering auxiliary contact side space S3.

Further, also in the present exemplary embodiment, fifth side wall 135 is formed integrally with third side wall 133 and partition wall extended portion 1342b.

Here, in the present exemplary embodiment, second side wall 132 has extension part 1321 extending outward in the Y direction (width direction: second direction). Extension part 1321 extends in the Y direction (width direction: second direction) and the Z direction (vertical direction: axial direction) from a rear end of second side wall 132 in the X direction (front-rear direction: first direction). In the present exemplary embodiment, extension part 1321 is formed from the lower end to the upper end side (the middle of the upper end) of second side wall 132. That is, extension part 1321 is connected to second side wall 132 in a state where the upper end of second side wall 132 protrudes upward from the upper end of extension part 1321 (see FIGS. 36 and 37).

In the present exemplary embodiment, the lower end of third side wall 133 is located near the upper end of extension part 1321. Thus, third side wall 133 is provided on case 120 so as to overlap second side wall 132 when viewed from the Y direction (width direction: second direction) and to overlap extension part 1321 of second side wall 132 when viewed from the Z direction (vertical direction: axial direction).

Fifth side wall 135 is provided on case 120 so as to overlap with extension part 1321 of second side wall 132 in a state of being viewed from the X direction (front-rear direction: first direction) (see FIGS. 41 and 42).

Such electromagnetic relay 1 can also achieve substantially the same operations and effects as those of electromagnetic relay 1 described in the first exemplary embodiment.

(Conclusion)

Hereinafter, the characteristic configuration of the electromagnetic relay described in the above-described exemplary embodiment and the effects obtained thereby will be described.

The following aspects are disclosed from the above-described exemplary embodiments and the like.

An electromagnetic relay 1 according to a first aspect includes housing 10 including base 110 and case 120 that opens downward and is provided on base 110, housing 10 having an internal space surrounded by base 110 and case 120; main fixed contact 411; main moving contact 421 configured to come into contact with and separates from main fixed contact 411; auxiliary fixed contact 611; auxiliary moving contact 621 configured to come into contact with and separates from auxiliary fixed contact 611; coil 210 provided in housing 10 such that a central axis extends in a vertical direction; iron core 230 disposed so as to be surrounded by coil 210; yoke 240 disposed around coil 210; armature 310 that is disposed to face an upper end of iron core 230 and swings in response to excitation/non-excitation of coil 210; movable part 50 provided with main moving contact 421 and configured to move in response to swinging of armature 310; and partition wall 130 that divides internal space S1 of housing 10 into main contact side space S2 and auxiliary contact side space S3. Main fixed contact 411 and main moving contact 421 are disposed in front of coil 210. Main fixed contact 411, main moving contact 421, auxiliary fixed contact 611, auxiliary moving contact 621, coil 210, iron core 230, yoke 240, armature 310, movable part 50, and partition wall 130 are provided in internal space S1 of housing 10. Main fixed contact 411 and main moving contact 421 are disposed in main contact side space S2. Auxiliary fixed contact 611 and auxiliary moving contact 621 are disposed in an auxiliary contact side space. Partition wall 130 includes first side wall 131 connected to base 110, a pair of second side walls 132 connected to base 110, and a pair of third side walls 133. When viewed from above, first side wall 131 is disposed between: (1) main fixed contact 411 and main moving contact 421; and (2) coil 210, iron core 230, yoke 240, auxiliary fixed contact 611, and auxiliary moving contact 621. One of the pair of second side walls 132 is connected to one end of first side wall 131. An other of the pair of second side walls 132 is connected to an other end of first side wall 131. One of the pair of third side walls 133 is disposed so as to be adjacent to one of the pair of second side walls 132. An other of the pair of third side walls 133 is disposed so as to be adjacent to an other of the pair of second side walls 132. Coil 210 is surrounded by first side wall 131 and the pair of second side walls 132. (1) Each of the pair of third side walls 133 is located laterally over a side of auxiliary fixed contact 611 and auxiliary moving contact 621, or (2) each of the pair of third side walls 133 is located above the upper end of iron core 230.

According to the above configuration, internal space S1 can be more reliably divided into main contact side space S2 and auxiliary contact side space S3 by first side wall 131, the pair of second side walls 132, and the pair of third side walls 133. In other words, a gap connecting main contact side space S2 and auxiliary contact side space S3 can be made as small as possible by first side wall 131, the pair of second side walls 132, and the pair of third side walls 133. It is possible to suppress auxiliary contact portion 60 from being affected by the consumable powder or the like scattering when main contact portion 40 is brought into contact with or separated from.

Therefore, in electromagnetic relay 1 described above, even in electromagnetic relay 1 in which the current flowing

31

through main contact portion 40 (main fixed contact 411 and main moving contact 421) is large, it is possible to further reduce the possibility that auxiliary contact portion 60 is affected by the consumable powder or the like scattered when main contact portion 40 is brought into and out of contact with each other. As a result, it is possible to suppress a decrease in contact reliability of the auxiliary contact portion.

In electromagnetic relay 1 described above, it is possible to suppress a decrease in the contact reliability of auxiliary contact portion 60 not only when the current flowing through main contact portion 40 is small but also when the current flowing through main contact portion 40 is large.

In electromagnetic relay 1 according to the second aspect, auxiliary fixed contact 611 and auxiliary moving contact 621 are disposed above coil 210.

According to the above configuration, the insulation distance between auxiliary contact portion 60 (auxiliary fixed contact 611 and auxiliary moving contact 621) and coil 210 can be secured more reliably.

In electromagnetic relay 1 according to the third aspect, each of the pair of third side walls 133 is provided in the case so as to overlap the corresponding one of the pair of second side walls 132 when viewed from the side.

According to the above configuration, it is possible to minimize the gap communicating between main contact side space S2 and auxiliary contact side space S3. Therefore, it is possible to suppress auxiliary contact portion 60 (auxiliary fixed contact 611 and auxiliary moving contact 621) from being affected by the consumable powder and the like scattering when main contact portion 40 (main fixed contact 411 and main moving contact 421) comes into contact with and separates from each other.

In electromagnetic relay 1 according to the fourth aspect, each of third side walls 133 is formed integrally with the corresponding one of the pair of second side walls 132.

According to the above configuration, it is possible to suppress formation of a gap between second side wall 132 and third side wall 133. Therefore, it is possible to more reliably prevent auxiliary contact portion 60 (auxiliary fixed contact 611 and auxiliary moving contact 621) from being affected by the consumable powder or the like scattering when main contact portion 40 (main fixed contact 411 and main moving contact 421) comes into contact with or separates from each other.

In electromagnetic relay 1 according to the fifth aspect, auxiliary fixed contact 611 and auxiliary moving contact 621 are disposed between the pair of third side walls 133.

According to the above configuration, it is possible to prevent the consumable powder or the like scattering when main contact portion 40 (main fixed contact 411 and main moving contact 421) is brought into contact with or separated from adhering to auxiliary contact portion 60 (auxiliary fixed contact 611 and auxiliary moving contact 621). It is possible to suppress a decrease in contact reliability of the auxiliary contact portion.

In electromagnetic relay 1 according to the sixth aspect, partition wall 130 further includes fourth side wall 134 protruding downward from the lower surface of the upper portion of case 120, and the lower end of fourth side wall 134 faces armature 310.

According to the above configuration, the space formed in the upper portion of armature 310 can be partitioned by fourth side wall 134. Further, the movement of armature 310 can be suppressed by fourth side wall 134. As a result, it is possible to prevent the consumable powder or the like scattering when main contact portion 40 (main fixed contact

32

411 and main moving contact 421) is brought into contact with or separated from adhering to auxiliary contact portion 60 (auxiliary fixed contact 611 and auxiliary moving contact 621) along the space formed above armature 310. In addition, it is possible to suppress an increase in the gap between armature 310 and partition wall 130 due to movement of armature 310, and it is possible to suppress adhesion of consumable powder or the like scattering at the time of contact and separation of main contact portion 40 to auxiliary contact portion 60.

In electromagnetic relay 1 according to the seventh aspect, yoke 240 includes vertical wall 241 (upright portion) extending along first side wall 131 outside coil 210, fourth side wall 134 includes pressing wall 1341, and pressing wall 1341 is located between vertical wall 241 of yoke 240 and main fixed contact 411 when viewed from above, and can press armature 310.

According to the above configuration, armature 310 is held down by pressing wall 1341 at a place other than vertical wall 241 (upright portion) which is a relatively unstable portion. As a result, an increase in the gap between armature 310 and partition wall 130 is suppressed. Therefore, it is possible to prevent the consumable powder or the like scattering when main contact portion 40 comes into contact with or separates from the auxiliary contact from adhering to auxiliary contact portion 60.

In electromagnetic relay 1 according to the eighth aspect, pressing wall 1341 has a pair of protrusions (pressing wall extended portions 1341c), and armature 310 is disposed between the pair of protrusions (pressing wall extended portions 1341c).

According to the above configuration, it is possible to suppress an increase in the gap between armature 310 and partition wall 130.

In electromagnetic relay 1 according to the ninth aspect, yoke 240 has vertical wall 241 (upright portion) extending along first side wall 131 outside coil 210, fourth side wall 134 has partition wall 1342, and partition wall 1342 is located above vertical wall 241 (upright portion) of yoke 240 and divides a space above armature 310 in internal space S1.

According to the above configuration, the space above armature 310 can be partitioned by partition wall 1342. As a result, it is possible to prevent the consumable powder or the like scattering at the time of contact and separation of main contact portion 40 from adhering to auxiliary contact portion 60 along the space above armature 310.

In electromagnetic relay 1 according to the tenth aspect, partition wall 1342 has a pair of protrusions (partition wall extended portions 1342b), and armature 310 is disposed between the pair of protrusions (partition wall extended portions 1342b).

According to the above configuration, it is possible to suppress an increase in the gap between armature 310 and partition wall 130.

In electromagnetic relay 1 according to the eleventh aspect, a part of each of the pair of protrusions (partition wall extended portions 1342b) is located outside second side wall 132.

According to the above configuration, it is possible to suppress an increase in the gap between second side wall 132 and the protrusion (partition wall extended portions 1342b).

In electromagnetic relay 1 according to the twelfth aspect, each of the pair of second side walls 132 has extension part 1321 bent outward.

33

According to the above configuration, extension part **1321** can be disposed in the space for extending the arc generated between main moving contact **421** and main fixed contact **411** so as to extend in the direction intersecting the extending direction of the arc. As a result, extension part **1321** can prevent the consumable powder or the like scattering together with the extended arc from moving toward auxiliary contact portion **60**. Therefore, it is possible to suppress a decrease in contact reliability of auxiliary contact portion **60**.

In electromagnetic relay **1** according to the thirteenth aspect, each of the pair of third side walls **133** overlaps the corresponding one of the pair of second side walls **132** when viewed from the side, and each of the pair of third side walls **133** is provided in the case so as to overlap the corresponding one extension part **1321** of the pair of second side walls **132** when viewed from the above.

According to the above configuration, it is possible to minimize the gap communicating between main contact side space **S2** and auxiliary contact side space **S3**, and it is possible to suppress auxiliary contact portion **60** from being affected by the consumable powder or the like scattering when main contact portion **40** comes into contact with or separates from each other.

In electromagnetic relay **1** according to the fourteenth aspect, partition wall **130** further includes fifth side wall **135** protruding downward from the lower surface of the upper portion of case **120**, fifth side wall **135** is disposed on one outer side of the pair of second side walls **132**, and fifth side wall **135** is provided in case **120** so as to overlap one extension part **1321** of the pair of second side walls **132** when viewed from the front.

According to the above configuration, extension part **1321** provided continuously upward from base **110** and fifth side wall **135** provided continuously downward from the upper portion of case **120** are arranged side by side in the arc extending direction. As a result, the space in which the arc is extended can have a bent shape along the extension direction of the arc, and it is possible to suppress the consumable powder or the like scattering together with the extended arc from moving toward auxiliary contact portion **60**.

In electromagnetic relay **1** according to the fourteenth aspect, main fixed contact **411** and main moving contact **421** are brought into contact with and separated from each other by one end of armature **310**, and auxiliary fixed contact **611** and auxiliary moving contact **621** are brought into contact with and separated from each other by the other end of armature **310**.

According to the above configuration, main contact portion **40** (main fixed contact **411** and main moving contact **421**) can be brought into and out of contact with each other and auxiliary contact portion **60** (auxiliary fixed contact **611** and auxiliary moving contact **621**) can be brought into and out of contact with each other only by swinging one armature **310**, so that the configuration can be simplified.

[Others]

Although the contents of the electromagnetic relay according to the present disclosure have been described above, the present disclosure is not limited to these descriptions, and it is obvious to those skilled in the art that various modifications and improvements can be made.

For example, it is possible to adopt a configuration in which the configurations described in the above exemplary embodiment are appropriately combined.

34

In the above exemplary embodiment, auxiliary contact portion **60** is disposed on the upper side of coil **210**, but auxiliary contact portion **60** may be disposed in the center or the lower side of coil **210**.

In the above exemplary embodiment, auxiliary contact portion **60** is turned off when main contact portion **40** is in the on state, but auxiliary contact portion **60** can also be turned on when main contact portion **40** is in the on state. At this time, main contact portion **40** and auxiliary contact portion **60** can be so-called normally closed contact portions in which the contacts are turned on in the initial state, and main contact portion **40** and auxiliary contact portion **60** can be so-called normally open contact portions in which the contacts are turned off in the initial state.

Also, specifications (including shape, size, and layout) of the base and the case, and other details can be appropriately changed.

REFERENCE MARKS IN THE DRAWINGS

1: electromagnetic relay
10: housing
110: base
111a: bottom surface
112a: protrusion
113: partition wall
120: case
130: partition wall
131: first side wall
132: second side wall
1321: extension part
133: third side wall
1331: extension part
134: fourth side wall
1341: pressing wall
1341a: protrusion
1341c: pressing wall extended portion
1342: partition wall
1342b: partition wall extended portion
135: fifth side wall
20: electromagnetic device (drive unit)
210: coil
210a: side surface
230: iron core
232: head
240: yoke
241: vertical wall (upright portion)
310: armature
312: vertical wall
40: main contact portion
411: main fixed contact
421: main moving contact
50: movable part
60: auxiliary contact portion
611: auxiliary fixed contact
621: auxiliary moving contact
S1: internal space
S2: main contact side space (first space)
S3: auxiliary contact side space (second space)
X: front-rear direction (direction intersecting axial direction: first direction)
Y: width direction (direction intersecting axial direction and first direction: second direction)
Z: vertical direction (axial direction)

35

The invention claimed is:

1. An electromagnetic relay comprising:

a housing including a base and a case that opens downward and is provided above the base, the housing having an internal space surrounded by the base and the case;

a main fixed contact;

a main moving contact configured to come into contact with and separate from the main fixed contact;

an auxiliary fixed contact;

an auxiliary moving contact configured to come into contact with and separate from the auxiliary fixed contact;

a coil provided in the housing, a central axis of the coil extending in a vertical direction;

an iron core disposed to be surrounded by the coil;

a yoke disposed around the coil;

an armature that is disposed to face an upper end of the iron core and swings in response to excitation/non-excitation of the coil;

a movable part provided with the main moving contact and configured to move in response to swinging of the armature; and

a partition wall that divides the internal space of the housing into a first space and a second space, wherein:

the main fixed contact, the main moving contact, the auxiliary fixed contact, the auxiliary moving contact, the coil, the iron core, the yoke, the armature, the movable part, and the partition wall are provided in the internal space of the housing,

the main fixed contact and the main moving contact are disposed in the first space,

the auxiliary fixed contact and the auxiliary moving contact are disposed in the second space,

the partition wall includes a first side wall connected to the base, a pair of second side walls connected to the base, and a pair of third side walls,

when viewed along the vertical direction, the first side wall is disposed between, in a front-back direction crossing the vertical direction,

(1) the main fixed contact and the main moving contact, and

(2) the coil, the iron core, the yoke, the auxiliary fixed contact, and the auxiliary moving contact,

one of the pair of second side walls is connected to one end of the first side wall,

an other of the pair of second side walls is connected to another end of the first side wall,

one of the pair of third side walls is disposed to be adjacent to one of the pair of second side walls,

an other of the pair of third side walls is disposed to be adjacent to another of the pair of second side walls,

the coil is partially surrounded by the first side wall and the pair of second side walls, and

each of the pair of third side walls is located laterally over a side of the auxiliary fixed contact and the auxiliary moving contact.

2. The electromagnetic relay according to claim 1, wherein the auxiliary fixed contact and the auxiliary moving contact are disposed above the coil.

3. The electromagnetic relay according to claim 1, wherein each of the pair of third side walls is provided in the case to overlap corresponding one of the pair of second side walls when viewed from a side of the electromagnetic relay.

36

4. The electromagnetic relay according to claim 1, wherein each of the pair of third side walls is formed integrally with corresponding one of the pair of second side walls.

5. The electromagnetic relay according to claim 1, wherein the auxiliary fixed contact and the auxiliary moving contact are disposed between the pair of third side walls.

6. The electromagnetic relay according to claim 1, wherein

the partition wall further includes a fourth side wall protruding downward from a lower surface of an upper portion of the case, and

a lower end of the fourth side wall faces the armature.

7. The electromagnetic relay according to claim 6, wherein

the yoke includes an upright portion extending along the first side wall outside the coil,

the fourth side wall includes a pressing wall, and

the pressing wall is located between the upright portion of the yoke and the main fixed contact when viewed along the vertical direction and is configured to press the armature.

8. The electromagnetic relay according to claim 7, wherein

the pressing wall includes a pair of pressing wall extended portions, and

the armature is disposed between the pair of pressing wall extended portions.

9. The electromagnetic relay according to claim 6, wherein

the yoke includes an upright portion extending along the first side wall outside the coil,

the fourth side wall includes a partition wall, and

the partition wall is located above the upright portion of the yoke and divides a space above the armature in the internal space.

10. The electromagnetic relay according to claim 9, wherein

the partition wall includes a pair of partition wall extended portions, and

the armature is disposed between the pair of partition wall extended portions.

11. The electromagnetic relay according to claim 10, wherein a part of each of the pair of partition wall extended portions is located outside the pair of second side walls.

12. The electromagnetic relay according to claim 1, wherein each of the pair of second side walls further includes an extension part bent outward.

13. The electromagnetic relay according to claim 12, wherein

each of the pair of third side walls overlaps corresponding one of the pair of second side walls when viewed from a side of the electromagnetic relay, and

each of the pair of third side walls is provided in the case to overlap the extension part of the corresponding one of the pair of second side walls when viewed along the vertical direction.

14. The electromagnetic relay according to claim 12, wherein

the partition wall further includes a fifth side wall protruding downward from a lower surface of an upper portion of the case,

the fifth side wall is disposed outside one of the pair of second side walls, and

the fifth side wall is provided in the case to overlap the extension part of one of the pair of second side walls when viewed along the front-back direction.

15. The electromagnetic relay according to claim 1,
wherein

the main fixed contact and the main moving contact are
brought into contact with and separated from each other
by one end of the armature, and

the auxiliary fixed contact and the auxiliary moving
contact are brought into contact with and separated
from each other by another end of the armature.

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